



56XX/Personas/662x/663x Self-Service Financial Terminal

Diagnostic Status Code
Notebook - Volume 1

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Issue 1

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REVISION RECORD

Date	Page	Description
Nov. 2004	All	New manual - created from part of B006-6273-D Issue 5
Nov. 2006	Title and Back Cover	Update Revision to F
	1-117	Amend Bit 7 of Byte 29 Pick Interface - Top pick module to mention only implemented on Integrated electronics and Pre-integrated electronics dispensers.
May 2007	1-23 to 1-27	Add in USB Smart Dip Card Reader
	1-28 to 1-37	Add in USB Integrated Motorised Card Reader Writer
	1-39 to 1-47	Update and correct Bape/Hi-Bape/EPPB-Legacy and Ekc/Ekcsn/EPPE-Legacy
	1-50 to 1-57	Update and correct EPP-Enhanced
	1-58 to 1-75	Add USB EPP and USB EPP2
	1-96 to 1-100	Add USB Thermal Printers
	1-129 to 1-150	Updated Currency Dispenser to cover P86-style dispensers and USB S1 Dispenser
June 2007	1-24	Update to USB Smart DIP Reader
	1-29 to 1-30	Update to Integrated Motorised Card Reader Writer/USB Integrated Motorised Card Reader Writer
	1-40 to 1-41	Convert BAPE/HI-BAPE/EPPB-LEGACY Values to Decimal
	1-42 to 1-45	Convert EKC/EKCSN/EPPE-Legacy Values to Decimal

Revision Record

Date	Page	Description
	1-51 to 1-55	Convert EPP Enhanced Values to Decimal
	1-59 to 1-64	Updated USB EPP MStatus values and change to show in Decimal
	1-67 to 1-70	Updated USB EPP2 MStatus values and change to show in Decimal
	1-131	Updated MStatus for USB S1 Dispenser
	1-157 to 1-162	Add USB MISC I/F and related devices
March 2008	1-28 to 1-44	Added RS232 Contactless Card Reader
	1-134 and 1-135	Added Graphical Operator Panel Keyboard and Graphical Operator Panel Display
	All	Title of document changed to include 662x/663x ATMs
	1-20 to 1-24	Updated Personal Bunch Note Acceptor
July 2008	1-45 to 1-61	Added USB Contactless Card Reader
	page 1-112 to 1-132	Added USB Statement Printer
	page 1-129 to 1-139	Updated USB Thermal Printers
	1-150 to 1-155	Added USB Passbook Printer
	1-202 to 1-206	Added USB Biometric Finger Print Reader
June 2009	1-123	Added Mstatus 1 to USB Statement Printer
	1-131	Corrected Mdata Byte 14 for USB Statement Printer
	1-140 to 1-144	Modified USB Thermal Printers
	1-190	Byte 0 - changed to Divert Gate Test Command

Date	Page	Description
	1-191	Add command 1C to list
August 2009	1-128 to 1-132	Updated Mdata for Bytes 5, 15, 16 and 18
August 2010	1-106	Added 2012 Printer
	1-168	Added Intelligent Fraud Detector
	1-213	Added USB Mini Miscellaneous Interface
	1-214	Added USB Barcode Reader

Preface

This publication is Volume 1 of the *NCR 56XX/Personas/662x/663x Self-Service Financial Terminal Diagnostic Status Code Notebook*. It documents the M_STATUS and M_DATA returned in error log reports or diagnostic reports although the diagnostics tests themselves are described in the *56XX/Personas Services Aids Mini Manual (SAMM)*, (publication ID: FM-0547) or *NCR 662x/3x ATMs Service Manual* (publication ID: B006-6606).

STATUS REPORTING FORMATS

ERROR LOG REPORTS

- Hardware and High Order Comms
- Current time
- Date and time log last cleared
- Sequence number of log entry (or start and end sequence numbers if multiple entry, plus number of occurrences)
- Date and time error was logged (or date/time of first and last entry when multiple entries)
- Service name
- Severity - The M_STATUS value
- M_DATA - Up to 11 bytes of M_DATA
- If output is to the display, number of entries still to be displayed

Example

HARDWARE MODULE ERROR LOG 10:15:45
LAST CLEARED 05/07/00 12:35:40

SEQ 0062
LOGGED 07/07 15:42
NAME: PASSBOOK-PRINTER
SEVERITY 01 M_STATUS 07

M_DATA 0E 04 00 10 08 00 42 B1 44 81

*** 01 MORE ENTRIES ***
< ERROR LOG MENU MORE ENTRIES >

DIAGNOSTIC REPORTS

- Device name
- Test name
- Prompt for operator action (if any)
- Error message (if any)
- One or more of:
 - M_CODE translation
 - M_STATUS value
 - M_STATUS translation
 - M_DATA (if any)

Example 1

CURRENCY DISPENSER
PURGE

EVENT EXPECTED

M_STATUS=00
GOOD

TEST PASSED
M_STATUS=00
GOOD

M_DATA
06 00 00

Example 2

CURRENCY DISPENSER
PRESENT

TEST FAILED
M_STATUS=34
NO BILLS PROVIDED FOR OP.

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TRACK 1, 2 (2, 3) DIP MSR

M_STATUS

M_STATUS	Meaning
00	No error
01	Blank track
02	Track not supported
03	Read error (Parity LRC no end sentinel)
05	No data available
146	SDC link failure

M_DATA

- Byte 0:
 - 00H - Reset command
 - 01H - Accept
 - 02H - Read
 - 04H - Determine reader state
 - 07H - Disable
 - 0AH - Cleaning cycle
 - 0BH - Card insertion
 - 0CH - Not used
 - 0DH - Card removal
- Byte 1:
 - Bit 7 - 0 Always
 - Bit 6 - 0 Always
 - Bit 5 - 0 Always
 - Bit 4 - 0 Always
 - Bit 3 - 1 Read error (Parity/LRC/End sentinel)
 - Bit 2 - 1 Track 3 (0 always for track 1, 2 Dip MSR)
 - Bit 1 - 1 Track 2
 - Bit 0 - 1 Track 1 (0 always for track 2, 3 Dip MSR)
- Byte 2: 0 Always
- Byte 3:
 - Bit 7 - 0 Always
 - Bit 6 - 0 Always
 - Bit 5 - 0 Always

TRACK 1, 2 (2, 3) DIP MSR

- Bit 4 - 0 Always
- Bit 3 - 0 Always
- Bit 2 - 0 Always
- Bit 1 - 1 Front Switch (FRS) blocked
- Bit 0 - 1 Rear Switch (RES) blocked

T_DATA

The T_DATA returned, for the Track 1, 2 (2, 3) Dip MSR, is card data which includes start and end sentinels:

- Track 1 data will be true ASCII in the range 20H to 5FH with a start sentinel of 25H (%) and an end sentinel of 3FH.
- Track 2 data will be pseudo ASCII in the range 30H to 3FH with a start sentinel of 3BH (;) and an end sentinel of 3FH.

MCR/MCRW

M_STATUS

M_STATUS	Meaning
00	No error
01	Blank track
02	Track not supported
03	Read error
04	Write error
05	No card in reader at start of command
06	Card removed by customer during capture
07	Shutter jammed open
08	Shutter switch failure detected
09	Error recovery successful
10	Shutter jammed closed
11	Card jam
12	Card captured after unsuccessful eject
13	Possible customer tampering
14	Too many consecutive read errors or blank tracks on any track
15	Too many consecutive write errors
16	Too many consecutive card jams
17	Too many consecutive shutter jammed closed
18	Card jam during capture
19	Too many consecutive invalid cards
20	Device still inoperative
21	No card entered during cleaning cycle test
22	Speed out of specification fast
23	Speed within +3%
24	Speed within +/-1%
25	Speed within -3%
26	Speed out of specification slow
27	Invalid card detected (diagnostic only)
37	CROPF feature not supported

M_STATUS	Meaning
38	LFCC bin full soon/full
39	LFCC bin shutter mechanism fail
40	LFCC bin removed/not configured
41	CROPF feature failure
146	SDC link failure

M_DATA

- Byte 0:
 - 00H - Reset command
 - 01H - Accept
 - 02H - Read
 - 03H - Write
 - 04H - Determine reader state
 - 05H - Eject
 - 06H - Capture
 - 07H - Disable
 - 08H - Speed test
 - 09H - Shutter/Sensor test
 - 0AH - Cleaning cycle test
 - 0BH - Card fully inserted
 - 0CH - Card fully removed
 - 0DH - Card detection - being inserted or not fully removed
 - 20H - Battery/Relay test
 - 21H - LFCC bin shutter test
 - 28H - Write buffer
 - 29H - Write card
- Byte 1:
 - Bit 7 - 1 Write error (invalid data)
 - Bit 6 - 1 Write error (comms fail)
 - Bit 5 - 1 Read after write error
 - Bit 4 - 1 Read error (comms fail)
 - Bit 3 - 1 Read error (Parity or LRC)
 - Bit 2 - 1 Track 3
 - Bit 1 - 1 Track 2
 - Bit 0 - 1 Track 1

Bits 3–6 will be set for any detected occurrence of the error independent of successful retries.

- Byte 2 - For all standard commands with the exception of the battery/relay test:
 - Bit 7 - 1 Long card detected
 - Bit 6 - 1 Wrote HiCo
 - Bit 5 - 1 Card jam during capture
 - Bit 4 - 1 Card jam in reverse direction (towards shutter)
 - Bit 3 - 1 Card jam in forward direction (towards aperture bin)
 - Bit 2 - 1 Shutter switch malfunction
 - Bit 1 - 1 Shutter jammed closed
 - Bit 0 - 1 Shutter jammed open

Bits 0–7 will be set for any detected occurrence of the condition, independent of successful error recovery.

- Byte 2 - Battery/Relay test:
 - Bit 7 - 0 Always
 - Bit 6 - 0 Always
 - Bit 5 - 0 Always
 - Bit 4 - 0 Always
 - Bit 3 - 0 Always
 - Bit 2 - 1 Relay 1 failure
 - Bit 1 - 1 Relay 2 failure
 - Bit 0 - 1 Battery low
- Byte 3:
 - Bit 7 - 1 Reader is HiCo Capable (1 = true, 0 = false)
 - Bit 6 - 1 LFCC bin full
 - Bit 5 - 1 Width switch blocked
 - Bit 4 - 1 Shutter switch open
 - Bit 3 - 1 PD1 or PD2 or PD3 busy
 - Bit 2 - 1 PD3 blocked
 - Bit 1 - 1 PD2 blocked
 - Bit 0 - 1 PD1 blocked

T_DATA

The T_DATA returned, for the MCR/MCRW, is card data which includes start and end sentinels:

- Track 1 data will be true ASCII in the range 20H to 5FH with a start sentinel of 25H (%) and an end sentinel of 3FH.
- Track 2 and track 3 data will be pseudo ASCII in the range 30H to 3FH with a start sentinel of 3BH (;) and an end sentinel of 3FH.

MCR/MCRW TAMPER INDICATOR

M_DATA

- Byte 0:
 - 01 - TI reporting enabled
 - 02 - TI reporting disabled
- Byte 1:
 - 01 - LFCC bin inserted
 - 02 - LFCC bin removed

SMART CARD READER/WRITER

M_STATUS

M_STATUS values, and their meaning for the SCRW, are shown in the following table. The M_STATUS codes for the associated MCRW are given in the appropriate sections in this document.

M_STATUS	Meaning
50	Card not in SCRW stage position
51	Smart card stopper pin lowered, unable to capture. This is only in response to card capture
52	Invalid smart card command code error
53	Invalid smart card command data error
54	SCIF not configured/not responding
55	SCIF response timeout exceeded
56	Not in MCRW mode
57	Invalid smart card response length

Additional M_STATUS codes may be returned by the SCIF as shown in the following table.

M_STATUS	Meaning
00	Good
64	Invalid command code
65	Invalid command format
67	Invalid key ID
68	Invalid key type
69	Invalid host key type
70	Invalid key - does not exist
71	Invalid password - not decimal
72	Invalid password - no match
73	Key integrity error
74	Key parity error
75	Invalid data format
76	Stopper pin failure

M_STATUS	Meaning
77	Contact unit failure
78	Card movement failure
79	SC switch off failure
80	Smart card removed error
81	No smart card response
82	Smart card transmission error
83	Smart card switched off
84	Illegal card type/parameter
85	Power up time exceeded
86	ANSI X9.8 pin block error

M_DATA

In addition to the standard responses in bytes 2 and 3, entries are returned in Bytes 0, 1 and 4 as follows:

- Byte 0:
 - 70H - SCIF Command
- Byte 1:
 - SCIF Sub-command code
- Byte 4:
 - Bit 7 - Not used
 - Bit 6 - Not used
 - Bit 5 - Not used
 - Bit 4 - Not used
 - Bit 3 - Not used
 - Bit 2 - Not used
 - Bit 1 - PD5/6 (1 - stopper pin/contacts down)
 - Bit 0 - PD7 (1 - blocked)

T_DATA - Smart

The T_DATA returned for the Smart Card Reader, is intended to contain information attained from the Integrated Chip on the card.

- Byte 0 - Smart Card Interface M_Status
 - 00 (GOOD)
 - Else Hexidecimal Representation of M_Status
- Byte 1 - Smart Card Severity Code
 - S_CODE
 - 0 - Good
 - 2 - Warning
 - 4 - Fatal
- Byte 2 - Dependent on Smart card sub-command
 - for 10H (Stage)
 - 00 = Card is smart
 - 01 = Card is not smart
 - for 20H (Power on)
 - Answer To Reset (ATR) from card
 - 22H and 27H
 - Response from card

CARD IDENTIFICATION MODULE (CIM86)

M_STATUS

M_STATUS	Meaning
00	No error
01	Blank track
02	Track not supported
03	Read error
04	Write error
05	No card in reader at start of command
06	Card removed by customer during capture
07	Shutter jammed open
08	Shutter switch failure detected.
09	Error recovery successful
10	Shutter jammed closed
11	Card jam
12	Card captured after unsuccessful eject
13	Possible customer tampering
14	Too many consecutive read errors or blank tracks on any track
15	Too many consecutive write errors
16	Too many consecutive card jams
17	Too many consecutive shutter jammed closed
18	Card jam during capture
19	Too many invalid cards
20	Device still inoperative
21	No card entered during cleaning cycle test
22	Speed out of specification fast
23	Speed within +3%
24	Speed within +/-1%
25	Speed within -3%
26	Speed out of specification slow
27	Invalid card detected (diagnostic only)
33	CIM service test failed

CARD IDENTIFICATION MODULE (CIM86)

M_STATUS	Meaning
35	Error in track 3 data
37	CROPF feature not supported
38	LFCC bin full soon/full
40	LFCC bin removed/not configured
41	CROPF feature failure
87	CIM self test failed
88	CIM security module defective
89	CIM security module communications failure
90	CIM Interface pcb defective
91	CIM Interface pcb communications failure

M_DATA

- Byte 0:
 - 00H - Reset
 - 01H - Accept
 - 02H - Read
 - 03H - Write
 - 04H - Determine reader state
 - 05H - Eject
 - 06H - Capture
 - 07H - Disable
 - 08H - Speed test
 - 09H - Shutter/Sensor test
 - 0AH - Cleaning cycle test
 - 0BH - Card insertion
 - 0CH - Card removal
 - 0DH - Card detection
 - 0EH - CIM verify
 - 0FH - CIM turnaround test
 - 10H - CIM service test
 - 11H - CIM self test
 - 20H - Battery/Relay test
 - 21H - TICC bin shutter test
 - 28H - Write buffer
 - 29H - Write card
- Byte 1:
 - Bit 7 - 1 Write error (Invalid data)
 - Bit 6 - 1 Write error (Comms fail)

- Bit 5 - 1 Read after write error
- Bit 4 - 1 Read error (Comms fail)
- Bit 3 - 1 Read error (Parity or LRC)
- Bit 2 - 1 Track 3
- Bit 1 - 1 Track 2
- Bit 0 - 1 Track 1

Bits 3–6 will be set for any detected occurrence of the error, independent of successful retries.

- Byte 2 - For all commands with the exception of the battery/ relay test:
 - Bit 7 - 1 Long card detected
 - Bit 6 - 0 Always
 - Bit 5 - 1 Card jam during capture
 - Bit 4 - 1 Card jam moving towards shutter
 - Bit 3 - 1 Card jam moving towards capture bin
 - Bit 2 - 1 Shutter switch malfunction
 - Bit 1 - 1 Shutter jammed closed
 - Bit 0 - 1 Shutter jammed open

Bits 0–7 will be set for any detected occurrence of the condition, independent of successful error recovery.

- Byte 2 - Battery/relay test:
 - Bit 7 - 0 Always
 - Bit 6 - 0 Always
 - Bit 5 - 0 Always
 - Bit 4 - 0 Always
 - Bit 3 - 0 Always
 - Bit 2 - 1 Relay 2 failure
 - Bit 1 - 1 Relay 1 failure
 - Bit 0 - 1 Battery low

CARD IDENTIFICATION MODULE (CIM86)

- Byte 3:
 - Bit 7 - 0 (Unused)
 - Bit 6 - 1 LFCC bin full
 - Bit 5 - 1 Width switch blocked
 - Bit 4 - 1 Shutter switch open
 - Bit 3 - 1 PD1 or PD2 or PD3 busy
 - Bit 2 - 1 PD3 blocked
 - Bit 1 - 1 PD2 blocked
 - Bit 0 - 1 PD1 blocked
- Byte 4 (as returned to VERIFY_MM or a CIM Self Test):
 - 00H - No error
 - 11H-19H - Self test error 1 - 9
 - 40H - CIM module permanently “error”
 - 41H - CIM module permanently “busy”
 - 70H - Wrap connector attached
 - 71H - Comms timeout or open line
- Byte 5 (as returned to VERIFY_MM or a CIM Self Test):
 - 00H - No error
 - 01H - comms error
 - 09H - CIM module or interface pcb defective
 - 0AH - Error in track 3 data

The following table lists the M_DATA values which can occur in bytes 4 and 5 in the response to the level 1 SERVICE_TEST. The test cards used for this test are numbered from T1 to T11.

Card No. (Byte 4)	Parameter to be Tested	Test Result (Byte 5)	Interpretation
0	-	59H	Not a test card
1	Internal parameter OS of measurement	50H-51H	CIM Module
		52H-54H	defect
		55H-57H	OK
		58H	CIM Module defect Test failed
2	Internal parameter D of measurement (May be influenced by ambient light)	50H-53H	OK
		54H-57H	CIM Module
		58H	defect Test failed

Card No. (Byte 4)	Parameter to be Tested	Test Result (Byte 5)	Interpretation
3	External electrical spikes	50H-53H	Good
		54H-57H	Too high
		58H	Test failed
4	External electrical noise or hum	50H-53H	Good
		54H-57H	Too high
		58H	Test failed
5	Average velocity deviation of card transport	50H-53H	Good
		54H, 56H	Too fast
		55H, 57H	Too slow
		58H	Test failed
6	Short range velocity deviation of card transport	50H-55H	Good
		56H	Too fast
		57H	Too slow
		58H	Test failed
7	Distance between MM sensor and card surface	50H-53H	Good
		54H-57H	Too narrow or too far
		58H	Test failed
8	Lateral distance between MM sensor and edge of the card	50H-51H	Sensor too far left
		52H-55H	OK
		56H-57H	OK
		58H	Sensor too far right
9	Lateral distance between MM sensor and edge of the card	50H-51H	Sensor too far left
		52H-55H	left
		56H-57H	OK
		58H	Sensor too far right
10		50H	Always
11	Reflection in card transport	50H-53H	OK
		54H	Acceptable
		55H-57H	Too high
		58H	Test failed

T_DATA - CIM86 Commands

- CIM Verify

CARD IDENTIFICATION MODULE (CIM86)

Byte 0-8 (Validation Data) for CIM Service Test

- CIM Service Test

Byte 0-8 Track 3 card data

SWIPE CARD READER

M_STATUS

M_STATUS	Meaning
00	No error
01	Blank track
02	Track not supported
03	Read error - Parity error
05	No data available
95	RS232 Comms error

M_DATA

- Byte 0:
 - 00H - Reset command
 - 01H - Accept
 - 02H - Read
 - 03H - Write
 - 04H - Determine reader state
 - 05H - Eject
 - 06H - Capture
 - 07H - Disable
 - 08H - Speed test
 - 09H - Shutter/Sensor test
 - 0AH - Cleaning cycle
 - 0BH - Card swiped and fully removed
 - 0CH - Not used
 - 0DH - Card detected being swiped
 - 13H - Turnaround test
 - 28H - Write buffer
 - 29H - Write card

SWIPE CARD READER

- Byte 1:
 - Bit 7 - 0 Always
 - Bit 6 - 0 Always
 - Bit 5 - 0 Always
 - Bit 4 - 0 Always
 - Bit 3 - 1 Read error (Parity/LRC/End sentinel)
 - Bit 1 - 1 Track 2
 - Bit 0 - 1 Track 1
- Byte 2: 00 Always
- Byte 3: 00 Always

T_DATA

The T_DATA returned for the Swipe Card Reader, is card data which includes start and end sentinels. In the event of any error the T-LEN field will be 1 (not diagnostics):

- Track 1 data will be true ASCII in the range 20H to 5FH with a start sentinel of 25H (%) and an end sentinel of 3FH.
- Track 2 data will be pseudo ASCII in the range 30H to 3FH with a start sentinel of 3BH (;) and an end sentinel of 3FH.

In diagnostic mode, the card data up to any error (parity or LRC) will be returned and M_CODE, M_STATUS, and SEVERITY will reflect the error.

MIMIC CARD READER

M_STATUS

M_STATUS	Meaning
00	No error
01	Blank track
02	Track not supported
03	Read error - Parity error
05	No data available
13	Card stuck in throat
20	Device still inoperable
27	Invalid card detected (Card not fully inserted or removed)
50	Card not in correct SCRW stage position
52	Invalid smart card command code error
54	SCIF not configured/not responding
55	SCIF response time-out exceeded
56	Reader in smart card mode
57	Invalid smart card response length
95	RS232 Comms error

M_DATA

- Byte 0:
 - 00H - Reset command
 - 01H - Accept
 - 02H - Read
 - 03H - Write
 - 04H - Determine reader state
 - 05H - Eject
 - 06H - Capture
 - 07H - Disable
 - 08H - Speed test
 - 09H - Shutter/Sensor test
 - 0AH - Cleaning cycle
 - 0BH - Card fully inserted
 - 0CH - Card fully removed
 - 0DH - Card detected being inserted
 - 13H - Turnaround test
 - 28H - Write buffer
 - 29H - Write card
 - 70H - Smart Card command
- Byte 1 contents depend on the test being run, as follows.
- Byte 1 (for READ tests):
 - Bit 7 - 0 Always
 - Bit 6 - 0 Always
 - Bit 5 - 0 Always
 - Bit 4 - 0 Always
 - Bit 3 - 1 Read error (Parity/LRC/End sentinel)
 - Bit 2 - 1 Track 3
 - Bit 1 - 1 Track 2
 - Bit 0 - 1 Track 1
- Byte 1 (for ENTER tests):
 - Bits 7–1 - 0
 - Bit 1 - 0 for “Enter Card” test, 1 for “Latched Enter” test

- Byte 1 (for SMART CARD tests) indicates the sub-command used, as follows:
 - 10H - Stage
 - 11H - Release
 - 20H - Power On
 - 21H - Power Off
 - 22H - Direct
- Byte 2: 00 Always
- Byte 3: MIMIC card position switch status:
 - 30H - Front and rear switches both clear
 - 31H - Only the rear switch blocked (error condition)
 - 32H - Only the front switch blocked
 - 33H - Front and rear switches both blocked
- Byte 4: MIMIC track data available status.

Byte 4	ISO #1	ISO#2	ISO#3
30H	No data	No data	No data
31H	Data exists	No data	No data
32H	No data	Data exists	No data
33H	Data exists	Data exists	No data
34H	No data	No data	Data exists
35H	Data exists	No data	Data exists
36H	No data	Data exists	Data exists
37H	Data exists	Data exists	Data exists

- Bytes 3 and 4: On a bad response, bytes 3 and 4 contain the MIMIC error codes as follows:

Byte 3	Byte 4	Error Definition
40H	41H	Undefined command
40H	42H	Unable to execute command
40H	44H	Command data error

MIMIC CARD READER

Byte 3	Byte 4	Error Definition
41H	40H	Sensor failure
41H	49H	Other failure
42H	40H	Card has not been read yet or other errors
42H	41H	No start sentinel
42H	42H	VRC error
42H	43H	No end sentinel; too much data
42H	44H	LRC error
42H	45H	No mag. stripe
42H	47H	No data
46H	40H	IC card error
46H	41H	No ATR
46H	42H	The card does not accept this protocol
46H	43H	IC card communication error (No answer)
46H	44H	IC card communication error (Others)
46H	45H	IC card is not activated
46H	46H	ICRW can not operate this card
46H	47H	IC card needs VPP

T_DATA

The T_DATA returned for the Track 1, 2, 3 MIMIC Card Reader, is card data which includes start and end sentinels:

- Track 1 data will be true ASCII in the range 20H to 5FH with a start sentinel of 25H (%) and an end sentinel of 3FH.
- Track 2 and track 3 data will be pseudo ASCII in the range 30H to 3FH with a start sentinel of 3BH (;) and an end sentinel of 3FH.

DASH CARD READER/ USB SMART DIP
CARD READER

M_STATUS

M_STATUS	Meaning
00	Operations successful
01	Blank track
02	Not supported
03	Read error qualified by M_DATA Byte 1
05	No data available - no card inserted since last enabled to read
06	Latch mechanism failure
13	Card stuck in throat
20	Device still inoperative
27	Invalid card detected
42	Not Authorised - Device hardware has changed
81	No Smart card response, card powered off
82	Smart card communications failure, card powered off
83	Smart card drawing excess current, card powered off
84	Smart card returned unexpected data or illegal procedure byte, card powered off
85	Smart card failed to respond within timeout, card powered off
95	RS232 communications error/Comms Error (USB Smart Dip Card Reader)
96	Front Tamper Switch Triggered (USB Smart Dip Card Reader)
97	Rear Tamper Switch Triggered (USB Smart Dip Card Reader)
98	Reserved (USB Smart Dip Card Reader)
99	USB Recovered (USB Smart Dip Card Reader)

M_DATA

- Byte 0:
 - 00H - Reset command
 - 01H - Accept
 - 02H - Read
 - 03H - Write
 - 04H - Determine reader state
 - 05H - Eject
 - 06H - Capture
 - 07H - Disable
 - 08H - Speed test (not applicable to USB Smart Dip Card Reader)
 - 09H - Shutter/Sensor test
 - 0AH - Cleaning cycle (not applicable to USB Smart Dip Card Reader)
 - 0BH - Card fully inserted
 - 0CH - Card fully removed
 - 0DH - Card detected being inserted or card not fully removed
 - 13H - Turnaround test (not applicable to USB Smart Dip Card Reader)
 - 14H - LED control
 - 15H - Control latch
 - 16H - Exit read
 - 28H - Write buffer
 - 29H - Write card
 - 70H - Smart Card command

Byte 1 contents depend on the test being run, as follows:

- Byte 1 (for READ tests):
 - Bit 7 - 0 Always
 - Bit 6 - 0 Always
 - Bit 5 - 0 Always
 - Bit 4 - 0 Always
 - Bit 3 - 1 Read error (Parity/LRC/End sentinel)
 - Bit 2 - 1 Track 3
 - Bit 1 - 1 Track 2
 - Bit 0 - 1 Track 1
- Byte 1 (for ACCEPT tests):
 - Bits 7–1 - 0
 - Bit 0 - 0 for “Latch Option not selected” test
1 for “Latch Option selected” test
- Byte 1 (for DETERMINE STATE tests - not applicable to USB Smart Dip Card Reader):
 - Bit 7 - 0 Always
 - Bit 6 - 0 Always
 - Bit 5 - 0 Always
 - Bit 4 - 0 Always
 - Bit 3 - Selected Baud rate (see below)
 - Bit 2 - Selected Baud rate (see below)
 - Bit 1 - Selected Baud rate (see below)
 - Bit 0 - Selected Baud rate (see below)

Bit 0–3	Baud Rate
0	1200
1	2400
2	4800
3	9600
4	19200
5	38400

DASH CARD READER/ USB SMART DIP CARD READER

- Byte 1 (for SMART CARD tests) indicates the sub-command used, as follows:
 - 10H - Stage
 - 11H - Release
 - 20H - Power On
 - 21H - Power Off
 - 22H - Direct
- Byte 2: 00 Always
- Byte 3: DASH card position switch status/ USB Smart Dip Reader switch status:
 - 30H - Front and rear switches both clear
 - 31H - Rear switch only blocked (error condition)
 - 32H - Front switch only blocked
 - 33H - Front and rear switches both blocked
- Byte 4: DASH/USB Smart Dip track data available status.

Byte 4	ISO #1	ISO#2	ISO#3
30H	No data	No data	No data
31H	Data exists	No data	No data
32H	No data	Data exists	No data
33H	Data exists	Data exists	No data
34H	No data	No data	Data exists
35H	Data exists	No data	Data exists
36H	No data	Data exists	Data exists
37H	Data exists	Data exists	Data exists

- Bytes 3 and 4: On a bad response, bytes 3 and 4 contain the following DASH error codes/ USB Smart Dip Card Reader error codes:

Byte 3	Byte 4	Error Definition
40H	41H	Undefined command/ Card Reader not Found (for USB Smart Dip)
40H	42H	Unable to execute command/ Cable disconnected (for USB Smart Dip)
40H	44H	Command data error/ Power Down (for USB Smart Dip)

Byte 3	Byte 4	Error Definition
41H	40H	Sensor failure/ Communication no response(for USB Smart Dip)
41H	49H	Other failure/ Device Busy (for USB Smart Dip)
42H	40H	Card has not been read yet or other errors/ Lock Fail (for USB Smart Dip)
42H	41H	No start sentinel
42H	42H	VRC error
42H	43H	No end sentinel; too much data
42H	44H	LRC error
42H	45H	No magnetic stripe
42H	47H	No data
46H	40H	IC card error
46H	41H	No ATR
46H	42H	The card does not accept this protocol
46H	43H	IC card communication error (No answer)
46H	44H	IC card communication error (Others)
46H	45H	IC card is not activated
46H	46H	ICRW cannot operate this card/ No Chip Media (for USB Smart Dip)
46H	47H	IC card needs VPP/ IC card contact fail (for USB Smart Dip)

T_DATA

The T_DATA returned in response to a READ command, for the Track 1, 2, 3 DASH Card Reader and USB Smart Dip Card Reader is card data which includes start and end sentinels:

- Track 1 data will be true ASCII in the range 20H to 5FH with a start sentinel of 25H (%) and an end sentinel of 3FH.
- Track 2 and track 3 data will be pseudo ASCII in the range 30H to 3FH with a start sentinel of 3BH (;) and an end sentinel of 3FH.

RS232 CONTACTLESS CARD READER

M_STATUS

M_STATUS	Meaning
0	The device is fully operational.
5	The card is not present to perform the card command.
81	There has been a communications failure between device and card.
82	A command has been sent to the card out of sequence.
84	The device is reporting a command is being requested illegally.
96	Communication failure.

M_DATA

- Byte 0: Transaction Command Code
- Byte 1: Most Recent Device Command Code
- Byte 2: Most Recent Device Command Parameter Code (00 if not applicable)
- Byte 3: Most Recent VivoTech Device Status Code
- Byte 4: Most Recent VivoTech Error Code
- Byte 5: Most Recent APDU Status Word SW1 Code returned by the Card
- Byte 6: Most Recent APDU Status Word SW2 Code returned by the Card
- Byte 7: Most Recent RF State Code indicating exactly where the error occurred in reader\card transaction flow.
- Bytes 8-11: NCR Communications Library Status & Errors

Byte[3] VivoTech Device Status Code

Status Code	Status
00H	OK
01H	Incorrect Header Tag
02H	Unknown Command
03H	Unknown Sub-Command
04H	CRC Error in Packet
05H	Incorrect Parameter
06H	Parameter Not Supported
07H	Mal-formatted Data
08H	Timeout
0AH	Failed / Nak
0BH	Command not Allowed
0CH	Sub-Command not Allowed
0DH	Buffer Overflow (Data Length too large for Reader Buffer)
23H	Request Online Authorization

Byte[4] VivoTech Error Code

The error codes returned are specific to EMV transactions.

Error Code	Description	Reason for Error and Suggested Error Handling
00H	No Error	None.
01H	Out of Sequence Command	Terminal application is sending serial commands in the wrong sequence. Terminal should send commands in the correct sequence in terminal application code.
02H	Go to Contact Interface	If the reader supports contact interface and transaction failed in the contactless interface then can use contact interface to perform the transaction.

Error Code	Description	Reason for Error and Suggested Error Handling
03H	Transaction Amount is Zero	If the transaction amount is zero and the terminal is offline only terminal then reader need terminate the transaction.
20H	Card returned Error Status	See detailed description that follows this table.
21H	Collision Error	If more than one card in the field will generate this kind of error.
22H	Amount Over Maximum Limit	If the Transaction Amount is greater than Maximum Offline Spending Amount
23H	Request Online Authorization	If the Transaction Amount is greater than the Balance on the card but is less than the Maximum Offline Spending Amount then reader send this error code back to the terminal along with other information needed by the acquirer to format an online authorization request
25H	Card Not Supported	If the card is not supported by the reader according to the value of parameter Application Capability (FFF3) this error code will send to the terminal.
30H	Card did not respond	See detailed description that follows this table.
40H	Unknown Data Element	This Error Code is reserved for future use.
41H	Required Data Element(s) Missing	This Error Code is reserved for future use.

Error Code	Description	Reason for Error and Suggested Error Handling
42H	Card Generated AAC	The card declined the transaction by sending an AAC instead of a TC. The reason why the card declines the transaction is not known to the Reader. For MXI (Cash) transactions: At this point the amount may not have been deducted from the Card Balance. However, in this case too the terminal must ask the customer to represent the card and retry the complete transaction again (from the Activate command). This will allow the Reader to carry out exception processing. This is specially important for ticketing applications. If all goes well a Clearing Record will be sent back. If the Reader is not allowed to do exception handling, then the balance on the card may not reflect the balance expected by the customer.

Error Code	Description	Reason for Error and Suggested Error Handling
43H	Card Generated ARQC	The card declined the transaction by sending an ARQC instead of a TC. The reason why the card declines the transaction is not known to the Reader. For MXI (Cash) transactions: □ At this point the amount may not have been deducted from the Card Balance. However, in this case too the terminal must ask the customer to represent the card and retry the complete transaction again (from the Activate command). This will allow the Reader to carry out exception processing. This is especially important for ticketing applications. If all goes well a Clearing Record will be sent back. If the Reader is not allowed to do exception handling, then the balance on the card may not reflect the balance expected by the customer.

Error Code	Description	Reason for Error and Suggested Error Handling
50H	SDA / DDA / CDDA Failed (CA Public Key)	Data Authentication failed due to missing CA Public Key. Retrying the transaction will not help since the transaction will fail at this point each time, unless the missing CA Public Key problem is corrected via Key Management commands. The terminal would have to know the correct key that is to be used before it can set it in the Reader. For MXI (Cash) transactions: At this point the amount has been deducted from the Card Balance.
51H	SDA / DDA / CDDA Failed (Issuer Public Key)	Data Authentication failed due to a problem in recovering the Issuer Public Key from the card data. This could be due to incorrect data from the card or due to incorrect CA Public Key data set in the Reader. Retrying the transaction will not help since the transaction will fail at this point each time, unless the problematic data is corrected. For MXI (Cash) transactions: At this point the amount has been deducted from the Card Balance.

Error Code	Description	Reason for Error and Suggested Error Handling
52H	SDA Failed (SSAD)	Data Authentication failed during SSAD. Retrying the transaction will not help since the transaction will fail at this point each time. For MXI (Cash) transactions: At this point the amount has been deducted from the Card Balance.
53H	DDA / CDDA Failed (ICC Public Key)	Data Authentication failed during attempted recovery of ICC Public Key. Retrying the transaction will not help since the transaction will fail at this point each time. For MXI (Cash) transactions: At this point the amount has been deducted from the Card Balance.
54H	DDA / CDDA Failed (Dynamic Signature Verification)	Data Authentication failed during Dynamic Signature Verification. Retrying the transaction will not help since the transaction will fail at this point each time. For MXI (Cash) transactions: At this point the amount has been deducted from the Card Balance.

Error Code	Description	Reason for Error and Suggested Error Handling
55H	Processing Restrictions Failed	The Processing Restrictions step as defined in EMV Specifications failed. This could be due to incorrectly set EMV parameters. Retrying the transaction with the same EMV parameters will not help since the transaction will fail at this point each time. For MXI (Cash) transactions: At this point the amount has been deducted from the Card Balance.
56H	Terminal Risk Management (TRM) Failed	The Terminal Risk Management step as defined in EMV Specifications failed. This could be due to incorrectly set EMV parameters. Retrying the transaction with the same EMV parameters will not help since the transaction will fail at this point each time. For MXI (Cash) transactions: At this point the amount has been deducted from the Card Balance.

Error Code	Description	Reason for Error and Suggested Error Handling
57H	Cardholder Verification Failed	The Cardholder Verification step as defined in EMV Specifications failed. This could be due to incorrectly set EMV parameters. Retrying the transaction with the same EMV parameters will not help since the transaction will fail at this point each time. For MXI (Cash) transactions: At this point the amount has been deducted from the Card Balance.
58H	Terminal Action Analysis (TAA) Failed	The Terminal Action Analysis step as defined in EMV Specifications failed. This could be due to incorrectly set EMV parameters. Retrying the transaction with the same EMV parameters will not help since the transaction will fail at this point each time. For MXI (Cash) transactions: At this point the amount has been deducted from the Card Balance.
61H	SD Memory Error	This error will be reported only when trying to retrieve Transaction Logs. This error will never be reported during a transaction.

Value 20H

Card returned SW1SW2 not equal to 9000 hex. Value of the SW1SW2 bytes returned by Card will be returned in the Data portion of the response packet. Details of what the SW1SW2 codes mean for each RF State are Card dependent and are outwith the scope of this document.

How the terminal handles this error would depend on when in the transaction flow the error occurred. The specific transaction state during which the error occurred is indicated by the RF State Code (see section on RF State Codes) that will be reported in the data section of the response packet. Suggested error handling for individual RF State Codes is given below:

- If RF State Code = PPSE:
- If RF State Code = SELECT:
- If RF State Code = GPO:
- If RF State Code = READ RECORD:
- If RF State Code = GET DATA (Ticket):
- If RF State Code = GET DATA (Ticketing Profile):
- If RF State Code = GET DATA (Balance):
- If RF State Code = PUT DATA (Ticket):
 - The terminal could retry the transaction or abandon it.

For MXI (Cash) transactions:

- The Card amount is not deducted at this point.
- If RF State Code = GEN AC:

For Credit transactions:

- The terminal could retry the transaction or abandon it.

For MXI (Cash) transactions:

- At this point the amount may or may not have been deducted from the Card Balance. In this case the terminal **must** ask the customer to represent the card and retry the complete transaction again (from the Activate command). This will allow the Reader to carry out exception processing. If amount was already deducted, it will not be deducted again. If the amount had not been deducted, it will be deducted. If all goes well a Clearing Record will be sent back. If the Reader is not allowed to do exception handling, then the balance on the card may not reflect the balance expected by the customer.

Value 30H:

Card was removed from the field or there was a Comm Error as a result of which the card response did not reach

the Reader. How the terminal handles this error would depend on when in the transaction the error occurred. The specific transaction state during which the error occurred is given by the RF State Code. Suggested error handling for each RF State Code is given below:

- RF State Code = PPSE:
 - The terminal could retry the transaction or abandon it.

For MXI (Cash) transactions:

- The Card amount is not deducted at this point.
- If RF State Code = SELECT:

The terminal could retry the transaction or abandon it.

For MXI (Cash) transactions:

- The Card amount is not deducted at this point.
- If RF State Code = GPO:

The terminal could retry the transaction or abandon it.

For MXI (Cash) transactions:

- The Card amount is not deducted at this point.
- If RF State Code = READ RECORD:

The terminal could retry the transaction or abandon it.

For MXI (Cash) transactions:

- The Card amount is not deducted at this point.
- If RF State Code = GEN AC:

For Credit transactions:

- The terminal could retry the transaction or abandon it.

For MXI (Cash) transactions:

- At this point the amount may or may not have been deducted from the Card Balance. The terminal **must** ask the customer to represent the card and retry the transaction again (from the Activate command). This will allow the Reader to carry out exception processing. If the amount has already been deducted, it will not be deducted again. If the amount has not been deducted, it will be deducted. On success a Clearing Record will be sent back. If the Reader is not allowed to do exception handling, then the balance on the card may not reflect the balance expected by the customer.

- If RF State Code = GET DATA (Ticket):
For MXI (Cash) transactions:
 - The terminal could retry the transaction or abandon it. The Card amount is not deducted at this point.
- If RF State Code = GET DATA (Ticketing Profile):
For MXI (Cash) transactions:
 - The terminal could retry the transaction or abandon it. The Card amount is not deducted at this point.
- If RF State Code = GET DATA (Balance):
For MXI (Cash) transactions:
 - The terminal could retry the transaction or abandon it. The Card amount is not deducted at this point.
- If RF State Code = PUT DATA (Ticket):
For MXI (Cash-Ticketing) Payment at Exit-Exit transactions only:
 - In this case the terminal **must** ask the customer to represent the card and retry the complete transaction again (from the Activate command). This will allow the Reader to carry out exception processing.

Note: If an error occurs during a transaction and the terminal determines that exception processing needs to be done by the Reader, then the terminal **must** keep on retrying the transaction until the transaction has been completed successfully or the terminal decides to abort it. The retries must be continued even if successive transactions fail with conditions that do not require exception processing. This must be done to allow the Reader to complete exception processing (even if there are failures during exception processing).

Under certain conditions the terminal may want to abort the retries even if the Reader has not been able to complete exception processing, for example the customer walks away or there is a problem with the card.

ViVOTech Device RF Status Codes

For some Error Codes, the RF State Code will indicate the exact Reader-Card command that failed. This will help determine the exact place where the failure occurred.

RS232 CONTACTLESS CARD READER

For MXI (Cash) transaction, the RF State Codes together with the Error Codes will allow the terminal to determine when exception processing needs to be done by Reader.

RF State Code	RF State	Description
00H	None	RF State Code not available
01H	PPSE	Error occurred during PPSE command
02H	SELECT	Error occurred during SELECT command
03H	GPO	Error occurred during GET PROCESSING OPTIONS command
04H	READ RECORD	Error occurred during READ RECORD command
05H	GEN AC	Error occurred during GEN AC command
06H	CCC	Error occurred during CCC command
07H	IA	Error occurred during IA command
08H	SDA	Error occurred during SDA processing
09H	DDA	Error occurred during DDA processing
0AH	CDA	Error occurred during CDA processing
0BH	TAA	Error occurred during TAA processing
0CH	UPDATE RECORD	Error occurred during UPDATE RECORD command
10H	GET DATA (Ticket)	Error occurred during GET DATA command to retrieve the Ticket

RF State Code	RF State	Description
11H	GET DATA (Ticketing Prof)	Error occurred during GET DATA command to retrieve the Ticketing Profile
12H	GET DATA (Balance)	Error occurred during GET DATA command to retrieve the Balance
13H	GET DATA (All)	Error occurred during GET DATA command to retrieve all data
20H	PUT DATA (Ticket)	Error occurred during PUT DATA command to retrieve the Ticket

USB CONTACTLESS CARD READER

M_STATUS

M_STATUS	Meaning
0	The device is fully operational.
5	The card is not present to perform the card command.
42	The device is not authorised.
81	There has been a communications failure between device and card.
82	A command has been sent to the card out of sequence.
84	The device is reporting a command is being requested illegally.
96	Communication failure.

M_DATA

- Byte 0: Transaction Command Code
- Byte 1: Most Recent Device Command Code
- Byte 2: Most Recent Device Command Parameter Code (00 if not applicable)
- Byte 3: Most Recent VivoTech Device Status Code
- Byte 4: Most Recent VivoTech Error Code
- Byte 5: Most Recent APDU Status Word SW1 Code returned by the Card
- Byte 6: Most Recent APDU Status Word SW2 Code returned by the Card
- Byte 7: Most Recent RF State Code indicating exactly where the error occurred in reader\card transaction flow.

- Bytes 8-11: NCR Communications Library Status & Errors

Byte[3] VivoTech Device Status Code

Status Code	Status
00H	OK
01H	Incorrect Header Tag
02H	Unknown Command
03H	Unknown Sub-Command
04H	CRC Error in Packet
05H	Incorrect Parameter
06H	Parameter Not Supported
07H	Mal-formatted Data
08H	Timeout
0AH	Failed / Nak
0BH	Command not Allowed
0CH	Sub-Command not Allowed
0DH	Buffer Overflow (Data Length too large for Reader Buffer)
23H	Request Online Authorization

Byte[4] VivoTech Error Code

The error codes returned are specific to EMV transactions.

Error Code	Description	Reason for Error and Suggested Error Handling
00H	No Error	None.
01H	Out of Sequence Command	Terminal application is sending serial commands in the wrong sequence. Terminal should send commands in the correct sequence in terminal application code.

USB CONTACTLESS CARD READER

Error Code	Description	Reason for Error and Suggested Error Handling
02H	Go to Contact Interface	If the reader supports contact interface and transaction failed in the contactless interface then can use contact interface to perform the transaction.
03H	Transaction Amount is Zero	If the transaction amount is zero and the terminal is offline only terminal then reader need terminate the transaction.
20H	Card returned Error Status	See detailed description that follows this table.
21H	Collision Error	If more than one card in the field will generate this kind of error.
22H	Amount Over Maximum Limit	If the Transaction Amount is greater than Maximum Offline Spending Amount
23H	Request Online Authorization	If the Transaction Amount is greater than the Balance on the card but is less than the Maximum Offline Spending Amount then reader send this error code back to the terminal along with other information needed by the acquirer to format an online authorization request
25H	Card Not Supported	If the card is not supported by the reader according to the value of parameter Application Capability (FFF3) this error code will send to the terminal.
30H	Card did not respond	See detailed description that follows this table.

Error Code	Description	Reason for Error and Suggested Error Handling
40H	Unknown Data Element	This Error Code is reserved for future use.
41H	Required Data Element(s) Missing	This Error Code is reserved for future use.
42H	Card Generated AAC	The card declined the transaction by sending an AAC instead of a TC. The reason why the card declines the transaction is not known to the Reader. For MXI (Cash) transactions: At this point the amount may not have been deducted from the Card Balance. However, in this case too the terminal must ask the customer to represent the card and retry the complete transaction again (from the Activate command). This will allow the Reader to carry out exception processing. This is specially important for ticketing applications. If all goes well a Clearing Record will be sent back. If the Reader is not allowed to do exception handling, then the balance on the card may not reflect the balance expected by the customer.

Error Code	Description	Reason for Error and Suggested Error Handling
43H	Card Generated ARQC	The card declined the transaction by sending an ARQC instead of a TC. The reason why the card declines the transaction is not known to the Reader. For MXI (Cash) transactions: □ At this point the amount may not have been deducted from the Card Balance. However, in this case too the terminal must ask the customer to represent the card and retry the complete transaction again (from the Activate command). This will allow the Reader to carry out exception processing. This is especially important for ticketing applications. If all goes well a Clearing Record will be sent back. If the Reader is not allowed to do exception handling, then the balance on the card may not reflect the balance expected by the customer.

Error Code	Description	Reason for Error and Suggested Error Handling
50H	SDA / DDA / CDDA Failed (CA Public Key)	Data Authentication failed due to missing CA Public Key. Retrying the transaction will not help since the transaction will fail at this point each time, unless the missing CA Public Key problem is corrected via Key Management commands. The terminal would have to know the correct key that is to be used before it can set it in the Reader. For MXI (Cash) transactions: At this point the amount has been deducted from the Card Balance.
51H	SDA / DDA / CDDA Failed (Issuer Public Key)	Data Authentication failed due to a problem in recovering the Issuer Public Key from the card data. This could be due to incorrect data from the card or due to incorrect CA Public Key data set in the Reader. Retrying the transaction will not help since the transaction will fail at this point each time, unless the problematic data is corrected. For MXI (Cash) transactions: At this point the amount has been deducted from the Card Balance.

Error Code	Description	Reason for Error and Suggested Error Handling
52H	SDA Failed (SSAD)	Data Authentication failed during SSAD. Retrying the transaction will not help since the transaction will fail at this point each time. For MXI (Cash) transactions: At this point the amount has been deducted from the Card Balance.
53H	DDA / CDDA Failed (ICC Public Key)	Data Authentication failed during attempted recovery of ICC Public Key. Retrying the transaction will not help since the transaction will fail at this point each time. For MXI (Cash) transactions: At this point the amount has been deducted from the Card Balance.
54H	DDA / CDDA Failed (Dynamic Signature Verification)	Data Authentication failed during Dynamic Signature Verification. Retrying the transaction will not help since the transaction will fail at this point each time. For MXI (Cash) transactions: At this point the amount has been deducted from the Card Balance.

Error Code	Description	Reason for Error and Suggested Error Handling
55H	Processing Restrictions Failed	The Processing Restrictions step as defined in EMV Specifications failed. This could be due to incorrectly set EMV parameters. Retrying the transaction with the same EMV parameters will not help since the transaction will fail at this point each time. For MXI (Cash) transactions: At this point the amount has been deducted from the Card Balance.
56H	Terminal Risk Management (TRM) Failed	The Terminal Risk Management step as defined in EMV Specifications failed. This could be due to incorrectly set EMV parameters. Retrying the transaction with the same EMV parameters will not help since the transaction will fail at this point each time. For MXI (Cash) transactions: At this point the amount has been deducted from the Card Balance.

Error Code	Description	Reason for Error and Suggested Error Handling
57H	Cardholder Verification Failed	The Cardholder Verification step as defined in EMV Specifications failed. This could be due to incorrectly set EMV parameters. Retrying the transaction with the same EMV parameters will not help since the transaction will fail at this point each time. For MXI (Cash) transactions: At this point the amount has been deducted from the Card Balance.
58H	Terminal Action Analysis (TAA) Failed	The Terminal Action Analysis step as defined in EMV Specifications failed. This could be due to incorrectly set EMV parameters. Retrying the transaction with the same EMV parameters will not help since the transaction will fail at this point each time. For MXI (Cash) transactions: At this point the amount has been deducted from the Card Balance.
61H	SD Memory Error	This error will be reported only when trying to retrieve Transaction Logs. This error will never be reported during a transaction.

Value 20H

Card returned SW1SW2 not equal to 9000 hex. Value of the SW1SW2 bytes returned by Card will be returned in the Data portion of the response packet. Details of what the SW1SW2 codes mean for each RF State are Card dependent and are outwith the scope of this document.

How the terminal handles this error would depend on when in the transaction flow the error occurred. The specific transaction state during which the error occurred is indicated by the RF State Code (see section on RF State Codes) that will be reported in the data section of the response packet. Suggested error handling for individual RF State Codes is given below:

- If RF State Code = PPSE:
- If RF State Code = SELECT:
- If RF State Code = GPO:
- If RF State Code = READ RECORD:
- If RF State Code = GET DATA (Ticket):
- If RF State Code = GET DATA (Ticketing Profile):
- If RF State Code = GET DATA (Balance):
- If RF State Code = PUT DATA (Ticket):
 - The terminal could retry the transaction or abandon it.

For MXI (Cash) transactions:

- The Card amount is not deducted at this point.
- If RF State Code = GEN AC:

For Credit transactions:

- The terminal could retry the transaction or abandon it.

For MXI (Cash) transactions:

- At this point the amount may or may not have been deducted from the Card Balance. In this case the terminal **must** ask the customer to represent the card and retry the complete transaction again (from the Activate command). This will allow the Reader to carry out exception processing. If amount was already deducted, it will not be deducted again. If the amount had not been deducted, it will be deducted. If all goes well a Clearing Record will be sent back. If the Reader is not allowed to do exception handling, then the balance on the card may not reflect the balance expected by the customer.

Value 30H:

Card was removed from the field or there was a Comm Error as a result of which the card response did not reach

USB CONTACTLESS CARD READER

the Reader. How the terminal handles this error would depend on when in the transaction the error occurred. The specific transaction state during which the error occurred is given by the RF State Code. Suggested error handling for each RF State Code is given below:

- RF State Code = PPSE:
 - The terminal could retry the transaction or abandon it.

For MXI (Cash) transactions:

- The Card amount is not deducted at this point.
- If RF State Code = SELECT:

The terminal could retry the transaction or abandon it.

For MXI (Cash) transactions:

- The Card amount is not deducted at this point.
- If RF State Code = GPO:

The terminal could retry the transaction or abandon it.

For MXI (Cash) transactions:

- The Card amount is not deducted at this point.
- If RF State Code = READ RECORD:

The terminal could retry the transaction or abandon it.

For MXI (Cash) transactions:

- The Card amount is not deducted at this point.
- If RF State Code = GEN AC:

For Credit transactions:

- The terminal could retry the transaction or abandon it.

For MXI (Cash) transactions:

- At this point the amount may or may not have been deducted from the Card Balance. The terminal **must** ask the customer to represent the card and retry the transaction again (from the Activate command). This will allow the Reader to carry out exception processing. If the amount has already been deducted, it will not be deducted again. If the amount has not been deducted, it will be deducted. On success a Clearing Record will be sent back. If the Reader is not allowed to do exception handling, then the balance on the card may not reflect the balance expected by the customer.

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- If RF State Code = GET DATA (Ticket):
For MXI (Cash) transactions:
 - The terminal could retry the transaction or abandon it. The Card amount is not deducted at this point.
- If RF State Code = GET DATA (Ticketing Profile):
For MXI (Cash) transactions:
 - The terminal could retry the transaction or abandon it. The Card amount is not deducted at this point.
- If RF State Code = GET DATA (Balance):
For MXI (Cash) transactions:
 - The terminal could retry the transaction or abandon it. The Card amount is not deducted at this point.
- If RF State Code = PUT DATA (Ticket):
For MXI (Cash-Ticketing) Payment at Exit-Exit transactions only:
 - In this case the terminal **must** ask the customer to represent the card and retry the complete transaction again (from the Activate command). This will allow the Reader to carry out exception processing.

Note: If an error occurs during a transaction and the terminal determines that exception processing needs to be done by the Reader, then the terminal **must** keep on retrying the transaction until the transaction has been completed successfully or the terminal decides to abort it. The retries must be continued even if successive transactions fail with conditions that do not require exception processing. This must be done to allow the Reader to complete exception processing (even if there are failures during exception processing).

Under certain conditions the terminal may want to abort the retries even if the Reader has not been able to complete exception processing, for example the customer walks away or there is a problem with the card.

ViVOTech Device RF Status Codes

For some Error Codes, the RF State Code will indicate the exact Reader-Card command that failed. This will help determine the exact place where the failure occurred.

USB CONTACTLESS CARD READER

For MXI (Cash) transaction, the RF State Codes together with the Error Codes will allow the terminal to determine when exception processing needs to be done by Reader.

RF State Code	RF State	Description
00H	None	RF State Code not available
01H	PPSE	Error occurred during PPSE command
02H	SELECT	Error occurred during SELECT command
03H	GPO	Error occurred during GET PROCESSING OPTIONS command
04H	READ RECORD	Error occurred during READ RECORD command
05H	GEN AC	Error occurred during GEN AC command
06H	CCC	Error occurred during CCC command
07H	IA	Error occurred during IA command
08H	SDA	Error occurred during SDA processing
09H	DDA	Error occurred during DDA processing
0AH	CDA	Error occurred during CDA processing
0BH	TAA	Error occurred during TAA processing
0CH	UPDATE RECORD	Error occurred during UPDATE RECORD command
10H	GET DATA (Ticket)	Error occurred during GET DATA command to retrieve the Ticket

RF State Code	RF State	Description
11H	GET DATA (Ticketing Prof)	Error occurred during GET DATA command to retrieve the Ticketing Profile
12H	GET DATA (Balance)	Error occurred during GET DATA command to retrieve the Balance
13H	GET DATA (All)	Error occurred during GET DATA command to retrieve all data
20H	PUT DATA (Ticket)	Error occurred during PUT DATA command to retrieve the Ticket

INTEGRATED MOTORISED CARD READER WRITER (RS232 OR USB)

M_STATUS

M_STATUS	Meaning
00	No error - Operation successful
01	Blank track
02	Track not supported - SCIF write not supported
03	Read error
04	Write error - write data error
05	No card in reader (at stage position) at start of command
06	Card removed by customer during capture
07	Shutter jammed open - after card staged
08	Shutter switch failure detected
09	Error recovery successful - invalid card ejected
10	Shutter jammed closed - on entry or eject
11	Card jam - or invalid card in throat at startup
12	Card captured after unsuccessful eject
13	Possible customer tampering - card jammed in throat, unable to pull card into reader
14	Too many consecutive read errors or blank tracks on any track
15	Too many consecutive write errors
16	Too many consecutive card jams
17	Too many consecutive shutter jammed closed (not shutter test)
18	Card jam during capture - unsuccessful eject
19	Too many consecutive invalid cards
20	Device still inoperative (not diagnostics)
21	No card entered within 10 seconds
22	Speed out of specification fast
23	Speed within +3%

M_STATUS	Meaning
24	Speed within +/-1%
25	Speed within -3%
26	Speed out of specification slow
27	Invalid card detected
28	Fraud throat detected obstruction or fraud throat failure
29	Manual card insertion fraud detected
30	MM invalid
31	MM Module defective
33	For CIM/MM: service test failed For Smart command: contact unit failure
34	CIM/MM module communications failure - service test
35	Error in track 3 data - CIM/MM
37	CROPF feature not supported (diagnostic)
38	LFCC bin full soon/full
40	LFCC bin removed/not configured
41	CROPF feature failure - invalid command format (diagnostic)
42	Device not authorised (UIMCRW only)
50	Smart/SAM/Memory card not in correct SCRW stage position
51	Smart card contacts lowered - unable to capture
52	Invalid Smart/SAM/Memory card command code error - transmission error
53	Invalid Smart/SAM/Memory card command data error - switched off
54	SCIF not configured/not responding - invalid card type/parameter
55	SCIF response time-out exceeded
56	Reader is in Smart/SAM/Memory code (SCRW) mode (Not in MCRW mode)
57	Invalid Smart/SAM/Memory card response code (length)
65	Invalid Smart Card Command Format

M_STATUS	Meaning
77	Contact unit failure
78	Smart/SAM/Memory card movement failure
79	Smart/SAM/Memory card switch off failure
80	Smart/SAM/Memory card removed error
81	Smart/SAM/Memory card - no response
82	Smart/SAM/Memory card transmission error
83	Smart/SAM/Memory card switched off
84	Smart/SAM/Memory card - illegal type/ parameter
87	CIM self test failed
88	CIM security module defective
89	CIM security module communications failure
90	CIM Interface pcb defective
91	CIM Interface pcb communications failure
96	Device Communications Inoperative (UIMCRW only)
97	Device Communications Operative (UIMCRW only)
98	Device ReConnection (State Lost) (UIMCRW only)
99	Device ReConnected (New Device) [SOH/ Tallies Cleared] (UIMCRW only)

M_DATA

The following table gives an overview of the M_DATA information:

M_DATA	Meaning
Byte 0	Command code
Byte 1	Mag track errors: Read error, Write error or Smart commands: sub function code
Byte 2	Shutter jams, card jams, HiCo write, invalid cards or Battery status
Byte 3	Sensors, HiCo capability

INTEGRATED MOTORIZED CARD READER WRITER (RS232 or USB)

M_DATA	Meaning
Byte 4	ICC block information or CIM overall status
Byte 5	0 or CIM detailed status
Byte 6	Sankyo library function code
Byte 7	0 or Sankyo library error code

INTEGRATED MOTORISED CARD READER WRITER (RS232 or USB)

- Byte 0: - Command Code
 - 00H - Reset command
 - 01H - Accept
 - 02H - Read
 - 03H - Write
 - 04H - Determine reader state
 - 05H - Eject
 - 06H - Capture
 - 07H - Disable
 - 08H - Speed test
 - 09H - Shutter/Sensor test
 - 0AH - Cleaning cycle test
 - 0BH - Card fully inserted
 - 0CH - Card fully removed
 - 0DH - Card detection - being inserted or not fully removed
 - 0EH - CIM verify
 - 0FH - CIM turnaround test
 - 10H - CIM service test
 - 11H - CIM self test
 - 20H - Battery/Relay test
 - 21H - TICC bin shutter test
 - 28H - Write buffer
 - 29H - Write card
 - 70H - Smart card (SCIF) command
- Byte 1

Byte 1 can take one of two meanings:

- Magnetic track errors

or

- SCIF sub-commands (if Byte 0 = 70H)

- Mag Track Errors

- Bit 7 - 1 Write error (invalid data)
- Bit 6 - 1 Write error (Comms fail)
- Bit 5 - 1 Read after write error
- Bit 4 - 1 Read error (Comms fail)
- Bit 3 - 1 Read error (Parity or LRC)
- Bit 2 - 1 Track 3
- Bit 1 - 1 Track 2
- Bit 0 - 1 Track 1

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Bits 3–6 will be set for any detected occurrence of the error, independent of successful retries.

- SCIF Sub-commands

- 10H - Stage
 - 11H - Release
 - 20H - Power on
 - 21H - Power off
 - 22H - Smart Card Direct
 - 25H - SCIF initialise
 - 27H - EMV Smart Card Direct
 - 28H - EMV Power on
 - 30H - Contact block
 - 31H - Firmware ID
-
- Byte 2

Byte 2 can take one of two meanings:

- for commands with the exception of the Battery/Circuit test

or

- for commands for the Battery/Charge test only

- For commands with the exception of the Battery/Circuit test:

- Bit 7 - 1 Long card detected
- Bit 6 - 0 HiCo available (only when HiCo capable and HiCo enabled)
- Bit 5 - 1 Card jam during capture
- Bit 4 - 1 Short card detected
- Bit 3 - 1 Card jam
- Bit 2 - 1 Shutter switch malfunction
- Bit 1 - 1 Shutter jammed closed
- Bit 0 - 1 Shutter jammed open

Bits 0–7 will be set for any detected occurrence of the condition, independent of successful error recovery.

- For commands for the Battery/Charge test only:

- Bit 1 - 1 CROPF circuit failure
- Bit 0 - 1 Battery Low/Battery Charging
- Byte 3 - Sensor and HiCo
 - Bit 7 - Reader is HiCo Capable - 1 = true, 0 = false

INTEGRATED MOTORISED CARD READER WRITER (RS232 or USB)

- Bit 6 - 1 LFCC bin full soon/full
- Bit 5 - 1 Width switch blocked
- Bit 4 - 1 Shutter switch open
- Bit 3 - 1 PD1 or PD2 or PD3 busy
- Bit 2 - 1 PD3 blocked
- Bit 1 - 1 PD2 blocked
- Bit 0 - 1 PD1 blocked
- Byte 4

Byte 4 has two meanings:

- ICC Block Information (only returned for SCIF sub-commands if byte 0 = 70H)

or

- CIM/MM overall status (returned to VERIFY_MM or a CIM_Self_Test).

- SCIF sub-commands:

- Bit 1 - Smart card block position - 1 = contacts down
- Bit 0 - Unused

- CIM Overall Status:

- 00H - No error
- 11H-19H - Self test error 1 - 9
- 40H - CIM module permanently “error”
- 41H - CIM module permanently “busy”
- 70H - Wrap connector attached
- 71H - Comms timeout or open line
- Byte 5 - CIM Detailed Status

Returned to VERIFY_MM or a CIM Self Test:

- 00H - CIM: No error or MM: Comms error
- 01H - CIM:Comms error or MM: Quality 1
- 02H - MM Only, Quality 2
- 03H - MM Only, Quality 3
- 04H - MM Only, Quality 4
- 06H - MM Only, Quality 5
- 07H - MM Only, No MM
- 08H - MM/CIM Module Defective
- 09H - CIM module or interface pcb defective
- 0AH - Error in track 3 data

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INTEGRATED MOTORISED CARD READER WRITER (RS232 or USB)

The following table lists additional M_DATA values for CIM/MM commands which can occur in bytes 4 and 5 in the response to the level 1 SERVICE_TEST. The test cards used for this test are numbered from T1 to T11.

Card No. (Byte 4)	Parameter to be Tested	Test Result (Byte 5)	Interpretation
0	-	59H	Not a test card
1	Internal parameter OS of measurement	50H-51H 52H-54H 55H-57H 58H	CIM Module defect OK CIM Module defect Test failed
2	Internal parameter D of measurement (may be influenced by ambient light)	50H-53H 54H-57H 58H	OK CIM Module defect Test failed
3	External electrical spikes	50H-53H 54H-57H 58H	Good Too high Test failed
4	External electrical noise or hum	50H-53H 54H-57H 58H	Good Too high Test failed
5	Average velocity deviation of card transport	50H-53H 54H, 56H 55H, 57H 58H	Good Too fast Too slow Test failed
6	Short range velocity deviation of card transport	50H-55H 56H 57H 58H	Good Too fast Too slow Test failed
7	Distance between MM sensor and card surface	50H-53H 54H-57H 58H	Good Too narrow or too far Test failed

Card No. (Byte 4)	Parameter to be Tested	Test Result (Byte 5)	Interpretation
8	Lateral position of MM sensor in respect to edge of card	50H-51H	Sensor too far
		52H-55H	left
		56H-57H	OK
		58H	Sensor too far right Test failed
9	Lateral position of MM sensor in respect to edge of card	50H-51H	Sensor too far
		52H-55H	left
		56H-57H	OK
		58H	Sensor too far right Test failed
10	Internal Parameter I	50H-54H	OK
		55H	OK but sensor has low
		56H-57H	sensitivity
		58H	Sensor Defect
		59H	Test Failed No Test Card
11	Reflection in card transport	50H-53H	OK, low
		54H	reflectivity
		55H-57H	Acceptable
		58H	Reflectivity too high Test failed

- Byte 6 - Sankyo Library Function Code
If a failure occurs, this byte will contain the function number of the Sankyo library where the error occurred.
- Byte 7 - Sankyo Library Error Code
If a failure occurs, this byte will contain the Sankyo detailed error code.

T_DATA - Magnetic Tracks Read Commands

The T_DATA returned, for the IMCRW, is card data which includes start and end sentinels:

- Track 1 data will be true ASCII in the range 20H to 5FH with a start sentinel of 25H (%) and an end sentinel of 3FH.

INTEGRATED MOTORISED CARD READER WRITER (RS232 or USB)

- Track 2 and track 3 data will be pseudo ASCII in the range 30H to 3FH with a start sentinel of 3BH (;) and an end sentinel of 3FH.

T_DATA - Smart Card Commands

- Byte 0 - Smart Card Interface M_Status
 - 00 (GOOD)
 - Else hexadecimal representation of M_Status
- Byte 1 - Smart Card Severity Code
 - S_CODE
 - 0 - Good
 - 2 - Warning
 - 4 - Fatal
- Byte 2 - Dependent on Smart card sub-command
 - for 10H (Stage)
 - 00 = Card is smart
 - 01 = Card is not smart
 - for 20H (Power on)
 - Answer To Reset (ATR) from card
 - 28H (EMV Power on)
 - Answer To Reset (ATR) from card
 - 22H and 27H
 - Response from card

T_DATA - CIM Commands

- CIM Verify
 - Byte 0-8 (Validation Data) for CIM Service Test
- CIM Service Test
 - Byte 0-8 Track 3 card data

IMCRW TAMPER INDICATOR

M_DATA

- Byte 0:
 - 01 - TI reporting enabled
 - 02 - TI reporting disabled
- Byte 1:
 - 01 - LFCC bin inserted
 - 02 - LFCC bin removed

BAPE/HI-BAPE/EPPB-LEGACY

M_STATUS

M_STATUS	Meaning
00	No error
01	This error is returned with a FATAL severity if the parity of key data to be taken from the keystore is wrong. Alternatively, it is returned with a ROUTINE severity if parity checking has been selected and any byte of data supplied in C_DATA has wrong parity.
02	Refer to M_DATA for a specific instance
05	Invalid command. Refer to specific command section
13	For NBS commands, this error code is returned if there has been no key loaded in the DEU prior to requesting a decryption operation.
32	Invalid key ID, outside the maximum range of key IDs
38	Invalid key ID, the key does not exist
39	Invalid key ID, the key ID was not defined as an IV
40	Invalid key, the key ID was not defined as a key
54	Invalid mode for a particular function. Refer to specific command section.
55	Invalid left shift, value of range. Used in PIN Verification Commands.
56	Invalid length, value out of range. Used in PIN Verification Commands.
58	Invalid POST PIN PAD digit, for example > 0FH. Used in PIN Verification Commands.
59	Invalid PRE-PIN PAD data length, for example > 0CH. Used in PIN Verification Commands.
60	Invalid PIN Verification Data

M_STATUS	Meaning
75	Key ID has a parity error
77	Invalid PIN, no encryption allowed (truncated PIN does not contain decimal digits). Used in PIN Verification Commands.
146	SDC link failure

M_DATA

- Bit 0 - 1 Tamper detected
- Bit 1 - 1 Encryption chip failure
- Bit 2 - 1 Keystore failure

EKC/EKCSM/EPPE-LEGACY

M_STATUS

M_STATUS	Meaning
KEYBOARD	
00	No error
ENCRYPTOR	
00	No error
01	Parity error
02	Hardware error detected during a key or data transmission or during an encryptor operation
03	Invalid command, not all keyspaces are closed
04	Invalid command, no keyspace open
05	Invalid command as keyboard is currently enabled
06	Communications Failure - Recovered
07	Communications failure
08	Security Module reset was successful
11	Invalid command data
12	Data mismatch
13	No key loaded in DEU prior to requesting an encryption or decryption operation
15	Set up failure - key space ID1 already exists and is not a programmer parent with E/D attribute and 17 keys
16	Invalid keyspace ID format, outside range 0 to 15
17	Keyspace already exists
18	Keyspace does not exist
19	Keyspace ID1 has wrong attribute or class
20	Keyspace ID2 has wrong attribute or class
22	Invalid keyspace, wrong access right
23	Only two RAR access spaces allowed, keyspace not created

M_STATUS	Meaning
24	Secret keyspace creation disabled
25	Keyspace is not a parent
26	Key ID2 not a keyspace specified or keyspace specified not a child of key ID2 keyspace
27	Invalid parent format
28	Invalid parent ID, keyspace not a parent or keyspace does not exist
29	Parent keyspace is a programmer, can not have secret child
30	Key spaces do not match
32	Key ID outside maximum range of keys
33	Key ID1 has wrong attribute
34	Key ID2 has wrong attribute
37	Invalid key ID, no PIN encryption allowed
38	Invalid key, key does not exist
39	Key ID2 does not specify an IV
40	Invalid key ID, no key
41	Invalid key ID, key may not be exclusively ORed with key from a different keyspace
42	Invalid double key, key should be single length
43	Invalid single key, key should be double length
48	Invalid attribute field
49	Attribute already in family
50	Invalid parameter field
51	Undefined parameter for keyspace class
52	Variation index outside range 00 to 7F hex
53	Variation index already in family
54	Invalid mode for this function
55	Invalid left shift, value out of range
56	Invalid length, or length and offset outside range
57	Invalid access right
58	Invalid POST PIN PAD digit (> 0F hex)

M_STATUS	Meaning
59	Invalid PRE PIN PAD data length (> 0C hex)
60	Modulus bit length > 576 or Exponent bit length > 576 or Modulus bit length <= exponent bit length
64	Password does not contain decimal digits
65	Invalid password, no match
66	Keyspace inaccessible, password error count exceeded
69	Invalid key position code
70	PIN verification failed
73	Invalid key code table
74	Invalid PIN encrypt type function (encryption mode changed)
75	Key integrity error
76	Invalid PIN, no encryption allowed (truncated PIN does not contain decimal digits)
77	Signature Verify Failed
97	Timeout error
98	Communications error
99	TC <> 41, or TRS <> 00 00 00 00, when decrypted
128	BRS <> previous BRS, TRS = NTRS.
129	BRS <> previous BRS, TRS = OTRS.
130	BRS <> previous BRS, TRS OTRS or NTRS
131	BRS = previous BRS, TRS = NTRS.
132	BRS = previous BRS, TRS = OTRS.
133	BRS = previous BRS, TRS OTRS or NTRS
134	Invalid TC (<> 41) decrypted, or TRS <> 00 00 00 00.
146	SDC link failure

M_DATA - EKC

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ATM Self-Service Diagnostic Status Code Notebook

EKC/EKCSM/EPPE-LEGACY

- Byte 0 - Command code
- Byte 1 - USART Status (the current state of the USART):
 - Bit 0 - TxRdy, set when ready to receive a character for transmission
 - Bit 1 - RxRdy, set when a character has been received
 - Bit 2 - TxEmpty, set when USART has no characters to transmit
 - Bit 3 - 0
 - Bit 4 - Overrun error, set when a character is not read before the next one becomes available
 - Bit 5 - Framing error, set when a valid stop bit is not detected at the end of every character
 - Bit 6 - 0
 - Bit 7 - Data Set Ready (DSR), set when security module has passed its level 0
- Byte 2 - TX Failures:
 - Bit 0 set - DSR fail
 - Bit 1 set - TxEmpty not empty
- Byte 3 - RX Failures:
 - Bit 0 set - on receiving a command
 - Bit 1 set - on receiving a message
 - Bit 2 set - timeout/rubbish returned continually
 - Bit 3 set - NAKs returned continually
- Byte 4 - A string of up to eight bytes of mismatched hexadecimal data returned on an unsuccessful communications echo command

M_DATA - EKCSM

- Byte 0 - Command code:
- Byte 1:
 - 00H - Good
 - 02H - Tx Timeout
 - 03H - Rx Timeout
 - 04H - Checksum Invalid
 - 05H - Not Ready
 - 06H - GCD Hung
 - 07H - GCD NAK
 - 08H - GCD Busy
 - 09H - Invalid character
 - 0AH - GCD Fault
- Byte 2:

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- 0 - Clear
- 1 - Set

Bit	7	6	5	4	3	2	1	0
	1	Secure	Key	Light	Temp	Gore	Gore	Gore
	Always	Mode	board			3	2	1

- Byte 3 - 00H
- Byte 4 - 00H

T_DATA - EKC

The T_DATA returned for the Identity test on the EKC are:

- 8 bytes EKC serial number
- 7 bytes ROM firmware number (‘SMF XXYY’, variant number
XX = 00 to 99, version number YY = 00 to 99)
- 7 bytes EKC firmware number (‘EKC XXYY’, variant number
XX = 00 to 99, version number YY = 00 to 99)
- PEC parameter:
 - 00H - Not set
 - 01H - Set
- IPE flag:
 - 00H - Not set
 - 01H - Set

T_DATA - EKCSM

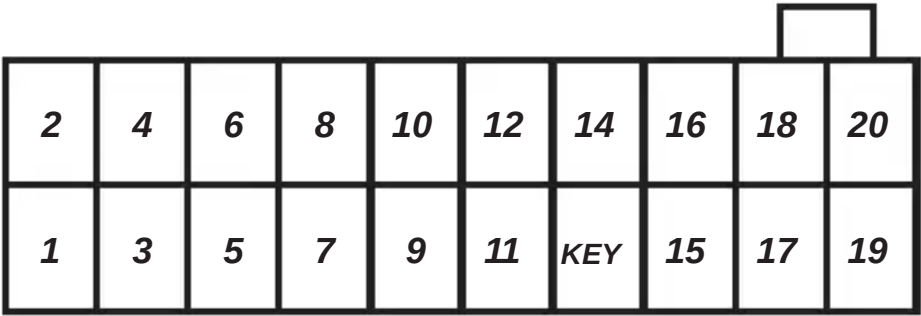
The T_DATA returned for the Identity test on the EKCSM are:

- 8 bytes GCD serial number
- 7 bytes SDC driver version (‘SDCXXYY’, version number
XX = 01 to 99, release number YY = 00 to 99)
- 7 bytes GCD version (‘SMXXYYY’, version number
XX = 01 to 99, release number YYY = 000 to 999)

BANKSYS SECURITY MODULE

TURNAROUND PLUG

A turnaround plug is required for the edge of the board turnaround test.



M_STATUS

M_STATUS	Meaning
00	Good
01	SCM sends AKO (CRC error) or service receives corrupt data from SCM (CRC error)
02	SCM sends incorrect character
03	SCM does not respond
04	SCM not ready to receive/send
05	RS232\TTL Interface not present
06	Serial port error
07	Not configured
97	Timeout error
98	Communications error

M_DATA

- Byte 0:
 - 00H - Reset
 - 02H - Send Command
 - 03H - Receive response
 - 60H - Turnaround test

- Byte 1:
 - Bit 0 - CRC error
 - Bit 1 -Parity error
 - Bit 2 - Overrun error
 - Bit 3 - Framing error
 - Bit 4 - DSR/CTS error. This error is returned if the RS232/TTL converter board is not present
 - Bit 5 - TX timeout
 - Bit 6 - RX timeout
 - Bit 7 - Not configured. This error is returned if the serial communications driver failed to open the selected serial port during configuration.

EPP-ENHANCED

Refer to the sections for EPPB-Legacy and EPPE-Legacy for the EPP M-Status information when the EPP is in a BAPE or EKCSM emulation mode.

M_STATUS

M_STATUS	Meaning
00	OK, command executed without detecting errors
01	Parity Error or Service/class (undefined command)
02	Invalid command format, length of command not according definition
03	Invalid command, not all key spaces closed
04	Invalid command, no key space open
05	Invalid sequence, keyboard is currently enabled
06	Communications Failure - Recovered
07	Communications Failure
08	Security Module Reset Successful
11	Invalid Command Data
12	Data Mismatch
13	No Key Loaded in DEU
15	Cannot perform emulation because key space ID1 already exists and is not a programmer parent with E/D attribute and 17 keys.
16	Invalid key space ID format, outside range 0 - 17
17	Invalid key space ID, key space exists already
18	Invalid key space ID, key space does not exist
19	Invalid key space ID1, wrong attribute or class

M_STATUS	Meaning
20	Invalid key space ID2, wrong attribute or class
22	Invalid key space ID, wrong access right
23	Invalid key space, only two RAR spaces allowed
24	Invalid key space, secret key space creation disabled
25	Invalid key space, no parent
26	Invalid key space, no child
27	Invalid parent ID format, outside range 0 - 17
28	Invalid parent ID, no parent key space, or does not exist
29	Invalid parent ID, programmer key space may not have secret child
30	Key spaces do not match
32	Invalid key ID format, outside range 0 - max no. of keys
33	Invalid key ID1, wrong attribute
34	Invalid key ID2, wrong attribute
37	Invalid key ID, no PIN encryption allowed (PKS differs)
38	Invalid key, key does not exist
39	Invalid key ID, no IV
40	Invalid key ID, no key
41	Invalid key ID, key may not be XOR with key from different key space
42	Invalid key, double length should be single
43	Invalid key, single length should be double
48	Invalid attribute field format
49	Invalid attribute, same attribute already in family
50	Invalid parameter field format
51	Invalid parameter, undefined parameter for this key space class
52	Invalid variation index format (range 00H - 7FH)

M_STATUS	Meaning
53	Invalid variation index, exists already in this family
54	Invalid mode for this function
55	Invalid Offset or Left Shift, outside range
56	Invalid Length or Length+Offset, outside range
57	Invalid access right
58	Invalid post PIN pad
59	Invalid pre PIN pad
60	Invalid data, values out of range
64	Invalid password, no decimal digits
65	Invalid password, password does not match
66	Inaccessible key space, password error count exceeded
69	Invalid key position code
70	PIN verification failed
71	PIN entry inhibited
73	Invalid key code table
74	Invalid PIN Encrypt Type function
75	Key integrity error
76	Invalid PIN, no encryption allowed
77	Signature Verify Failed
80	Invalid PKCS Packing
81	Invalid PKCS Unpacking
82	Public Key too big
83	Secret Key too big
85	Exponent too big
86	Modulus too big
87	Input parameter too big
88	Input parameter too small
89	Unknown Pad Block type
90	Fixed Header Decrypt Pad invalid
91	Block missing before Null Pad
92	Pad Byte Count invalid

M_STATUS	Meaning
93	Import Symmetric Key length mismatches mode
94	Invalid Serial Number
95	Invalid Random Number
96	Invalid Signature
97	Timeout Error
98	Communications Error
99	TC <> 41 or TRS <> 00 00 00 00 when decrypted
101	Root Public Key does not exist
102	Command not allowed in this mode
103	Master Key Deleted EPP Inoperable
128	BRS <> previous BRS, TRS = NTRS
129	BRS <> previous BRS, TRS = OTRS
130	BRS <> previous BRS, TRS <> OTRS or NTRS
131	BRS = previous BRS, TRS = NTRS
132	BRS = previous BRS, TRS = OTRS
133	BRS = previous BRS, TRS <> OTRS or NTRS
134	Invalid TC (<> 41) decrypted, or TRS <> 00 00 00 00
145	Invalid Key, Key not active
146	Invalid component, all zero not allowed
147	Invalid key, only one component has been entered
148	Invalid Serial Number
149	Invalid Random Number
150	Invalid signature
151	Key already exists
153	Invalid command sequence
154	Master key deleted - EPP inoperable
253	Hardware or Firmware error not covered by any response code
254	Key Stuck
255	Unknown Error

M-DATA

The M-Data returned depends on the EPP emulation mode.

M_Data EPPB

If the EPPB is being run in legacy mode, i.e. purely emulating the HI-BAPE then Byte 0 will always be 0.

If the EPPB is being run in enhanced mode (Triple DES) then the EPPB will return the same M_Data as EPPE.

M_Data EPPE

Non-direct Encryptor Commands

- Byte 0 - Command code
- Byte 1 - EPP/SDC Communications Status:
 - 00H - Good
 - 01H - Not ready
 - 02H - Null on enquire
 - 03H - Invalid byte on enquire
 - 04H - Nakd frame
 - 05H - Null on frame
 - 06H - Invalid byte on frame
 - 07H - Has no data
 - 08H - Null on poll
 - 09H - Front end nakd frame
- Byte 2:
 - 0 - Clear
 - 1 - Set

Bit	Sensor
7	Membrane Plus
6	Membrane Ground
5	Temp High
4	Temp Low
3	Battery Low
2	Battery High
1	Keyboard Switch
0	Security Mode: 0 - Secure, 1- Normal

Bits 7 -1 are active if 1.

- Byte 3 - 00H

Direct Encryptor Commands

- Byte 0 - 70H Direct Command
- Byte 1 - Command Code
- Byte 2 - EPP/SDC Communications Status:
 - 00H - Good
 - 01H - Not ready
 - 02H - Null on enquire
 - 03H - Invalid byte on enquire
 - 04H - Nakd frame
 - 05H - Null on frame
 - 06H - Invalid byte on frame
 - 07H - Has no data
 - 08H - Null on poll
 - 09H - Front end nakd frame
- Byte 3 - Security Sensors:
 - 0 - Clear
 - 1 - Set

Bit	Sensor
7	Membrane Plus
6	Membrane Ground
5	Temp High
4	Temp Low
3	Battery Low
2	Battery High
1	Keyboard Switch
0	Security Mode: 0 - Secure, 1- Normal

Bits 7 -1 are active if 1.

- Byte 4 - 00H

T_DATA

The T_Data returned for the Identity test on the EPPE are:

- 8 bytes EPP serial number
- 7 bytes SDC firmware number (‘EPPE XXX’, where XXX = 000 to 999)
- 7 bytes security module number (‘DAPI YXX’ where Y = 1 to F, XX = 00 to 99)
- PEC parameter:

EPP-ENHANCED

- 00H - Not set
 - 01H - Set
- IPE Flag:
 - 00H - Not set
 - 01H - Set

USB EPP

M_STATUS

KEYBOARD

M_STATUS	Meaning
0	Good
06	Security Module reset was successful
07	Security Module reset was unsuccessful

M_DATA

- Byte 0 - Hardware matrix code of key entered, FFH if secure data, 0 for invalid T_CODE
- or Byte 0 - 00H, no FDK is depressed, 01H - FDK is depressed

ENCRYPTOR

M_STATUS	Meaning
00	Good
01	Parity Error or Invalid Command
02	Hardware error
03	Invalid command as not all key spaces are closed
04	Invalid command no key space is currently open
05	Invalid command as keyboard is currently enabled
06	Communications Failure - Recovered
07	Communications Failure
08	The Security Module reset was successful.
11	Not all key spaces closed or Invalid Command Data
12	Data Mismatch
13	No Key Loaded in DEU

M_STATUS	Meaning
15	Cannot perform emulation because key space ID1 already exists and is not a programmer parent with E/D attribute and 17 keys.
16	Key space ID > 15, Invalid key space ID format
17	Key space already exists
18	Key space doesn't exist
19	Key space ID1 has wrong attribute or class
20	Key space ID2 has wrong attribute or class
22	Invalid key space, wrong access right
23	Only 2 RAR access spaces allowed, key space not created
24	Secret key space creation disabled
25	Key space is not a parent
26	Key ID2, not in key space specified or key space specified not a child of key ID2 key space
27	Invalid parent ID format
28	Invalid parent ID, key space not a parent or key space doesn't exist
29	Parent key space is a programmer, cannot have secret child
30	Key spaces do not match
32	Key ID outside maximum range of keys
33	Key ID1 has wrong attribute
34	Key ID2 has wrong attribute
37	Invalid key id, pin encryption not allowed as pin key space indicator of pin buffer does not match key space. This will normally occur due to multiple commands on a once entered pin.
38	Invalid key id, XOR mode was specified & key does not exist
39	Key id2 does not specify an IV

M_STATUS	Meaning
40	Invalid key id, XOR mode was specified but key was not a key
41	Invalid key id, key may not be XOR'd with key from another key space
42	Key is double length, should be single
43	Key is single length, should be double
48	Invalid attribute field
49	Attribute already in family
50	Invalid parameter field
51	Undefined parameter for key space class
52	Variation index outside range 00-7FH
53	Variation index already in family
54	Invalid mode for this function
55	Invalid offset or left shift value
56	Invalid length
57	Invalid access right
58	Invalid Post PIN Pad digit, that is > 0FH
59	Invalid Pre PIN Pad data length, that is > 0CH
60	Modulus bit length > 576 or 1024 or exponent bit length > 576 or 1024 or modulus bit length <= exponent bit length, the total user data is > modulus bit length
64	Invalid password not all decimal digits
65	Invalid password no match
66	Key space inaccessible, password error count exceeded.
69	Invalid key position code
70	PIN verification failed
71	PIN entry inhibited
73	Invalid key code table
74	Invalid pin encrypt type function, (encryption mode changed)
75	Key integrity error

M_STATUS	Meaning
76	invalid pin, no encryption allowed (truncated pin doesn't contain decimal digits)
77	Signature Verify Failed
80	Invalid PKCS Packing or USB Communications Error
81	Invalid PKCS Unpacking or Not Authorised
82	Public Key too big
83	Secret Key too big
85	Exponent too big
86	Modulus too big
87	Input parameter too big
88	Input parameter too small
89	Unknown Pad Block type
90	Fixed Header Decrypt Pad invalid
91	Block missing before Null Pad
92	Pad Byte Count invalid
93	Import Symmetric Key length mismatches mode
94	Invalid Serial Number
95	Invalid Random Number
96	Invalid Signature
97	Timeout Error
98	Key Already Exists
99	TC <> 41 or TRS <> 00 00 00 00 when decrypted
101	Root Public Key does not exist
102	Command not allowed in this mode
103	Master Key Deleted EPP Inoperable
128	BRS <> previous BRS, TRS = NTRS
129	BRS <> previous BRS, TRS = OTRS
130	BRS <> previous BRS, TRS <> OTRS or NTRS
131	BRS = previous BRS, TRS = NTRS
132	BRS = previous BRS, TRS = OTRS

M_STATUS	Meaning
133	BRS = previous BRS, TRS <> OTRS or NTRS
134	Invalid TC (<> 41) decrypted, or TRS <> 00 00 00 00
145	Invalid Key, Key not active
146	Invalid component, all zero not allowed
147	Invalid key, only one component has been entered
148	Invalid Serial Number
149	Invalid Random Number
150	Invalid signature
151	Key already exists
152	USB Communications failure
153	Invalid command sequence
154	Master key deleted - EPP inoperable
253	Hardware or Firmware error not covered by any response code
254	Key Stuck
255	Unknown Error

M_DATA

Non-direct Encryptor Commands

- Byte 0 - Command code
- Byte 1 - EPP/SDC Communications Status:
 - 00H - Good
 - 01H - Not ready
 - 02H - Null on enquire
 - 03H - Invalid byte on enquire
 - 04H - Nakd frame
 - 05H - Null on frame
 - 06H - Invalid byte on frame
 - 07H - Has no data
 - 08H - Null on poll
 - 09H - Front end nakd frame
- Byte 2:
 - 0 - Clear

USB EPP

- 1 - Set

Bit	Sensor
7	Membrane Plus
6	Membrane Ground
5	Temp High
4	Temp Low
3	Battery Low
2	Battery High
1	Keyboard Switch
0	Security Mode: 0 - Secure, 1- Normal

Bits 7 -1 are active if 1.

- Byte 3 - 00H

Direct Encryptor Commands

- Byte 0 - 70H Direct Command
- Byte 1 - Command Code
- Byte 2 - EPP/SDC Communications Status:
 - 00H - Good
 - 01H - Not ready
 - 02H - Null on enquire
 - 03H - Invalid byte on enquire
 - 04H - Nakd frame
 - 05H - Null on frame
 - 06H - Invalid byte on frame
 - 07H - Has no data
 - 08H - Null on poll
 - 09H - Front end nakd frame
- Byte 3 - Security Sensors:
 - 0 - Clear
 - 1 - Set

Bit	Sensor
7	Membrane Plus
6	Membrane Ground
5	Temp High
4	Temp Low
3	Battery Low

Bit	Sensor
2	Battery High
1	Keyboard Switch
0	Security Mode: 0 - Secure, 1- Normal

Bits 7 -1 are active if 1.

- Byte 4 - 00H

T_DATA

The T_Data returned for the Identity test on the USB EPP in EPPE mode are:

- 8 bytes EPP serial number
- 7 bytes SDC firmware number (‘EPPE XXX’, where XXX = 000 to 999)
- 7 bytes security module number (‘DAPI YXX’ where Y = 1 to F, XX = 00 to 99)
- PEC parameter:
 - 00H - Not set
 - 01H - Set
- IPE Flag:
 - 00H - Not set
 - 01H - Set

T_DATA

The T_Data returned for the Identity UEPP or Security Module test on the USB EPP are:

- 8 bytes UEPP serial number
- 7 bytes UEPP firmware number (‘EPPXXYY’, where release number XX = 00 to 99, version number YY= 00 to 99)
- 7 bytes security module number (‘SMXXYYY’ where release number XX = 00 to 99, version number YYY = 00 to 999)
- PEC parameter:
 - 00H - Not set
 - 01H - Set
- IPE Flag:
 - 00H - Not set
 - 01H - Set

USB EPP2

M_STATUS

KEYBOARD

M_STATUS	Meaning
0	Good
06	Security Module reset was successful
07	Security Module reset was unsuccessful

M_DATA

- ▣ Byte 0 - Hardware matrix code of key entered, FFH if secure data, 0 for invalid T_CODE
- ▣ or Byte 0 - 00H, no FDK is depressed, 01H - FDK is depressed

ENCRYPTOR

M_STATUS	Meaning
00	Good
01	Parity Error or Invalid Command
02	Hardware error
03	Invalid command as not all key spaces are closed
04	Invalid command no key space is currently open
05	Invalid command as keyboard is currently enabled
06	Communications Failure - Recovered
07	Communications Failure
08	The Security Module reset was successful.
11	Not all key spaces closed or Invalid Command Data
12	Data Mismatch
13	No Key Loaded in DEU

M_STATUS	Meaning
15	Cannot perform emulation because key space ID1 already exists and is not a programmer parent with E/D attribute and 17 keys.
16	Key space ID > 15, Invalid key space ID format
17	Key space already exists
18	Key space doesn't exist
19	Key space ID1 has wrong attribute or class
20	Key space ID2 has wrong attribute or class
22	Invalid key space, wrong access right
23	Only 2 RAR access spaces allowed, key space not created
24	Secret key space creation disabled
25	Key space is not a parent
26	Key ID2, not in key space specified or key space specified not a child of key ID2 key space
27	Invalid parent ID format
28	Invalid parent ID, key space not a parent or key space doesn't exist
29	Parent key space is a programmer, cannot have secret child
30	Key spaces do not match
32	Key ID outside maximum range of keys
33	Key ID1 has wrong attribute
34	Key ID2 has wrong attribute
37	Invalid key id, pin encryption not allowed as pin key space indicator of pin buffer does not match key space. This will normally occur due to multiple commands on a once entered pin.
38	Invalid key id, XOR mode was specified & key does not exist
39	Key id2 does not specify an IV

M_STATUS	Meaning
40	Invalid key id, XOR mode was specified but key was not a key
41	Invalid key id, key may not be XOR'd with key from another key space
42	Key is double length, should be single
43	Key is single length, should be double
48	Invalid attribute field
49	Attribute already in family
50	Invalid parameter field
51	Undefined parameter for key space class
52	Variation index outside range 00-7FH
53	Variation index already in family
54	Invalid mode for this function
55	Invalid offset or left shift value
56	Invalid length
57	Invalid access right
58	Invalid Post PIN Pad digit, that is > 0FH
59	Invalid Pre PIN Pad data length, that is > 0CH
60	Modulus bit length > 576 or 1024 or exponent bit length > 576 or 1024 or modulus bit length <= exponent bit length, the total user data is > modulus bit length
64	Invalid password not all decimal digits
65	Invalid password no match
66	Key space inaccessible, password error count exceeded.
69	Invalid key position code
70	PIN verification failed
71	PIN entry inhibited
73	Invalid key code table
74	Invalid pin encrypt type function, (encryption mode changed)
75	Key integrity error

M_STATUS	Meaning
76	invalid pin, no encryption allowed (truncated pin doesn't contain decimal digits)
77	Signature Verify Failed
78	Data Not Found
80	USB Communications Error
81	Not Authorised
82	Master key deleted - EPP inoperable
83	USB EPP2 Pin pad key stuck
84	Tamper detected, device inoperable
85	Firmware update failed, device operating with previous software
86	Self Test Failed
87	Hardware or Firmware error not covered by any response code

M_DATA

Non-direct Encryptor Commands

- Byte 0 - Command code
- Byte 1 - USB Status:
 - 00H - Good
- Byte 2 - Security Sensor Field 1:
 - 0 - Clear
 - 1 - Set

Bit	Sensor
7	Reserved
6	Power fail (brown-out detected)
5	Removal Switch Interrupt Detected
4	Master Key Exchange Key Not Loaded
3	Reserved
2	Reserved
1	Reserved
0	Reserved

Bits 6-4 are active if 1.

- Byte 3 - Security Sensor Field 2:
 - 0 - Clear
 - 1 - Set

Bit	Sensor
7	Reserved
6	Reserved
5	RTC power fail
4	RTC clock fail
3	Internal temperature sensor
2	Internal shield sensor
1	Reserved
0	Reserved

Bits 5-2 are active if 1.

- Byte 4 - Security Sensor Field 3:

USB EPP2

- 0 - Clear
- 1 - Set

Bit	Sensor
7	Reserved
6	Reserved
5	Reserved
4	Reserved
3	Reserved
2	Reserved
1	Snake C
0	Snake B

Bits 1-0 are active if 1.

- Byte 5 - Security Sensor Field 4:
 - 0 - Clear
 - 1 - Set

Bit	Sensor
7	Snake A
6	Snake Main
5	Reserved
4	V temp
3	V bat
2	V usb
1	V3.3
0	V Core

Bits 7-0, excluding 5 are active if 1.

Direct Encryptor Commands

- Byte 0 - 70H Direct Command
- Byte 1 - Command Code
- Byte 2 - USB Status:
 - 00H - Good
 - 01H - Not ready
 - 02H - Null on enquire
 - 03H - Invalid byte on enquire

- 04H - Nakd frame
 - 05H - Null on frame
 - 06H - Invalid byte on frame
 - 07H - Has no data
 - 08H - Null on poll
 - 09H - Front end nakd frame
- Byte 3 - Security Sensor Field 1:
 - 0 - Clear
 - 1 - Set

Bit	Sensor
7	Reserved
6	Power fail (brown-out detected)
5	Removal Switch Interrupt Detected
4	Master Key Exchange Key Not Loaded
3	Reserved
2	Reserved
1	Reserved
0	Reserved

Bits 6-4 are active if 1.

- Byte 4 - Security Sensor Field 2:
 - 0 - Clear
 - 1 - Set

Bit	Sensor
7	Reserved
6	Reserved
5	RTC power fail
4	RTC clock fail
3	Internal temperature sensor
2	Internal shield sensor
1	Reserved
0	Reserved

Bits 5-2 are active if 1.

- Byte 5 - Security Sensor Field 3:
 - 0 - Clear

USB EPP2

- 1 - Set

Bit	Sensor
7	Reserved
6	Reserved
5	Reserved
4	Reserved
3	Reserved
2	Reserved
1	Snake C
0	Snake B

Bits 1-0 are active if 1.

- Byte 6 - Security Sensor Field 4:
 - 0 - Clear
 - 1 - Set

Bit	Sensor
7	Snake A
6	Snake Main
5	Reserved
4	V temp
3	V bat
2	V usb
1	V3.3
0	V Core

Bits 7-0, excluding 5 are active if 1.

GRAPHICS/VIDEO

M_STATUS

M_STATUS	Meaning
00	No error
01	Reserved
02	Disc player is not ready to accept commands. Disc table has not been inserted or the disc is incorrectly inserted.
03	Disc player is not responding to commands. Either the link between the disc player and the SDC controller is faulty, or the disc player has been powered off and on after it has been activated, but no reset command had been sent to the SDC firmware.
04	Disc player hardware or interface fault. Either the interface to the disc player is faulty or the disc player returned a hardware error status indicating a problem with the player.
05	Graphics display adapter fault. Graphics is not available because of a fault with the display adapter or its associated software.
06	Invalid frame number was detected. Either the frame number was not in the valid range or it does not exist on the disc.
07	Invalid disc command detected. Either a video command with illegal parameters has been sent or the command sent is not allowed when a section play command is in progress.
08	Invalid command sent to disc player. Transmission error on the disc player to SDC controller interface.
09	Hit end of disc in section play. End of disc reached before section play could be complete.
10	Receive error from disc player. Overrun or framing error.

M_STATUS	Meaning
11	Drop-out error on disc. Frame cannot be found due to drop-outs or a track jump command exceeds zero address on the disc.
12	Unrecognised data from disc player. May indicate a parity error has occurred on the link between the disc player and the SDC controller.
13	Reserved
14	Disc player has not been activated
15	Reserved
16	Graphics image file not found. Check file is on the currently defined image path.
17	Graphics image file format invalid
18	Graphics image file too big. In response to a Display command, indicates that there was not room on the screen to display the whole of an image at the requested position. In response to a Highlight command, indicates that the requested image exceeds the size limits for highlighting images.
19	Video not locked. Check video cable and mixer.
20	No video signal has been obtained
21	Reserved
22	Font not loaded. This may be because it does not exist, or because it is too big to fit in the available memory.
23	Font library not found
24	Palette file not found
25	Palette file format invalid
26	Reserved
27	Mixer parameter error. A request to set mixer parameters has been received, but the parameter values are invalid.
28	Mixer missing. A request to mix video has been received, but the machine does not contain a usable mixer.

M_STATUS	Meaning
29	Font library invalid. The font of the library header and/or individual font header is invalid.
30	Mixer parameter file error. An attempt to read or write the mixer parameter disk file has failed.
31	Not enough onboard memory. The command required dynamic allocation of memory on the graphics board, but not enough was available.
32	Animation file error. An error was encountered executing the animation file. This could be invalid data in the file, or invalid file length. Check contents of file.
33	Command not supported. The command received is not supported by the current hardware, for example, a unique penny command was sent by the Pepper Sgt hardware/software.
34	Too many echo fields. More than 25 echo fields have been received by the VGM in a multiple echo mode definition. Reduce the number of echo fields.
35	File write protected. An attempt to write over a write protected file. The error is associated with either Save Graphics Screen to disk command, or writing mixer parameter file to disk operation.
64	Section play timeout. The video driver has timed out awaiting a response from the video player to mark the end of a section play.
65	Disk read error on picture download
66	Disk write error on save graphics file or mixer file update
146	SDC link failure

DIGITAL VIDEO

M_STATUS

M_STATUS	Meaning
01	Hardware error
02	Out of memory
03	MCI driver not found
04	Cannot load MCI driver
05	Invalid media type
06	Mixer parameter invalid
07	Mixer file error
08	File not found
09	Invalid command format
10	No file loaded
11	Unsupported function
12	Invalid frame number
13	Illegal command class
14	Invalid colour
15	Invalid video mix option
18	Invalid capture
19	Invalid quadrant
20	Invalid VESA code
21	Video command terminated
22	Invalid device name
23	MCI driver error
24	Out of range
25	Invalid audio select option
26	Invalid display select
27	MCI error
28	Could not open digital video device
29	Invalid Read Mixer parameter
30	Invalid Write Mixer parameter

PC VIDEO CAMERA

M_STATUS

M_STATUS	Meaning
08	Communications error

M_DATA

None

AUDIO

M_STATUS

M_STATUS	Meaning
00	No error
02	File not found
03	Path not found
04	Too many open files
05	Access denied
06	Invalid file handle
08	Insufficient memory
11	Invalid format
15	Invalid drive (Not returned within the S4 environment. M_STATUS 03 will be returned instead)
18	No more files
50	Restart failed
51	Software not found
52	Hardware not found
53	Play did not start
97	Hardware failure

M_DATA For all M_STATUS other than 53

- Byte 0 - Line output attenuation level
- Byte 1 - Speaker output attenuation level

M_DATA For M_STATUS=53

- Byte 0 - Media Control Interface (MCI) Error Code (High part)
- Byte 1 - MCI Error Code (Low part)
- Byte 2 - Calling Function ID

The MCI error codes are used by the factory during error tracing.

ENHANCED AUDIO

M_STATUS

M_STATUS	Meaning
00	No error

M_DATA

- Byte 0 - Command Class:
 - 00H - IO
 - 01H - DIAGNOSTIC
- Byte 1 - Command Code:
 - 01H - INITIALISE
 - 02H - REPORT_STATUS
 - 03H - SELECT_SOUND_DESTINATION
 - 04H - ENABLE_JACK_SOCKET_MONITORING
 - 05H -
DISABLE_JACK_SOCKET_MONITORING
- Byte 2:
 - 00H - No Jack Plug in Jack Socket
 - 01H - Jack Plug in Jack Socket
- Byte 3:
 - 00H - PUBLIC output channel active
 - 01H - PRIVATE output channel active
- Byte 4:
 - 00H - Jack Socket event monitoring disabled
 - 01H - Jack Socket event monitoring enabled
- Byte 5:
 - 00H - Semi-automatic Channel Switching Mode disabled
 - 01H - Semi-automatic Channel Switching Mode enabled
 - 02H - Automatic Channel Switching Mode
- Byte 6 - Current state of switch pack
- Byte 7 - State of switch pack during last execution of IDENTIFY_TASK_SERVICES command

RECEIPT/JOURNAL (40-COLUMN)/2012
PRINTERS

M_STATUS

M_STATUS	Meaning
00	No error (good)
01	Head jam/Knife jam
02	Not used
03	Black mark error
04	Not used
05	Paper out
06	Not used
07	Paper not loaded or paper jam before transport
08	Communications error
09	Printer open
10	Ribbon needs to be replaced soon (not thermal printer)
11	Printhead needs to be replaced soon
12	Knife needs to be replaced soon
13	Ribbon needs to be replaced now (not thermal printer)
14	Printhead needs to be replaced now
15	Sideways communications error
16	Eject jam
146	SDC link failure

M_DATA

- Byte 0:
 - Bit 7 - 0 Always
 - Bit 6 - 1 Fault status
 - Bit 5 - 0 Always
 - Bit 4 - 1 Exit sensor (covered)
 - Bit 3 - 1 Paper out sensor (active)
 - Bit 2 - 1 Black mark sensor (covered)
 - Bit 1 - 1 Paper low sensor (active)
 - Bit 0 - 0 Always
- Byte 1:
 - Bit 7 - 0 Always
 - Bit 6 - 0 Always
 - Bit 5 - 0 Transport sensor (home) (Not thermal printer)
 - Bit 4 - 1 Printhead sensor (home) (Not thermal printer)
 - Bit 3 - 1 Unit open sensor (open)
 - Bit 2 - 1 Paper not loaded status.
 - Bit 1 - 1 Paper jam sensor (covered)
 - Bit 0 - 0 Knife sensor (open)
- Byte 2:
 - Bit 7 - 0 Always
 - Bit 6 - 0 Always
 - Bit 5 - 0 Always
 - Bit 4 - 0 Always
 - Bit 3 - 1 Communications failure status
 - Bit 2 - 1 Controller RAM error status
 - Bit 1 - 1 Controller ROM error status
 - Bit 0 - 1 Interface ROM/RAM error status
- Byte 3:
 - Bit 7 - 0 Always
 - Bit 6 - 0 Always
 - Bit 5 - 0 Always
 - Bit 4 - 0 Always
 - Bit 3 - 0 Always
 - Bit 2 - 1 Failed to detect black mark status
 - Bit 1 - 1 Paper jam status
 - Bit 0 - 1 Knife jam status

STATEMENT PRINTER

M_STATUS

M_STATUS	Meaning
00	Operation successful (good)
01	Mechanical jam
02	Communications error
03	Not used
04	Black mark error
05	Media jam
05	Exit sensor error
06	Not used
07	Paper not loaded
08	Ribbon needs to be replaced now
08	Ribbon needs to be replaced soon
09	Knife needs to be replaced soon
10	Printhead needs to be replaced now.
10	Printhead needs to be replaced soon
11	Printer left open
12	Not used
13	Not used
14	Not used
15	Not used
16	Capture jam
17	Statement removed by customer during capture
18	Statement stuck in throat, unable to pull into printer - possible customer tampering
19	No paper present at start of command
20	Printer malfunction - paper present
21	Bin overfill
22	Print attempted beyond end of statement
23	Printer fatal
30	Illegal media
33	Buncher communications failure

M_STATUS	Meaning
34	Clamp drive jammed open
35	Clamp drive jammed closed
36	Capture bin full
146	SDC link failure

M_DATA

- Byte 0 - Status and Sensors:
 - Bit 7 - 0 Always
 - Bit 6 - 1 Fault
 - Bit 5 - 0 Always
 - Bit 4 - 1 Exit sensor blocked
 - Bit 3 - 1 Paper out status
 - Bit 2 - 1 Black mark at sensor
 - Bit 1 - 1 Paper low status
 - Bit 0 - 0 Always
- Byte 1 - Command or Phase:
 - 00 - Reset
 - 00 - Report tally mnemonics (Diagnostics)
 - 01 - Action print stream
 - 02 - Special print
 - 03 - Not used
 - 04 - Exercise mechanism (Diagnostics)
 - 05 - Not used
 - 06 - Eject
 - 07 - Capture
 - 08 - Determine/Disable
 - 09 - Not used
 - 0A - Book removal
- Byte 2 - Status and Sensors:
 - Bit 7 - 1 Buncher comms error
 - Bit 6 - 1 Comms error
 - Bit 5 - 1 No document
 - Bit 4 - 1 Cutter jam
 - Bit 3 - 1 Head jam
 - Bit 2 - 1 Positioned at last line
 - Bit 1 - 1 Mechanical jam
 - Bit 0 - 1 Statement jam
- Byte 3 - Status:
 - Bit 7 - 1 Printer RAM failure
 - Bit 6 - 1 Printer ROM failure

STATEMENT PRINTER

- Bit 5 - 1 Statement jam during capture
 - Bit 4 - 1 Statement jam towards shutter
 - Bit 3 - 1 Statement jam towards capture bin
 - Bit 2 - 0 Always
 - Bit 1 - 0 Always
 - Bit 0 - 0 Always
- Byte 4 - Sensors:
 - Bit 7 - 0 Always
 - Bit 6 - 0 Always
 - Bit 5 - 0 Always
 - Bit 4 - 1 Print unit open
 - Bit 3 - 1 Capture bin NOT empty
 - Bit 2 - 1 Paper out detected
 - Bit 1 - 1 Paper low (X13) detected
 - Bit 0 - 1 Black mark (X14) detected
- Byte 5 - Sensors (Statement):
 - Bit 7 - 0 Always
 - Bit 6 - 0 Always
 - Bit 5 - 0 Always
 - Bit 4 - 0 Always
 - Bit 3 - 0 Always
 - Bit 2 - 1 Printhead at right side
 - Bit 1 - 1 Printhead at left side
 - Bit 0 - 1 Knife at home position
- Byte 5 - Sensors (Buncher):
 - Bit 7 - 1 Clamp drive jammed closed
 - Bit 6 - 1 Clamp drive jammed open
 - Bit 5 - 0 Always
 - Bit 4 - 1 Clamp drive raised
 - Bit 3 - 1 Clamp drive lowered
 - Bit 2 - 1 Printhead at right side
 - Bit 1 - 1 Printhead at left side
 - Bit 0 - 1 Knife at home position
- Byte 6 - Sensors (Statement):
 - Bit 7 - 0 Always
 - Bit 6 - 1 Statement transport (S3) sensor covered
 - Bit 5 - 1 Statement printhead sensor (S4) covered
 - Bit 4 - 0 Always
 - Bit 3 - 0 Always
 - Bit 2 - 0 Always
 - Bit 1 - 1 Statement feed sensor (X12) covered
 - Bit 0 - 1 Statement capture sensor (X17) covered

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- Byte 6 - Sensors (Buncher):
 - Bit 7 - 0 Always
 - Bit 6 - 0 Always
 - Bit 5 - 0 Always
 - Bit 4 - 0 Always
 - Bit 3 - 0 Always
 - Bit 2 - 0 Always
 - Bit 1 - 1 Statement feed sensor (X12) covered
 - Bit 0 - 1 Statement capture sensor (X17) covered
- Byte 7 - Sensors (Statement):
 - Bit 7 - 0 Always
 - Bit 6 - 0 Always
 - Bit 5 - 0 Always
 - Bit 4 - 0 Always
 - Bit 3 - 0 Always
 - Bit 2 - 1 Statement exit sensor (S1) covered
 - Bit 1 - 0 Always
 - Bit 0 - 1 Statement transport sensor (S2) covered
- Byte 7 - Sensors (Buncher):
 - Bit 7 - 0 Always
 - Bit 6 - 0 Always
 - Bit 5 - 0 Always
 - Bit 4 - 0 Always
 - Bit 3 - 1 Media exit sensor (B1) covered
 - Bit 2 - 0 Always
 - Bit 1 - 1 Stack area sensor (B2) covered
 - Bit 0 - 1 Primary transport sensor (B3) covered

USB STATEMENT PRINTER

MSTATUS

MStatus (Dec)	Meaning
0	Printer OK
1	Print Engine Jam
2	Comms Error
3	Passbook Transport Cover Open
4	Black Mark Error
5	Jam Near Eject/Transport Sensor
7	Paper Not Loaded/Jam before Transport
9	Knife Life Count Nearly Reached
10	Printhead Life Count Nearly Reached
11	Printer Head Open
16	Capture Jam
18	Exit Jam On Capture
19	No Media Present in Transport
36	Capture Bin Full
40	Paper Low
41	Paper Out
42	Head Jam / Knife Jam
43	Power Supply Voltage Error
44	Control Board Error
45	Transport Not Connected On Power Up
46	Print Head Over Temperature (Warning). 10s Transient status. Not reported.
47	Print Head Over Temperature (Error).
48	Print Engine Jam (CAM Motor)
49	Media In Transport
50	Dual Feed Mechanism Jam
51	Buncher Jam
52	Buncher Capture Jam
53	Buncher Present Jam
54	Buncher PS Home Jam
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MStatus (Dec)	Meaning
55	Buncher Shutter Home Error
56	Buncher Position 1 Home Jam
57	Buncher Flapper Error
58	Buncher Cover Open
59	Upper Cover Open
60	Lower Cover Open
61	Buncher Full
62	Buncher Too Much Data
63	Passbook Transport Busy
64	Passbook Shutter Jam
65	Different Black Mark Paper In Both Hoppers

M_DATA

Statement Printer (1 or 2-sided)

- Byte #1 : Sensor status-1
 - Bit 7 - BD sensor (Black Dot), 1 = Black
 - Bit 6 - PE sensor (Paper End of Print engine), 1 = Exhausted
 - Bits 1 - 5 Reserved, always 0
 - Bit 0 - Paper low (#1) sensor, 1 = Paper Low
- Byte #2 : Sensor status-2
 - Bit 7 - CAM Motor home position sensor, 1 = Home
 - Bit 6 - Reserved, always 0
 - Bit 5 - Cover Open Sensor, 1 = Open
 - Bit 4 - CBF sensor (Capture Bin Full), 1 = Full
 - Bit 3 - CB sensor (Capture Bin), 1 = Empty
 - Bit 2 - Exit sensor (Transport exit), 1 = Blocked
 - Bit 1 - TJ sensor (Transport Jam), 1 = Blocked
 - Bit 0 - HP sensor (Cutter Home Position), 1 = Home

USB STATEMENT PRINTER

- Byte #3 : Sensor status-3
 - Bit 7 - TOF Paper Jam, 1 = Jam
 - Bit 6 - CAM Motor Jam, 1 = Jam
 - Bit 5 - Print Engine Paper Jam, 1 = Jam
 - Bit 4 - Exit sensor, 1 = Blocked
 - Bit 3 - 2ST Paper Auto Detection, 1 = Black
 - Bit 2 - Top of Form sensor, 1 = Blocked
 - Bit 1 - Reserved, always 0
 - Bit 0 - Feed Switch (Standard mode), 1 = Pushed
- Byte #4 : Motor status
 - Bit 7 - CAM Motor, 1 = Going
 - Bit 6 - Printing, 1 = In progress
 - Bits 3 - 5 Reserved, always 0
 - Bit 2 - Transport motor, 1 = Going
 - Bit 1 - Cutter motor, 1 = Going
 - Bit 0 - Feed motor, 1 = Going
- Byte #5 : Printer status
 - Bit 7 - Fault state, 1 = Fault
 - Bit 6 - Warning due to Overheated, 1 = Overheated
 - Bit 5 - Non Transport Detection Error, 1 = Error
 - Bit 4 - Warning - Media not cleared after Manual feed , 1 = Remaining media
 - Bit 3 - Paper Type by Auto Paper Detection, 1 = Two side
 - Bit 2 - Paper Feed, 1 = Feed
 - Bit 1 - Reboot flag , 1 = After reboot
 - Bit 0 - Recovery running, 1 = Yes

- Byte #6 : Error status-1
 - Bit 7 - Black mark error, 1 = Error
 - Bit 6 - Thermal head temperature, 1 = Error
 - Bit 5 - Reserved, always 0
 - Bit 4 - Transport jam, 1 = Jam
 - Bit 3 - Cutter jam, 1 = Jam
 - Bit 2 - Reserved, always 0
 - Bit 1 - Flash Erase/Write error or Memory full, 1 = Error
 - Bit 0 - H/W error -unrecoverable- (ROM, RAM, etc), 1 = Error
- Byte #7 : Error status-2
 - Bit 7 - Capture Jam (inc. Customer tampering), 1 = Jam
 - Bit 6 - Cover Open, 1 = Open
 - Bit 5 - Paper End (Print engine), 1 = Exhausted
 - Bits 3 - 4 Reserved, always 0
 - Bit 2 - Unable to load paper, 1 = Unable
 - Bits 0 - 1 Reserved, always 0
- Byte #8 : Warning
 - Bit 7 - Illegal sequence or parameter, 1 = Present
 - Bit 6 - 24V Anomaly, 1 = Anomaly
 - Bit 5 - Capture - Customer tampering, 1 = Tampering
 - Bit 4 - Reserved, always 0
 - Bit 3 - Capture bin full, 1 = Full
 - Bit 2 - Executed Present, Purge or Retract w/o a receipt, 1 = Executed
 - Bit 1 - Reserved, always 0
 - Bit 0 - Paper low sensor, 1 = Paper Low

Statement Printer Bunch Feeder and Statement/Passbook variants

- Byte 1 - Sensor status 1
 - Bit 7 - BD1 sensor (Black Dot - Left side, Kinetics), 1 = Black
 - Bit 6 - PE sensor (Paper End of Print engine), 1 = Exhausted
 - Bit 5 - DF_2b sensor (Upper feeder positioning sensor), 1 = Blocked

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USB STATEMENT PRINTER

- Bit 4 - DF_1b sensor (Lower feeder positioning sensor), 1 = Blocked
- Bit 3 - DF_2a sensor (Upper feeder PE sensor), 1 = Empty
- Bit 2 - DF_1a sensor (Lower feeder PE sensor), 1 = Empty
- Bit 1 - Paper low sensor for Upper feeder, 1 = Paper Low
- Bit 0 - Paper low sensor for Lower feeder, 1 = Paper Low

- Byte 2 - Sensor status 2
 - Bit 7 - CAM Motor Home position sensor, 1 = Home
 - Bit 6 - always 0
 - Bit 5 - Cover Open Sensor for Print Engine, 1 = Open
 - Bit 4 - CBF sensor (Capture Bin Full), 1 = Full
 - Bits 1 - 3 Reserved, always 0
 - Bit 0 - HP sensor (Cutter Home Position), 1 = Home

- Byte 3 - Sensor status 3
 - Bit 7 - Top of Form Paper Jam, 1 = Jam
 - Bit 6 - CAM Motor Jam, 1 = Jam
 - Bit 5 - Print Engine Paper Jam, 1 = Jam
 - Bit 4 - Exit sensor, 1 = Blocked
 - Bit 3 - 2ST Paper Auto Detection, 1 = Black
 - Bit 2 - TOF Sensor status, 1 = Blocked
 - Bit 1 - Upper Feed Switch, 1 = Pushed
 - Bit 0 - Lower Feed Switch, 1 = Pushed

- Byte 4 - Motor status
 - Bit 7 - CAM Motor, 1 = Going
 - Bit 6 - Printing , 1 = In progress
 - Bit 5 - always 0
 - Bit 4 - Upper feed motor, 1 = Going
 - Bit 3 - Lower feed motor, 1 = Going
 - Bit 2 - Reserved, always 0
 - Bit 1 - Cutter motor, 1 = Going
 - Bit 0 - Feed motor, 1 = Going

- Byte #5 : Printer status
 - Bit 7 - Fault state, 1 = Fault
 - Bit 6 - Warning due to Overheated, 1 = Overheated
 - Bit 5 - Non Transport Detection Error, 1 = Error
 - Bit 4 - Warning - Media not cleared after Manual feed , 1 = Remaining media
 - Bit 3 - Paper Type by Auto Paper Detection (on Buncher Variants for lower feeder), 1 = Two side
 - Bit 2 - Paper Feed, 1 = Feed
 - Bit 1 - Reboot flag , 1 = After reboot
 - Bit 0 - Recovery running, 1 = Yes

- Byte #6 : Error status-1
 - Bit 7 - Black mark error, 1 = Error
 - Bit 6 - Thermal head temperature, 1 = Error
 - Bits 4 - 5 Reserved, always 0
 - Bit 3 - Cutter jam, 1 = Jam
 - Bit 2 - Reserved, always 0
 - Bit 1 - Firmware download error, 1 = Error
 - Bit 0 - H/W error -unrecoverable- (ROM, RAM, etc), 1 = Error

USB STATEMENT PRINTER

- Byte #7 : Error status-2
 - Bit 7 - Reserved , always 0
 - Bit 6 - Cover Open, 1 = Open
 - Bit 5 - Paper End (Print engine), 1 = Exhausted
 - Bit 4 - Paper End Upper Feeder, 1 = Empty
 - Bit 3 - Paper End Lower Feeder, 1 = Empty
 - Bit 2 - Unable to load paper, 1 = Unable
 - Bit 1 - Unable to load paper in Upper Feeder, 1 = Unable
 - Bit 0 - Unable to load paper in Lower Feeder, 1 = Unable
- Byte #8 : Warning
 - Bit 7 - Illegal sequence or parameter, 1 = Present
 - Bit 6 - 24V Anomaly, 1 = Anomaly
 - Bit 5 - Capture - Customer tampering, 1 = Tampering
 - Bit 4 - Reserved, always 0
 - Bit 3 - Capture bin full, 1 = Full
 - Bit 2 - Executed Present, Purge or Retract w/o a receipt, 1 = Executed
 - Bit 1 - Upper Paper low sensor, 1 = Paper Low
 - Bit 0 - Lower Paper low sensor, 1 = Paper Low

Note Bytes 9 - 13 are applicable to the Statement Passbook variants. The Bunch Feeder variants report these bytes as zero.

- Byte #9, Sensor status on PB-1
 - Bit - 7 Upper Alignment Roller, 1 = Not Home
 - Bit - 6 Platen up/down, 1 = Up
 - Bit - 5 Reserved, always 0
 - Bit - 4 Front Pinch Up/Down, 1 = Up
 - Bit - 3 Reserved (HMS CA motor, Sensor) -
 - Bit - 2 Alignment Roller Up/Down, 1 = Up
 - Bit - 1 PB Right AL Shutter, 1 = Up
 - Bit - 0 Exit Pinch Up/Down, 1 = Up

- Byte #10: Sensor status on PB-2
 - Bit 7 - Reserved, always 0
 - Bit 6 - Shutter Open , 1 = Closed
 - Bit 5 - CA Home Position , 1 = not Home Position
 - Bit 4 - Front upper unit cover open , 1 = Closed
 - Bit 3 - PRT unit cover open , 1 = Closed
 - Bits 0 - 2 Reserved, always 0
- Byte #11: Sensor status on PB-3
 - Bit 7 - P6 Length check (max), 1 = paper present
 - Bit 6 - P5 Length check (min), 1 = paper present
 - Bit 5 - P4 Bottom edge of MS, 1 = paper present
 - Bit 4 - P3 Front Shutter, 1 = paper present
 - Bit 3 - P2B Media width check Left (B), 1 = paper present
 - Bit 2 - P2A Media width check Left (A), 1 = paper present
 - Bit 1 - P1 Media width check Right, 1 = paper present
 - Bit 0 - P0 Media width check Right, 1 = paper present
- Byte #12: Sensor status on PB-4
 - Bit 7 - Reserved, always 0
 - Bit 6 - Capture Bin Sensor, 1 = paper present
 - Bit 5 - Reserved, always 0
 - Bit 4 - P11 ST Capture Bin exit, 1 = paper present
 - Bit 3 - P10 ST Paper Jam, 1 = paper present
 - Bit 2 - P9 PLF Roller Up/down sensor, 1 = paper present
 - Bit 1 - P8 Right Align sensor (REAR), 1 = paper present
 - Bit 0 - P7 Right Align sensor (FR), 1 = paper present

USB STATEMENT PRINTER

- Byte #13: Error status on PB-5
 - Bit 7 - Paper on the Transport including Present position , 1 = Present
 - Bit 6 - Reserved, always 0
 - Bit 5 - Reserved, always 0
 - Bit 4 - Capture Jam, 1 = Jam
 - Bit 3 - Shutter Jam, 1 = Jam
 - Bit 2 - Transport Jam, 1 = Jam
 - Bit 1 - PB (Transport) Cover Open, 1 = Open
 - Bit 0 - PB (Transport) busy, 1 = busy

Note: Bytes 14 - 18 are applicable to the Bunch Feeder Variants, the Statement Passbook variants report these bytes as zero.

- Byte #14 : Sensor Status - Bunch and Dual Feeder
 - Bit 7 - Bunch Position1 HP sensor (S09), 1 = Not Home
 - Bit 6 - Bunch Paper Stopper (S07), 1 = Not Home
 - Bit 5 - Bunch Capture Bin sensor (S06), 1 = Present
 - Bit 4 - Bunch Paper Exit sensor (S05), 1 = Covered
 - Bit 3 - Bunch paper position sensor (S04), 1 = Covered
 - Bit 2 - Bunch timing sensor (S02), 1 = Down
 - Bit 1 - Cover open sensor for Upper feeder, 1 = Open
 - Bit 0 - Cover open sensor for Lower feeder, 1 = Open

- Byte #15 : Sensor Status - Bunch
 - Bit 7 - Paper Type by Auto Paper Detection for Upper Feeder, 1 = Two Side
 - Bit 6 - BD3 sensor (Black Dot) - Left side (Side-W position) , 1 = Black
 - Bit 5 - BD2 sensor (Black Dot) - Center, 1 = Black
 - Bit 4 - Bunch Cover open2 sensor (S15), 1 = Open
 - Bit 3 - Bunch Cover open1 sensor (S14), 1 = Open
 - Bit 2 - Bunch Capture Flapper2 (S13), 1 = Closed
 - Bit 1 - Bunch Capture Flapper1 (S12), 1 = Closed
 - Bit 0 - Bunch Shutter HP sensor (S11), 1 = Not Home

- Byte #16 : Motor status - Bunch
 - Bit 7 - Black Mark Paper type for Upper Feeder, 1 = Centre (Side-K)
 - Bit 6 - Black Mark Paper type for Lower Feeder, 1 = Centre (Side-K)
 - Bit 5 - Bunch Motor 06, 1 = Going
 - Bit 4 - Bunch Motor 05, 1 = Going
 - Bit 3 - Bunch Motor 04, 1 = Going
 - Bit 2 - Bunch Motor 03, 1 = Going
 - Bit 1 - Bunch Motor 02, 1 = Going
 - Bit 0 - Bunch Motor 01, 1 = Going

USB STATEMENT PRINTER

- Byte #17 : Error status - Bunch and Dual Feeder
 - Bit 7 - Bunch Position1 HP Error, 1 = Error
 - Bit 6 - Bunch Shutter HP Error, 1 = Error
 - Bit 5 - Bunch Paper Stopper HP Jam, 1 = Error
 - Bit 4 - Bunch Paper Jam - Capture motion, 1 = Error
 - Bit 3 - Bunch Paper Jam - Present motion, 1 = Error
 - Bit 2 - Bunch Paper Jam - Bunching motion, 1 = Error
 - Bit 1 - Cover Open Error - Upper Feeder, 1 = Error
 - Bit 0 - Cover Open Error - Lower feeder, 1 = Error
- Byte #18 : Error status - Bunch
 - Bit 7 - Paper at Present position, 1= Present
 - Bit 6 - Unable to initialize due to remaining paper, 1 = Unable
 - Bit 5 - Print data are beyond the next BM, 1 = Over page
 - Bit 4 - Unload Paper due to idle state (Warning), 1 = Unload position
 - Bit 3 - Reserved, always 0
 - Bit 2 - Black mark pitch is different between Upper and Lower Feeder, 1 = Different pitch
 - Bit 1 - Cover Open1 Error, 1 = Error
 - Bit 0 - Bunch Capture Flapper Error, 1 = Error

80-COLUMN THERMAL PRINTER

M_STATUS

M_STATUS	Meaning
00	Operation successful
01	Mechanical jam
02	Communications error (fatal)
03	Not used
04	Black mark error (fatal)
05	Media jam (fatal)
05	Exit sensor error (warning)
06	Not used
07	Paper out (fatal)
08	Ribbon needs to be replaced now (fatal)*
08	Ribbon needs to be replaced soon (warning)*
09	Knife needs to be replaced soon (warning)
10	Printhead needs to be replaced now (fatal)*
10	Printhead needs to be replaced soon (warning)*
11	Printer left open (fatal)
12	Not used
13	Not used
14	Not used
15	Not used
16	Capture jam (fatal)
17	Media removed by customer during capture (routine)
18	Media stuck in throat, unable to pull into printer (suspend)
19	No media present at start of command (routine)
20	Printer malfunction- media present (routine)
21	Bin overfill (fatal)
22	Print attempted beyond end of statement (routine)

M_STATUS	Meaning
23	Printer fatal (fatal)
30	Illegal media (routine)
33	Buncher Comms failure (fatal)#
34	Clamp Drive jammed open (fatal)#
35	Clamp Drive jammed closed (fatal)#
36	Capture Bin full (warning)

Applies to buncher only version.

* These error messages do not apply to the thermal printer. However, when they are generated, “*State of Health*” clearance in “*Supervisory Mode*” is required. Refer to Customer documentation for more information.

M_DATA

- Byte 0 - Status and Sensors:
 - Bit 7 - 0 Always*
 - Bit 6 - 1 Fault status
 - Bit 5 - 0 Always
 - Bit 4 - 1 Exit sensor (S1) ¹ covered
 - Bit 3 - 1 Paper set sensor (ES6) active
 - Bit 2 - 1 Black dot sensors (ES4&5) covered
 - Bit 1 - 1 Paper low status (S11/12) active
 - Bit 0 - 0 Always

¹ See heading “Sensor Locations”

* Signifies bit is set to zero

- Byte 1 - Command or Phase:
 - 0A= Not used
 - 09 - Not used
 - 08 - Determine/Disable
 - 07 - Capture
 - 06 - Eject
 - 05 - Not used
 - 04 - Exercise mechanism (Diagnostics Only)
 - 03 - Not used
 - 02 - Special print
 - 01 - Action print stream
 - 00 - Report tally mnemonics (Diagnostics)
 - 00 - Reset
- Byte 2 - Status and Sensors:
 - Bit 7 - 0 Buncher comms error
 - Bit 6 - 1 Comms error
 - Bit 5 - 1 No document
 - Bit 4 - 1 Cutter jam
 - Bit 3 - 0 Always
 - Bit 2 - 1 Positioned at last line
 - Bit 1 - 1 Mechanical jam
 - Bit 0 - 1 Media jam
- Byte 3 - Status:
 - Bit 7 - 1 Printer RAM failure
 - Bit 6 - 1 Printer ROM failure
 - Bit 5 - 1 Media jam during capture
 - Bit 4 - 1 Media jam towards exit
 - Bit 3 - 1 Media jam towards capture bin
 - Bit 2 - 0 Always
 - Bit 1 - 0 Always
 - Bit 0 - 0 Always

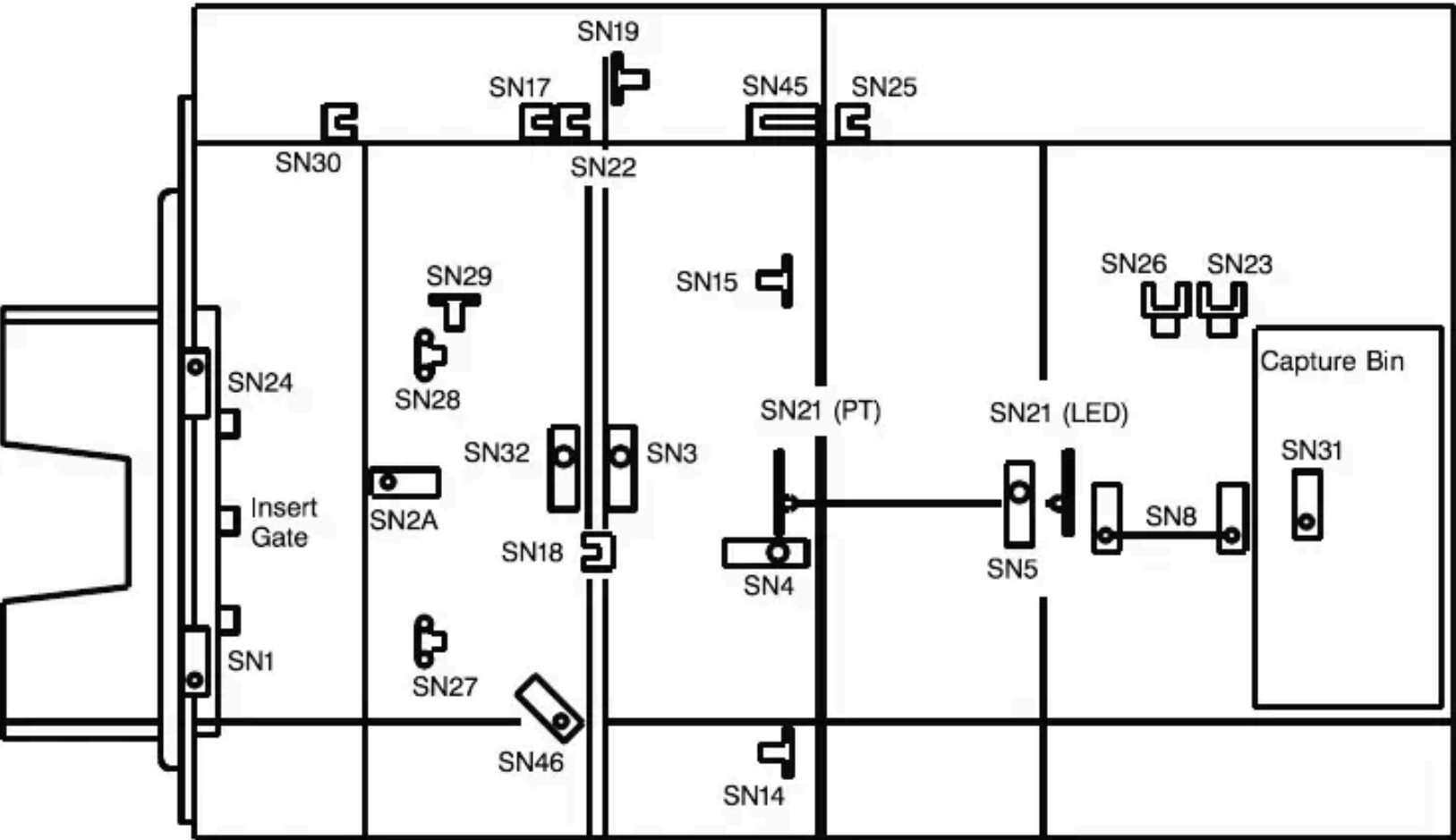
80-COLUMN THERMAL PRINTER

- Byte 4 - Sensors:
 - Bit 7 - 0 Always
 - Bit 6 - 0 Always
 - Bit 5 - 0 Always
 - Bit 4 - 1 Transport open - Transport open sensor (S10) or
Print Engine open - Head close sensors 1 & 2 (ES2) active
 - Bit 3 - 1 Capture bin NOT empty - Capture area sensors 1 or 2 (S8/S9) active
 - Bit 2 - 1 Paper out detected - Paper set sensor (ES6) active
 - Bit 1 - 1 Paper low detected - Paper low sensor (S11/S12) active
 - Bit 0 - 1 Black dot sensors (ES4&5) covered
- Byte 5 - Sensors (Statement only):
 - Bit 7 - 0 Always
 - Bit 6 - 0 Always
 - Bit 5 - 1 Transport open- Transport open sensor (S10) or Print Engine open- Head close sensors 1 & 2 (ES2) active
 - Bit 4 - 0 Always
 - Bit 3 - 0 Always
 - Bit 2 - 0 Always
 - Bit 1 - 0 Always
 - Bit 0 - 1 Knife at home position
- Byte 5 - Sensors (Buncher option):
 - Bit 7 - 1 Clamp drive jammed closed
 - Bit 6 - 1 Clamp drive jammed open
 - Bit 5 - 1 Transport open- Transport open sensor (S10) or (Print Engine open- Head close sensors 1 & 2 (ES2) active
 - Bit 4 - 1 Clamp drive raised - Pinch open sensor (S3) active
 - Bit 3 - 1 Clamp drive lowered - Pinch closed sensor (S4) active
 - Bit 2 - 0 Always
 - Bit 1 - 0 Always
 - Bit 0 - 1 Knife at home position

- Byte 6 - Sensors: (Statement and Buncher)
 - Bit 7 - 1 Capture entry sensor (S7) covered
 - Bit 6 - 1 Stack entry sensor (S6) covered
 - Bit 5 - 0 Always
 - Bit 4 - 0 Always
 - Bit 3 - 0 Always
 - Bit 2 - 0 Always
 - Bit 1 - 1 Paper set sensor (ES6) covered
 - Bit 0 - 1 Capture entry sensor (S7) covered
- Byte 7 - Sensors (Statement only):
 - Bit 7 - 0 Always
 - Bit 6 - 0 Always
 - Bit 5 - 0 Always
 - Bit 4 - 0 Always
 - Bit 3 - 1 Exit sensor (S1) covered
 - Bit 2 - 0 Always
 - Bit 1 - 0 Always
 - Bit 0 - 0 Always
- Byte 7 - Sensors (Buncher option):
 - Bit 7 - 0 Always
 - Bit 6 - 0 Always
 - Bit 5 - 0 Always
 - Bit 4 - 0 Always
 - Bit 3 - 1 Exit sensor (S1) covered
 - Bit 2 - 1 Stack area sensor (S5) covered
 - Bit 1 - 1 Stack entry sensor (S6) covered
 - Bit 0 - 1 Stack area sensor (S5) covered

Sensor Locations

The following figure shows the location and numbering of the sensors in the printer:



- SN1 - Media Insertion

SN3 - Print position

SN5 - Turn page position

SN14 - Print head right home position

SN17 - Platen shutter close

SN19 - Platen home position

SN22 - Platen shutter open

SN24 - Inserted media width

SN26 - Turn page book guide open

SN28 - Magnetic head left home

SN30 - Magnetic plate position

SN32 - Media at mid transport position

SN46 - Passbook alignment covered
- SN2 - Magnetic head read/write position

SN4 - Line find position

SN8 - Escape route

SN15 - Print head left home position

SN18 - Platen up/down position

SN21 - Turn page timing

SN23 - Turn page book guide close

SN25 - Turn page roller middle position

SN27 - Magnetic head right home

SN29 - Magnetic head encoder

SN31 - Capture bin

SN45 - Print unit open

USB THERMAL PRINTERS

JOURNAL PRINTER MSTATUS

M_STATUS	Meaning
0	Printer OK
5	Paper Out
7	Paper Not Loaded
8	Communications Error
9	Printer Open
11	Printhead Needs Replaced Soon
14	Printhead Needs Replaced Now
19	Paper Not Winding
20	Mechanism Jammed
24	Power Supply Voltage Error
25	Control Board Error
26	Head Over Heated (recoverable). 10s Transient status. Not reported.
27	Head Burnt Out (non-recoverable)
28	Paper Low

RECEIPT PRINTER MSTATUS

M_STATUS	Meaning
0	Printer OK
1	Head Jam / Knife Jam
3	Black Mark Error
5	Paper Out
7	Paper not loaded or Jam before transport
8	Communications Error
9	Printer Open
11	Printhead needs replaced soon
12	Knife needs replaced soon
14	Printhead needs replaced now
16	Eject Jam
17	Transport Jam

USB THERMAL PRINTERS

M_STATUS	Meaning
18	Capture Jam
21	No Receipt During Control
23	Capture Bin Full
24	Power Supply Voltage Error
25	Control Board Error
26	Head Over Heated (recoverable). 10s Transient status. Not reported.
27	Head Burnt Out (non-recoverable)
28	Paper Low
29	Exit Jam On Capture
30	Knife needs replaced now.
31	Receipt Transport Not Connected On Power Up
32	Media Not Cleared
34	Dual Feed Mechanism Jam

M_DATA

Receipt Printer

Bytes 0 to 8 refer to the Receipt Printer (1ST Receipt, 2ST Receipt, 1ST Dual Roll Receipt)

Bytes 0 to 18 refer to the Receipt Printer (2ST Dual Roll Receipt)

- Byte 0 - Printer Indication
 - 30H - Receipt Printer

- Byte 1 - Sensor status 1
 - Bit 7 - BD sensor (Black Dot), 1=Black
 - Bit 6 - PE sensor (Paper End of Print engine), 1=Exhausted
 - Bit 5 - DF_2b sensor (Dual feeder #2 positioning sensor), 1 = Blocked
 - Bit 4 - DF_1b sensor (Dual feeder #1 positioning sensor), 1 = Blocked
 - Bit 3 - DF_2a sensor (Dual feeder #2 PE sensor), 1 = Empty
 - Bit 2 - DF_1a sensor (Dual feeder #1 PE sensor), 1 = Empty
 - Bit 1 - Paper low (#2) sensor, 1 = Paper Low
 - Bit 0 - Paper low (#1) sensor, 1 = Paper Low
- Byte 2 - Sensor status 2
 - Bit 7 - always 0
 - Bit 6 - always 0
 - Bit 5 - Cover Open Sensor, 1 = Open
 - Bit 4 - CBF sensor (Capture Bin Full), 1 = Full
 - Bit 3 - CB sensor (Capture Bin), 1 = Empty
 - Bit 2 - Exit sensor (Transport exit), 1 = Blocked
 - Bit 1 - TJ sensor (Transport Jam), 1 = Blocked
 - Bit 0 - HP sensor (Cutter Home Position), 1 =Home
- Byte 3 - Sensor status 3
 - Bit 7 - Paper Jam for TOF Sensor, 1 = Jam
 - Bits 4 to 6 - always 0
 - Bit 3 - 2ST Paper Auto Detection Sensor, 1 = Black
 - Bit 2 - TOF sensor status, 1 = Blocked
 - Bit 1- always 0
 - Bit 0 - Feed Switch (Standard model or Feeder #1), 1 = Pushed
 - Other Bits, always 0

USB THERMAL PRINTERS

- Byte 4 - Motor status
 - Bit 7 - always 0
 - Bit 6 - Printing , 1 = In progress
 - Bit 5- always 0
 - Bit 4 - Dual feed motor #2, 1 = Going
 - Bit 3 - Dual feed motor #1, 1 = Going
 - Bit 2 - Transport motor, 1 = Going
 - Bit 1 - Cutter motor, 1 = Going
 - Bit 0 - Feed motor, 1 = Going
- Byte 5 - Printer status
 - Bit 7 - Fault state, 1 = Fault
 - Bit 6 - Warning due to Over Heated, 1 = Over Heated
 - Bit 5 - Non Transport Detection, 1 = Error
 - Bit 4 - Warning - Media not cleared after Manual feed, 1 = Remained
 - Bit 3 - Paper Type by Auto Paper Detection (FR Feeder), 1 = Two sided
 - Bit 2 - Paper Feed, 1 = Feed
 - Bit 1 - Reboot flag, 1 = After reboot
 - Bit 0 - Recovery running, 1 = Yes
 - Other Bits, always 0
- Byte 6 - Error status 1
 - Bit 7 - Black mark error, 1 = Error
 - Bit 6 - Thermal head temperature, 1 = Error
 - Bit 5 - always 0
 - Bit 4 - Transport jam, 1 = Jam
 - Bit 3 - Cutter jam, 1 = Jam
 - Bit 2 - Dual feeder jam, 1 = Jam
 - Bit 1 - Flash Erase/Write error or Memory full, 1 = Error
 - Bit 0 - H/W error -unrecoverable- (ROM, RAM, etc), 1 = Error

- Byte 7 - Error status 2
 - Bit 7 - Capture Jam (inc. Customer tampering), 1 = Jam
 - Bit 6 - Cover Open, 1 = Open
 - Bit 5 - PE (Print engine), 1 = Exhausted
 - Bit 4 - PE feeder #2, 1 = Empty
 - Bit 3 - PE feeder #1, 1 = Empty
 - Bit 2 - Unable to load paper (Std model), 1 = Unable
 - Bit 1 - Unable to load paper feeder #2, 1 = Unable
 - Bit 0 - Unable to load paper feeder #1, 1 = Unable
- Byte 8 - Warning
 - Bit 7 - illegal sequence or parameter, 1 = Present
 - Bit 6 - 24V Anomaly, 1 = Anomaly
 - Bit 5 - Capture - Customer tampering, 1 = Tampering
 - Bit 4 - always 0
 - Bit 3 - Capture bin full, 1 = Full
 - Bit 2 - Executed Present, Purge or Retract w/o a receipt, 1 = Executed
 - Bit 1 - Paper low (#2) sensor, 1 = Paper Low
 - Bit 0 - Paper low (#1) sensor, 1 = Paper Low
- Byte 9 - Reserved
 - Always 0
- Byte 10 - Reserved
 - Always 0
- Byte 11 - Reserved
 - Always 0
- Byte 12 - Reserved
 - Always 0
- Byte 13 - Reserved
 - Always 0
- Byte 14 - Reserved
 - Always 0

USB THERMAL PRINTERS

- Byte 15 - Sensor status
 - Bit 7 - Paper Type by Auto Paper Detection (BK Feeder), 1 = Two sided
 - Other Bits, always 0
- Byte 16 - Reserved
 - Always 0
- Byte 17 - Reserved
 - Always 0
- Byte 18 - Reserved
 - Always 0

Journal Printer

Unspecified bits in each byte are always 0.

- Byte 0 - Printer Indication
 - 31H - Journal Printer
- Byte 1 - Sensor status 1
 - Bit 6 - PE sensor (Paper End of Print engine), 1=Exhausted
 - Bit 0 - Paper low (#1) sensor, 1 = Paper Low
- Byte 2 - Sensor status 2
 - Bit 6 - Winding Jam sensor, 1 = High level
 - Bit 5 - Cover Open Sensor, 1 = Open
- Byte 3 - Sensor status 3
 - Bit 0 - Feed Switch (Standard model or Feeder #1), 1 = Pushed
- Byte 4 - Motor status
 - Bit 6 - Printing , 1 = In progress
 - Bit 5- Winding motor, 1 = Going
 - Bit 0 - Feed motor, 1 = Going
- Byte 5 - Printer status
 - Bit 7 - Fault state, 1 = Fault
 - Bit 6 - Warning due to Over Heated, 1 = Over Heated
 - Bit 1 - Reboot flag, 1 = After reboot
 - Bit 0 - Recovery running, 1 = Yes
 - Other Bits, always 0

- Byte 6 - Error status 1
 - Bit 7 - Black mark error, 1 = Error
 - Bit 6 - Thermal head temperature, 1 = Error
 - Bit 5 - Winding jam, 1 = Jam
 - Bit 1 - Flash Erase/Write error or Memory full, 1 = Error
 - Bit 0 - H/W error -unrecoverable- (ROM, RAM, etc), 1 = Error
- Byte 7 - Error status 2
 - Bit 6 - Cover Open, 1 = Open
 - Bit 5 - PE (Print engine), 1 = Exhausted
 - Bit 2 - Unable to load paper (Std model), 1 = Unable
 - Bit 1 - Unable to load paper feeder #2, 1 = Unable
 - Bit 0 - Unable to load paper feeder #1, 1 = Unable
- Byte 8 - Warning
 - Bit 7 - illegal sequence or parameter, 1 = Present
 - Bit 6 - 24V Anomaly, 1 = Anomaly
 - Bit 4 - Paper not wound, 1 = Not wound
 - Bit 0 - Paper low (#1) sensor, 1 = Paper Low

ENHANCED PAGE TURNING PASSBOOK PRINTER

M-STATUS

M_STATUS	Meaning
0	Operation successful
1	Blank track
2	Track not supported
3	Read error
4	Write error
5	No passbook in reader at start of command
6	Error recovery successful
7	Passbook jam
8	Too many consecutive read errors
9	Too many consecutive write errors
10	Too many consecutive passbook jams
11	Device inoperative
12	Format error
13	RS-232 communications error
14	Passbook jammed in throat - possible customer tampering
15	Book jammed - unable to capture
20	Book removed by customer during capture
22	Ribbon needs to be replaced soon
23	Printhead needs to be replaced soon
24	Barcode error
25	Ribbon needs to be replaced now
26	Printhead needs to be replaced now
27	Capture bin overfill
28	Mechanical jam
31	Blank page
36	Capture bin full
37	Page turn error
38	Too many consecutive page turn errors

M_STATUS	Meaning
40	Page turn book mechanism jammed
41	Main electronics board failure
44	Book present

M_DATA

- Byte 0 - Command or Phase:
 - 00 - Reset
 - 01 - Accept
 - 02 - Read
 - 03 - Write
 - 04 - Determine State
 - 05 - Eject
 - 06 - Print
 - 07 - Disable
 - 08 - Define Document
 - 09 - Not used
 - 0A - Page Number
 - 0B - Check Line
 - 0C - Report Configuration
 - 0D - Report Definition
 - 0E - Book Detection
 - 0F - Book Insertion
 - 10 - Book Removal
 - 11 - Exercise Mechanism
 - 12 - Not Used
 - 13 - Not Used
 - 14 - Capture
 - 15 - Multiple Line Print
 - 16 - Not Used
 - 17 - Define Scan
 - 18 - Define Magnetics
 - 19 - Not Used
 - 1A - Line Scan
 - 1B - Page Turn Forward
 - 1C - Page Turn Backward
 - 1D - Sensor Status test
 - 1E - Report Firmware Version
 - 1F - Download Firmware

ENHANCED PAGE TURNING PASSBOOK PRINTER

- Byte 1 - MAG Read/Write Status:
 - Bit 7 - 0 always
 - Bit 6 - 0 always
 - Bit 5 - 0 always
 - Bit 4 - 0 always
 - Bit 3 - 0 always
 - Bit 2 - 1 for ISO track
 - Bit 1 - 1 for IBM track
 - Bit 0 - 1 for DIN track
- Byte 2, Byte3, Byte 4 and Byte 5

M_DATA Byte				Error Description
(2)	(3)	(4)	(5)	
30h	30h	30h	31h	Data Format Error
31h	30h	30h	30h	Undefined ESC Sequence
31h	30h	30h	32h	Undefined Print Code
31h	30h	30h	33h	Incorrect Escape Sequence
31h	30h	30h	38h	Illegal Double Byte Sequence
31h	30h	32h	30h	Illegal LF/CR Sequence
31h	30h	32h	38h	Illegal Sequence
31h	30h	33h	31h	Undefined magnetic stripe data
31h	30h	33h	32h	Illegal Magnetic Stripe Format
31h	30h	34h	31h	No media Present
32h	33h	32h	30h	Illegal Passbook Type Detected
33h	31h	30h	31h	Power Supply Transistor Short Error
33h	31h	30h	32h	Printhead Transistor Short Error
33h	31h	30h	33h	Ground Side Transistor Short Error
33h	31h	30h	34h	MS Write Transistor Open Error
33h	32h	xxh	xxh	Printhead Power Supply Transistor Open Error
33h	33h	30h	31h	Solenoid Power Supply Open Error
33h	33h	30h	32h	Platen Solenoid Transistor Open Error
(2)	(3)	(4)	(5)	Error Description

M_DATA Byte				
33h	34h	30h	31h	
33h	34h	30h	31h	Printhead Transistor Open Error
33h	34h	30h	32h	Printhead Power Supply Transistor Open Error
33h	35h	xxh	xxh	Solenoid/Stepper Motor Transistor Open Error
33h	36h	30h	31h	Stepper Motor, Ground Side, Transistor Open Error
33h	37h	30h	31h	Printhead Dot Alarm
34h	31h	xxh	xxh	Sensor Check Error
34h	32h	xxh	xxh	Printhead Jam
34h	34h	30h	30h	Magnetic Head Jam
34h	35h	30h	30h	Shutter Jam
35h	30h	30h	30h	Protocol Error
36h	xxh	xxh	xxh	Paper Ejection Error
37h	xxh	xxh	xxh	Paper Retraction Error
38h	xxh	xxh	xxh	Media Jam
39h	xxh	xxh	xxh	Transport Jam, (Forward Direction)
3Bh	04h	xxh	xxh	Transport jam while attempting to Accept or Eject. Customer tampering
3Bh	08h	xxh	xxh	Illegal media - book too small/large
41h	xxh	xxh	xxh	Transport Jam, (Reverse Direction)
42h	xxh	xxh	xxh	Printhead Control Timeout
43h	xxh	xxh	xxh	Magnetic Head Control timeout
45h	30h	xxh	xxh	Page Turn Error
45h	38h	xxh	xxh	Read Error, (Blank Track)
45h	39h	xxh	xxh	Read Error
46h	38h	xxh	xxh	Write Error, (Blank Track)
46h	39h	xxh	xxh	Write Error

Sensor Values at End of Command

The remaining bytes of M_DATA (Bytes 6 to 10) report the sensor values at the completion of the command. These bytes are always generated except when there has been an RS-232 communications error or a main electronic board failure.

- Byte 6 - Sensors:
 - Bit 7 - Turn page detection sensor 1 S25
 - Bit 6 - Capture bin not empty S21
 - Bit 5 - 0 always
 - Bit 4 - 0 always
 - Bit 3 - 0 always
 - Bit 2 - Media at magnetic head position S3
 - Bit 1 - 0 always
 - Bit 0 - Media insertion sensor S2 covered
- Byte 7 - Sensors:
 - Bit 7 - 0 always
 - Bit 6 - 0 always
 - Bit 5 - 0 always
 - Bit 4 - Media at width sensor S5
 - Bit 3 - Media at last transport position S12
 - Bit 2 - Media at line find position S8
 - Bit 1 - 0 always
 - Bit 0 - Turn page detection sensor 2 S26

- Byte 8 - Sensors:
 - Bit 7 - 0 always
 - Bit 6 - 0 always
 - Bit 5 - 0 always
 - Bit 4 - 0 always
 - Bit 3 - 0 always
 - Bit 2 - 0 always
 - Bit 1 - 0 always
 - Bit 0 - 0 always
- Byte 9 - Sensors:
 - Bit 7 - 0 always
 - Bit 6 - 0 always
 - Bit 5 - 0 always
 - Bit 4 - 0 always
 - Bit 3 - Front shutter open S24
 - Bit 2 - Turn page mechanism sensor S10
 - Bit 1 - Magnetic head left home position
 - Bit 0 - Print head left home position S7
- Byte 10 - Sensors:
 - Bit 7 - 0 always
 - Bit 6 - 0 always
 - Bit 5 - 0 always
 - Bit 4 - 0 always
 - Bit 3 - 0 always
 - Bit 2 - 0 always
 - Bit 1 - 0 always
 - Bit 0 - 0 always

Printer Firmware Error Codes

The Enhanced Page Turning Passbook Printer does not return any firmware error codes.

USB PASSBOOK PRINTER

M-STATUS

M_STATUS	Meaning
0	Operation successful
5	No passbook in reader at start of command
6	Error recovery successful
7	Passbook jam
10	Too many consecutive passbook jams
11	Device inoperative
12	Format error
13	Communications error
14	Passbook jammed in throat - possible customer tampering
15	Book jammed - unable to capture
16	Shutter jammed open
17	Shutter jammed closed
20	Book removed by customer during capture
22	Ribbon needs to be replaced soon
23	Print head needs to be replaced soon
25	Ribbon needs to be replaced now
26	Print head needs to be replaced now
27	Capture bin overfill
28	Mechanical jam
29	Print head open
36	Capture bin full
41	Control board failure
44	Book present
45	Transport cover open
46	Transport jam
48	Statement operation in progress, printer busy

M_DATA

- Byte 0 - Command or Phase:
 - 00 - Reset
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Use pursuant to Company Instructions
- 142ATM Self-Service Diagnostic Status Code Notebook

- 01 - Accept
 - 04 - Determine State
 - 05 - Eject
 - 06 - Print
 - 07 - Disable
 - 08 - Define Document
 - 0D - Report Definition
 - 0E - Book Detection
 - 11 - Exercise Mechanism
 - 14 - Capture
 - 15 - Multiple Line Print
 - 1D - Sensor Status test
- Byte 1 - MAG Read/Write Status:
 - Always 0
 - Byte 2, Byte3, Byte 4, Byte 5, Byte 6, Byte 7

M_DATA Byte						Error Description
(2)	(3)	(4)	(5)	(6)	(7)	
33h	31h	30h	31h	30h	30h	RAM Error
33h	32h	30h	31h	30h	30h	ROM Error
33h	33h	xxh	xxh	xxh	xxh	Thermistor (Print Head) Error
33h	34h	xxh	xxh	xxh	xxh	Thermistor (Out side temperature) Error
33h	35h	xxh	xxh	xxh	xxh	Carriage Initialization Error
33h	36h	xxh	xxh	xxh	xxh	Exit Shutter Initialization Error
33h	37h	xxh	xxh	xxh	xxh	Right Alignment Initialization Error
33h	38h	xxh	xxh	xxh	xxh	Right Alignment Roller Initialization Error
33h	39h	xxh	xxh	xxh	xxh	Front Pinch Roller Initialization Error
33h	3ah	xxh	xxh	xxh	xxh	Platen up / down Initialization Error
34h	35h	xxh	xxh	xxh	xxh	Exit Shutter Jam

M_DATA Byte						Error Description
32h	33h	32h	30h	30h	30h	Passbook's dimensions are out with the media spec
36h	30h	xxh	xxh	xxh	xxh	Passbook jam reverse
37h	30h	xxh	xxh	xxh	xxh	Passbook jam forward
38h	30h	xxh	xxh	xxh	xxh	Passbook jam power on
39h	30h	xxh	xxh	xxh	xxh	Transport timeout feeding
3bh	30h	xxh	xxh	xxh	xxh	Transport jammed
3bh	01h	xxh	xxh	xxh	xxh	No passbook at exit
3bh	02h	xxh	xxh	xxh	xxh	Book is removed by customer
3bh	04h	xxh	xxh	xxh	xxh	Book is jammed in transport
3bh	08h	xxh	xxh	xxh	xxh	Passbook length is out of spec (short or long)
41h	30h	xxh	xxh	xxh	xxh	Transport timeout ejecting
42h	30h	xxh	xxh	xxh	xxh	Printhead jam
34h	36h	xxh	xxh	xxh	xxh	Cover Open Sensor
35h	30h	30h	30h	30h	30h	Data overrun
F0h	30h	30h	30h	30h	30h	Passbook printer is not selected by printer driver
30h	30h	30h	31h	30h	30h	Command received out of sequence
31h	30h	30h	30h	30h	30h	Unidentified command
31h	30h	30h	33h	30h	30h	Incorrect command length
31h	30h	34h	31h	30h	30h	Command requires media and no media present
40h	30h	30h	30h	30h	30h	No error

- Byte 8 - Sensors
 - Bit 7 - 0 always
 - Bit 6 - Platen up/down, 0 = Down, 1 = Up
 - Bit 5 - 0 always

- Bit 4 - Front Pinch Up/Down, 0 = Down, 1 = Up
- Bit 3 - 0 always
- Bit 2 - Alignment Roller Up/Down, 0 = Down, 1 = Up
- Bit 1 - Passbook Right Align Shutter, 0 = Down, 1 = Up
- Bit 0 - Shutter (close), Exit Pinch Up/Down, 0 = Down, 1 = Up
- Byte 9 - Sensors:
 - Bit 7 - 0 always
 - Bit 6 - Shutter Open, 0 = Open, 1 = Closed
 - Bit 5 - CA(carriage) Home Position, 0 = Home, 1 = Not Home
 - Bit 4 - Front upper unit cover open, 0 = Open, 1 = Closed
 - Bit 3 - Printer unit cover open, 0 = Open, 1 = Closed
 - Bit 2 - Reserved (Page Turn Table (R)), 0 always
 - Bit 1 - Reserved (Page Turn Flapper (F)), 0 always
 - Bit 0 - Reserved (Page Turn Table), 0 always

USB PASSBOOK PRINTER

- Byte 10 - Sensors:
 - Bit 7 - P6 Length check (max), 0 = no media, 1 = media exists
 - Bit 6 - P5 Length check (min), 0 = no media, 1 = media exists
 - Bit 5 - P4 Bottom edge of MS, 0 = no media, 1 = media exists
 - Bit 4 - P3 Front Shutter, 0 = no media, 1 = media exists
 - Bit 3 - P2B Media width check Left (B), 0 = no media, 1 = media exists
 - Bit 2 - P2A Media width check Left (A), 0 = no media, 1 = media exists
 - Bit 1 - P1 Media width check Right, 0 = no media, 1 = media exists
 - Bit 0 - P0 Media width check Right, 0 = no media, 1 = media exists
- Byte 11 - Sensors
 - Bit 7 - 0 always
 - Bit 6 - Capture Bin Sensor, 0 = no media, 1 = media exists
 - Bit 5 - 0 always
 - Bit 4 - P11 Capture Bin exit, 0 = no media, 1 = media exists
 - Bit 3 - P10 Paper Jam, 0 = no media, 1 = media exists
 - Bit 2 - 0 always
 - Bit 1 - P8 Right Align sensor (REAR), 0 = no media, 1 = media exists
 - Bit 0 - P7 Right Align sensor (FR), 0 = no media, 1 = media exists

OPEN DOCUMENT PRINTER

M_STATUS

M_STATUS	Meaning
00	Operation successful
01	Mechanical jam
02	Communications error / command specification error
03	Transport left open
04	Black mark error
05	Media jam
07	Paper out
11	Printer left open
16	Capture jam
18	Media stuck in throat, unable to pull into printer - possible customer tampering
34	Clamp drive jammed
36	Capture bin full

M_Data

- Byte 0 - Status and Sensors:
 - Bit 7 - Undefined
 - Bit 6 - 1 = fault
 - Bit 5 - Undefined
 - Bit 4 - 1 = Exit Sensor
 - Bit 3 - 1 = Paper out
 - Bit 2 - 1 = Operation beyond last line, or insufficient memory to complete operation
 - Bit 1 - 1 = Paper low
 - Bit 0 - Undefined
- Byte 1 - Undefined

OPEN DOCUMENT PRINTER

- Byte 2:
 - Bit 7 - Undefined
 - Bit 6 - 1 = Communication error
 - Bit 5 - 1 = No document
 - Bit 4 - 1 = Cutter jam
 - Bit 3 - Undefined
 - Bit 2 - 1 = Capture bin nearly full
 - Bit 1 - 1 = Mechanical jam
 - Bit 0 - 1 = Media jam
- Byte 3:
 - Bit 7 - 1 = Printer RAM failure
 - Bit 6 - 1 = Printer ROM failure
 - Bit 5 - 1 = Media jam during capture
 - Bit 4 - 1 = Media jam towards exit
 - Bit 3 - 1 = Media jam towards capture bin
 - Bit 2 - Undefined
 - Bit 1 - Undefined
 - Bit 0 - Undefined
- Byte 4:
 - Bit 7 - Undefined
 - Bit 6 - Undefined
 - Bit 5 - 1 = Transport open
 - Bit 4 - 1 = Print unit open
 - Bit 3 - 1 = Capture bin not empty
 - Bit 2 - 1 = Paper out
 - Bit 1 - 1 = Paper low
 - Bit 0 - 1 = Black mark 1
- Byte 5:
 - Bit 7 - Undefined
 - Bit 6 - 1 = Clamp drive jammed - refer to bits 3 & 4 (buncher only)
 - Bit 5 - Undefined
 - Bit 4 - 1 = Clamp drive raised (buncher only)
 - Bit 3 - 1 = Clamp drive lowered (buncher only)
 - Bit 2 - Undefined
 - Bit 1 - Undefined
 - Bit 0 - 1 = Knife at home position

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Use pursuant to Company Instructions

- Byte 6 - Sensors:
 - Bit 7 - Undefined
 - Bit 6 - Undefined
 - Bit 5 - Undefined
 - Bit 4 - Undefined
 - Bit 3 - Undefined
 - Bit 2 - Undefined
 - Bit 1 - 1 = Stack entry sensor
 - Bit 0 - 1 = Capture entry sensor
- Byte 7 - Sensors:
 - Bit 7 - Undefined
 - Bit 6 - Undefined
 - Bit 5 - Undefined
 - Bit 4 - Undefined
 - Bit 3 - 1 = Media exit sensor
 - Bit 2 - 1 = Media stage sensor
 - Bit 1 - 1 = Stack area sensor (buncher only)
 - Bit 0 - Undefined

PC DISKS

Items marked with an asterisk (*) are only reported when the migrated disk service is being used.

M_STATUS

M_STATUS	Meaning
00	No error
01	Write failure CRC error *
02	Read failure Disk error *
03	Drive not configured
04	Random read error Hardware error *
05	Seek error
08	Disk write protected *
09	Operator intervention required *
0A	Filename not found in directory *
0D	General error *

M_DATA

M_DATA	Meaning
00	No error
01	Invalid function number
02	File not found
03	Path not found
04	Out of handles
05	Access denied
06	Invalid handle Open/Create failed due to explicit fail command *
07	Memory control block destroyed
08	Insufficient memory
09	Invalid memory block address

M_DATA	Meaning
0A	Invalid environment
0B	Invalid format
0C	Invalid access code
0D	Invalid data
0E	Unknown unit
0F	Invalid disk drive
10	Cannot remove current directory
11	Not same device
12	No more files
13	Disk write protected
14	Unknown unit
15	Drive not ready (disk not in drive)
16	Unknown command
17	Data error (CRC)
18	Bad request structure length
19	Seek error
1A	Unknown type of medium
1B	Sector not found
1D	Write fault
1E	Read fault
1F	General failure
20	Sharing violation
21	Lock violation
22	Invalid disk change
23	FCB unavailable
24	Sharing buffer exceeded
50	File already exists
70	Disk is full
82	Disk is not formatted
FF	General error

SDC TOUCH SCREEN

M_STATUS

M_STATUS	Description
00	No error
03	Hardware error - Touch screen controller hardware error
04	Not calibrated - Touch screen has not been calibrated

M_DATA

M_DATA returned if M_STATUS is 03 (Hardware_Error):

- Byte 0 - Touch screen error code:
 - 00H - No error
 - 03H - A/D converter error
 - 04H - Failed beams detected
 - 05H - Communications error (controller/frame interface error)
 - 06 - Weak beams detected

CARDHOLDER KEYBOARDS

NUMERIC KEYBOARD

M_STATUS

The following M_STATUS are returned for the keyboard’s Key Detect test:

M_STATUS	Meaning
00	No error
06	Security Module (SM) required a reset and the reset was successful (EKC only)
07	SM required a reset and the reset was unsuccessful (EKC only)
146	SDC link failure

M_DATA

The Key Detect test returns the two-digit hexadecimal values shown in the following figure:

Left FDK	Numeric				Right FDK
07	10	11	12	13	03
06	14	15	16	17	02
05	24	25	26	27	01
04	20	21	22	23	00

ALPHANUMERIC KEYBOARD

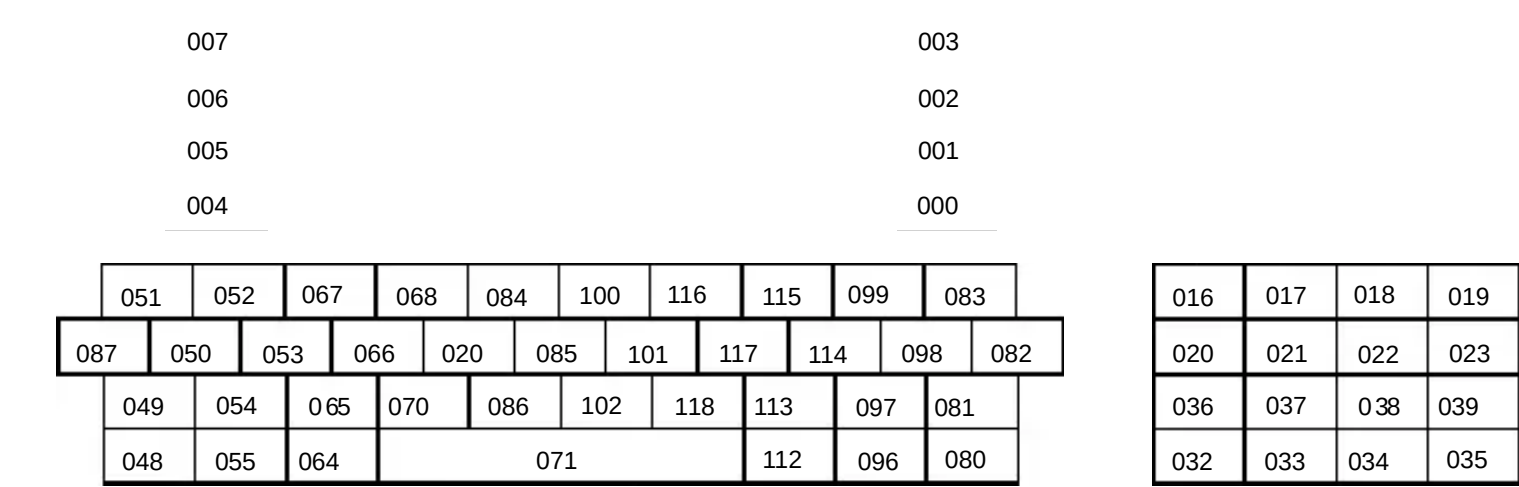
M_STATUS

The following M_STATUS are returned for the keyboard and FDK Key Detect test:

M_STATUS	Meaning
00	No error
06	If the Security Module (SM) reset was successful (EKC only)
07	If the SM reset was unsuccessful (EKC only)
146	SDC link failure

M_DATA

The Key Detect test returns the three-digit numbers shown in the following figure:



OPERATOR PANELS

BASIC OPERATOR PANEL

M_STATUS

M_STATUS	Meaning
00	No error
146	SDC link failure

M_DATA

The M_DATA returned for the basic operator panel keyboard KEY DETECT test, is the hardware position code for the last key entered (refer to “Hardware Position Codes”).

Early versions of the software treat the hardware position code as hexadecimal numbers and display three-digit decimal representations of the position code numbers, both sets of codes are shown in the following table.

Hardware Position Codes

The following hardware position codes are returned under M_DATA for the BOP keyboard KEY DETECT test:

Basic Operator Panel				
00	10	20	30	40
01	11	21	31	41
02	12	22	32	42
03	13	23	33	43

Decimal Representation of Hardware Position Codes

Basic Operator Panel				
000	016	032	048	064
001	017	033	049	065
002	018	034	050	066
003	019	035	051	067

ENHANCED OPERATOR PANEL KEYBOARD

M_STATUS

M_STATUS	Meaning
00	No error
146	SDC link failure

M_DATA

The M_DATA returned for the enhanced operator panel keyboard KEY DETECT test, is the hardware position code for the last key entered (refer to “Hardware Position Codes”).

Early versions of the software treat the hardware position code as hexadecimal numbers and display three-digit decimal representations of the position code numbers; both sets of codes are shown in the following table.

Hardware Position Codes

The following hardware position codes are returned under M_DATA for the EOP keyboard KEY DETECT test:

Left FDK	16 - Key Keyboard				Right FDK
00	10	20	30	40	50
01	11	21	31	41	51
02	12	22	32	42	52
03	13	23	33	43	53

Decimal Representation of Hardware Position Codes

Left FDK	16 - Key Keyboard				Right FDK
000	016	032	048	064	080
001	017	033	049	065	081
002	018	034	050	066	082
003	019	035	051	067	083

GRAPHICAL OPERATOR KEYBOARD

The M_STATUS, M_DATA and hardware position codes are the same as those for the Enhanced Operator Panel Keyboard.

BASIC OPERATOR PANEL DISPLAY

M_STATUS

M_STATUS	Meaning
00	No error
01	Controller error
02	Character ROM missing
146	SDC link failure

ENHANCED OPERATOR PANEL DISPLAY

M_STATUS

M_STATUS	Meaning
00	No error
146	SDC link failure

GRAPHICAL OPERATOR PANEL DISPLAY

M_STATUS

M_STATUS	Meaning
00	No error
01	Touch Screen fault
02	Display fault

OPERATOR PANEL MODE SWITCH

M_STATUS

M_STATUS	Meaning
00	No error

M_DATA

M_DATA	Meaning
30	Switch closed
31	Switch open

OPERATOR PANEL REAR INDICATORS

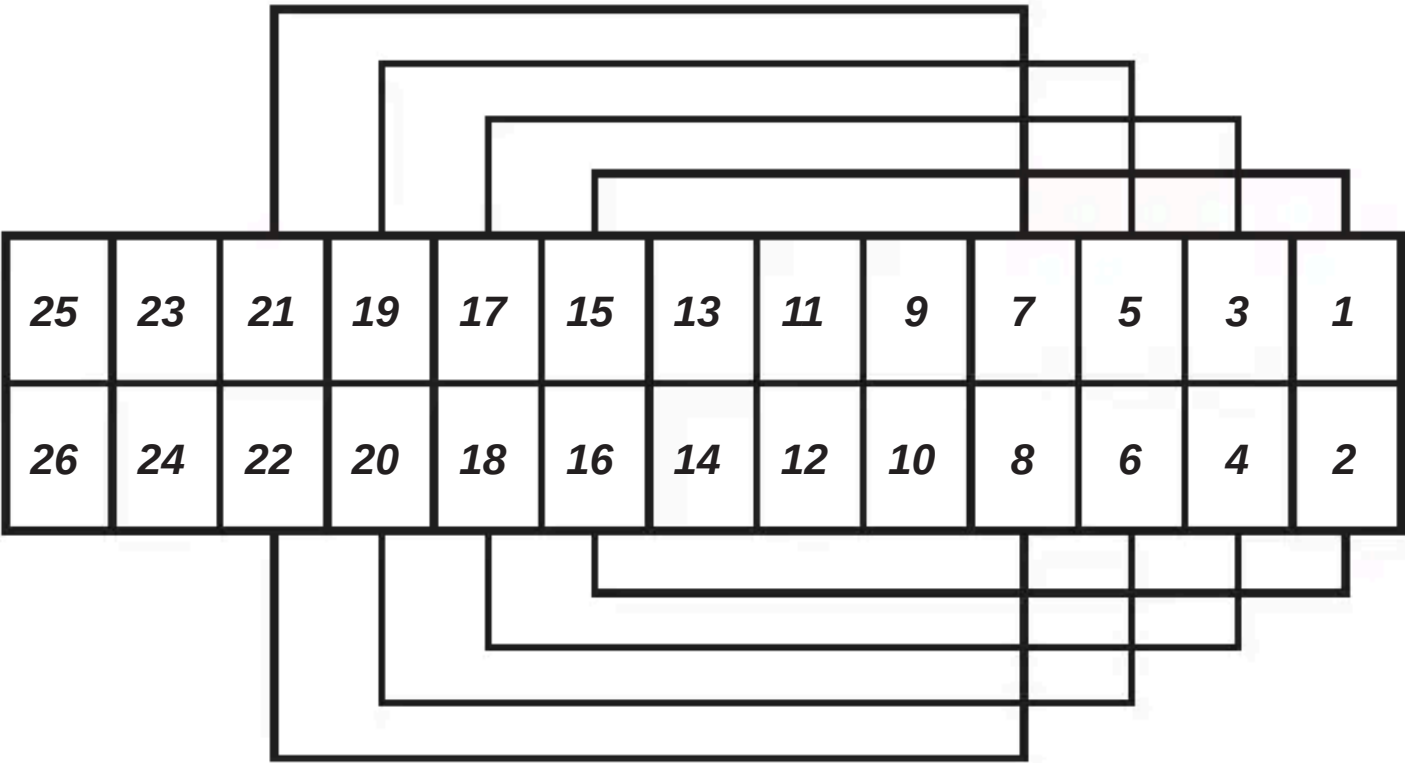
M_STATUS

M_STATUS	Meaning
00	No error

MISC I/F BOARD

Turnaround Plug

A turnaround plug, A/B (NCR Part No. 445-0593125), is required for the edge of board turnaround test.



M_STATUS

M_STATUS	Meaning
00	No error
03	Turnaround test failure
146	SDC link failure

M_DATA

- Byte 0 - Test state of input line 0:
 - 30H - Good
 - 31H - Stuck high
 - 32H - Stuck low
- Byte 1 - Test state of input line 1:
 - 30H - Good
 - 31H - Stuck high
 - 32H - Stuck low
- Byte 2 - Test state of output line 2:
 - 30H - Good
 - 31H - Stuck high
 - 32H - Stuck low
- Byte 3 - Test state of output line 3:
 - 30H - Good
 - 31H - Stuck high
 - 32H - Stuck low
- Byte 4 - Test state of output line 4:
 - 30H - Good
 - 31H - Stuck high
 - 32H - Stuck low
- Byte 5 - Test state of output line 5:
 - 30H - Good
 - 31H - Stuck high
 - 32H - Stuck low

NIGHTSAFE DEPOSITORY

M_STATUS

M_STATUS	Meaning	
	Standard Depository	5285 Business Depository
00	Bag drop switch is open	Bag not deposited within five seconds
01	Bag drop switch is closed	Bag deposited correctly (PASS)
02	Deposit not done and bag drop switch is open	Sensor failure or blocked sensor (FAIL)
03	Error in turnaround test	Error in turnaround test

M_DATA

M_DATA is only returned for the enhanced version of the nightsafe depository.

Byte 0 for all commands except turnaround test:

- Bit 0:
 - 0 - Door closed
 - 1 - Door open
- Bit 1:
 - 0 - Bag drop switch open
 - 1 - Bag drop switch closed

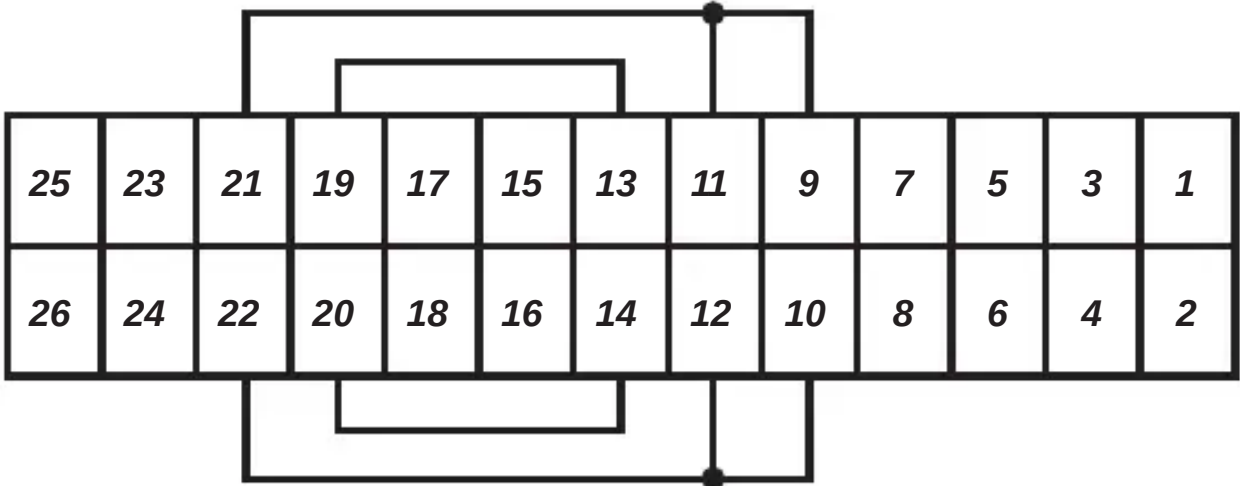
ALARMS

M_STATUS and M_DATA for the Alarms are detailed under NLX PC CORE - ALARMS.

MEDIA ENTRY INDICATORS

Turnaround Plug

A turnaround plug C, (NCR Part No. 445-0596570), is required for the media entry indicators turnaround test.



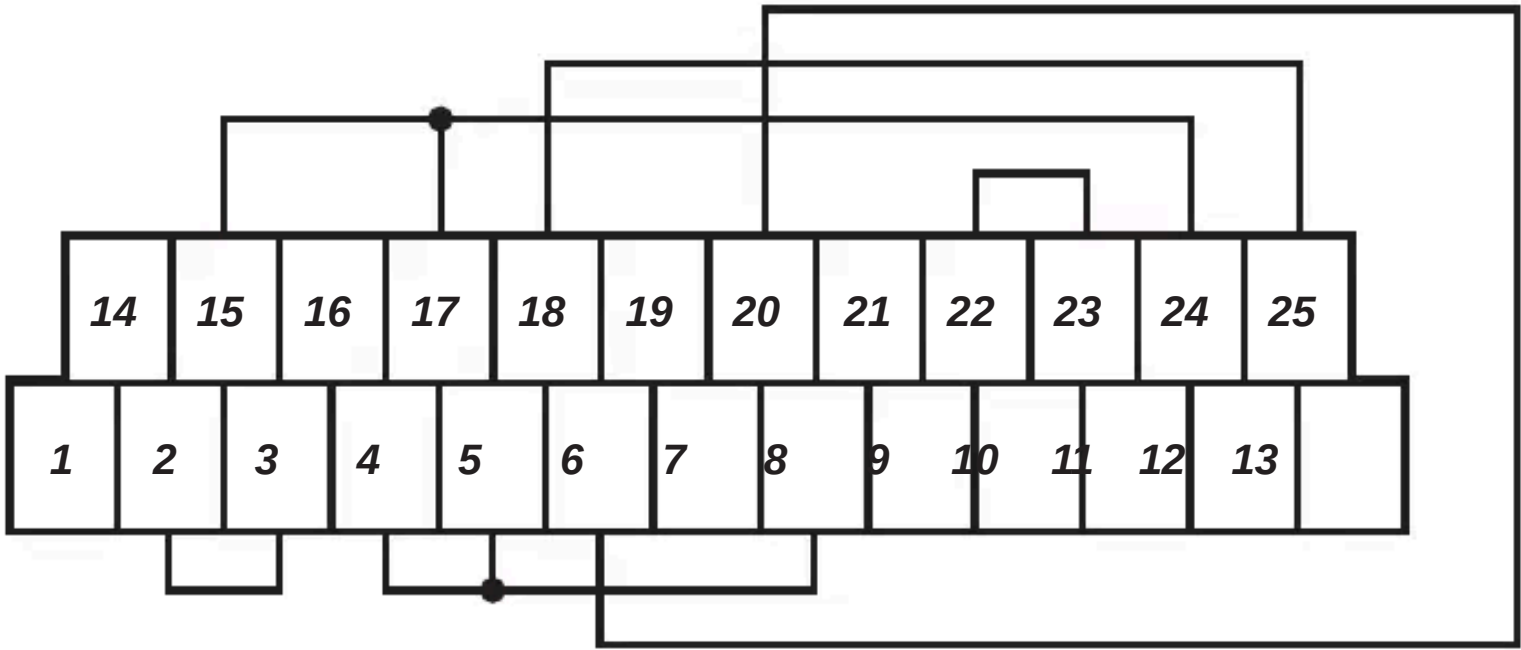
M_STATUS

M_STATUS	Meaning
00	Good
03	Turnaround test failed

DOOR ACCESS

Turnaround Plug

A turnaround plug is required for the door access unit RS232 turnaround test.



M_STATUS

M_STATUS	Meaning
00	No error
01	Bad read error
02	Timeout error (Open door test)
03	Communications error (Open door test)
97	Timeout error (Turnaround test)
98	Communications error (Turnaround test)
146	SDC link failure

M_DATA

- Byte 0:
 - 00 - Good
 - 01 - Parity error
 - 02 - Overrun error
 - 03 - Framing error
 - 04 - DTR or DSR timeout
 - 05 - Transmit timeout
 - 06 - Receive timeout

INTELLIGENT FRAUD DETECTION

M_STATUS

M_STATUS	Meaning
00	Device is fully operational
01	Device is faulty
02	Possible fraud detected
03	Fraud detected
04	A communications error has occurred between PC and device
05	Incompatible firmware loaded
06	Unauthorised device connected
07	The facia is open but REPLENISHMENT_TIMEOUT is not yet reached. This state is reported when ATM facia is opened for replenishment activities. If the replenishment timeout is reached, the SOH is changed to fraud
08	The attempt to configure the device and its sensor parameters failed
09	Invalid sensor ID
0A	Reset IFD Sensor command failed to reset the sensors. The possible causes for this response are: *ATM facia is open. *A skimming device has been added to the machine. *Analogue signals are not stable.
0B	IFD command failed. Firmware has rejected the given command. The possible causes for this response are: *The device is loaded with wrong firmware. *The device is wrongly configured, possibly using IFD Direct command.

M_STATUS Meaning

0C The sensor configuration failed to configure the sensors at system boot of the ATM machine.
The possible causes for this response are:
*ATM facia is open.
*A skimming device has been added to the machine.
*Analogue signals are not stable.
This M_Status is reported only if the “Force Sensor Reset” registry value of the key “Group3” within “Operational Parameter” is set to 1, and on the most recent reboot the driver was not able to reset the sensor defaults. The possible causes of this error are similar to M_Status 0AH error causes.

M_DATA

- Byte 0: Transaction Command Code
- Byte 1: Most Recent Device Command Code
- Byte 2: Most Recent Device Command Parameter Code (00 if not applicable)
- Byte 3-4: Most Recent Device Sensor Status
- Byte 5-6: Most Recent Device Error Code
- Byte 7-8: NCR Communications Library Status & Errors

The sensor data is stored in byte 3 and byte 4. Each sensor status is stored in two consecutive bits of a byte in the following format:

- 00 = Healthy
- 01 = Possible Fraud Detected
- 02 = Fraud Detected
- 03 = Module not configured to be monitored

- Byte 3:
 - Bit 0 - Media In Sensor data
 - Bit 1 - Media In Sensor data
 - Bit 2 - Media Out Sensor data
 - Bit 3 - Media Out Sensor data
 - Bit 4 - Cardholder Sensor data
 - Bit 5 - Cardholder Sensor data
 - Bit 6 - Service Sensor data
 - Bit 7 - Service Sensor data

INTELLIGENT FRAUD DETECTION

- Byte 4:
 - Bit 0 - Environment Sensor data
 - Bit 1 - Environment Sensor data
 - Bit 2 - Reserved, set to ‘0’
 - Bit 3 - Reserved, set to ‘0’
 - Bit 4 - Reserved, set to ‘0’
 - Bit 5 - Reserved, set to ‘0’
 - Bit 6 - Reserved, set to ‘0’
 - Bit 7 - Reserved, set to ‘0’

SECURITY CAMERA

M_STATUS

M_STATUS	Meaning
00	Picture taken
01	NAK received
03	Hardware error detected, bad communications
97	Timeout error
98	Communications error
146	SDC link failure

M_DATA

- Byte 0:
 - 00 - Good
 - 01 - Parity error
 - 02 - Overrun error
 - 03 - Framing error
 - 04 - DTR, DSR error
 - 05 - Tx error
 - 06 - Rx error
 - 30 - For the Take Picture/Controller Status tests
- Byte 1:
 - 30 - Ready
 - 31 - Busy
- Byte 2:
 - 30 - For the Take Picture/Controller Status tests
- Byte 3:
 - 31 - For the Take Picture/Controller Status tests

The 3M camera does not return data for bytes 1, 2 and 3.

T_DATA

The T_DATA returned for the Controller Status test consists of four fields separated by a unit separator as follows:

hhmm yymmdd aaaa bbbbbb

Where:

- hhmm - time (hour/minute)
- yymmdd - date (year/month/day)
- aaaa - transaction number (0000 if D/A General or Digital camera)
- bbbbbb - terminal number (123456 if D/A General or Digital camera)

CURRENCY DISPENSER

M_STATUS

A ¹ alongside the code shows that the M_STATUS applies to stacking dispensers only and a ² shows that the code is for spray dispensers only:

M_STATUS	Meaning
00	No error
01	Dispense requested from a cassette which is not installed or no cassettes installed during self test
02	Too many bills being rejected, for example, undersize, oversize, doubles, extra bills, or unable to learn bill width and singularity
03	Pick failure
04	Pick failure plus low (out of bills)
05	Sensor failure or currency jam in main transport or A short dispense was performed (bills dispensed fewer than bills requested) and no error was reported
06 ²	Divert gate fault caused by one of the following: * Divert gate in wrong position initially * Divert gate failed to move to correct position * Divert gate sensor failure
07	Purge bin not present (unit inoperative)
08 ¹	Purge bin overfill sensor blocked (unit inoperative)
08 ²	Purge bin full (unit inoperative)
09	Communications failure between Command and Execution firmware, or between host and device for USB S1 Dispenser. Current replenishment information may not have been determined

M_STATUS	Meaning
10	Operation not attempted because the device or a cassette type is FATAL from a previous operation. A reset of SOH to HEALTHY must be performed to clear this condition. Current replenishment information is not determined.
11	Operation not attempted due to one of the following conditions: * Bills are still stacked in the transport from a previous operation (stacking dispensers only) * Bills may not have been cleared from the transport due to an unrecovered error during a previous operation * The dispenser has been powered up and no CLEAR/DIAG_CLEAR has been issued A CLEAR or DIAG_CLEAR must be performed to clear this condition. Replenishment information is not determined
12 ¹	Presenter clamping mechanism (or bill alignment mechanism) failed or jammed
13 ¹	Exit shutter jammed open
14 ¹	Exit shutter jammed closed
15 ¹	Reserved
16	Main transport timing disk did not operate correctly or the main ac motor failed
17	Reserved
18 ¹	Currency jam in presenter transport or transport sensor failure
19 ¹	Exit sensor blocked initially on next operation after a good present (bills may not have been taken by cardholder). Current replenishment information is not determined. See “Replenishment Data”.
20 to 27	Reserved
28	An area of NVRAM cannot be accessed (SOH cannot be updated)
29 ¹	Presenter transport timing disk did not operate correctly or motor failed

M_STATUS	Meaning
30	One of the presenter transport sensors failed clear or blocked
31	Reserved
32	Heartbeat on execution processor failed
33 ¹	Communications failure on the I2C bus of the execution processor. On P86 style and USB S1 Dispensers this code denotes a communications failure on the pick interface to pick modules. Also used to denote a Fluditi attack activation.
34 ¹	A present bills operation has been attempted without any bills being stacked or a RETRACT with no bills presented
35 ¹	Operation incomplete due to ac and 24V dc power disconnection through interlock on ATM access door
36 ¹	Pick valve test not attempted because cassettes were installed
37	TI service switched ON
38	TI service switched OFF
39	The LVDT/NTS or associated circuit is malfunctioning
40	Bill parameters have not been set up. This is because this is the first operation since initial power-up or because there was an NVRAM corruption and the NVRAM was re-initialised
41	An attempt to pick from a cassette in a pick module which has been disabled for in-service replenishment
42	The dispenser has not been authenticated by the PC Core module. This code is only generated by USB S1 dispenser variants
43	The serial number of the dispenser control board has changed and has not yet been authorised for use. This code is only generated by USB S1 dispenser variants.

M_STATUS	Meaning
146	SDC link failure

M_DATA

- Dispenser - stacking and spray types
- Divert Gate - spray type only
- Presenter - stacking type only
- Main Motor * - stacking and spray types
- Exit Shutter * - stacking type only
- Presenter Bill Motor * - stacking type only
- Presenter Clamp * - stacking type only
- Sensor/Switch * - stacking and spray types

*** The M_DATA returned for these tests are unique to the tests.**

The number of transport sensors in the currency dispenser depends on the dispenser type as follows:

- Stacking Dispensers:
 - 5674/75 Front Access ATM - Transport Sensors 1 to 7
 - 5674/75 Rear Access ATM - Transport Sensors 1 to 5
 - 5670 ATM Front or Rear Access - Transport Sensors 1 to 5
 - 5684/85/88 ATMs (Through-The-Wall) - Transport sensors 1 to 8
 - Personas 86 ATM (Through-The-Wall) - Transport sensors 1 to 5
 - Personas 87 ATM (Through-The-Wall) - Transport sensors 1 to 5
 - UF-NID, NID, UF-P86 - Transport sensors 1 to 5
 - USB S1 ATMs - Transport sensors 1 to 5
- Spray Dispensers:
 - MCD1 Front Access ATM - Transport Sensors 1 (FEED) and 2 (EXIT)

The highest number in each of the above ranges is the exit sensor for that type.

The M_DATA field contains the execution level command executed and status information returned. The first byte (byte

0) is common to all responses. Byte 0 contains the execution firmware command description and this indicates the format of the status information in M_DATA bytes 1 to n.

In the following lists, a ¹ alongside the code shows that the M_DATA applies to stacking dispensers only and a ² shows that the code is for spray dispensers only.

- Byte 0 - Firmware Level Commands:
 - 00 - Reset
 - 01¹ - Stack
 - 01² - Dispense bills
 - 02¹ - Present
 - 03¹ - Move clamp to present
 - 04 - Clear main transport
 - 05¹ - Check if bills taken
 - 06¹ - Purge
 - 07 - Set bill widths
 - 08 - Set bill singularities
 - 09 - Set bill presentation order
 - 0A - Read dispense counts
 - 0B - Clear dispense counts
 - 0C - Read configured parameters
 - 0D - Read container status
 - 0E - Read (“Set” on P86 style and USB S1) virtual cassette type
 - 0F¹ - Rotate (“Move” on P86 style and USB S1) clamp to home position
 - 10¹ - Close shutter command
 - 11 - Main motor test
 - 12¹ - Pick valve test
 - 12² - Pick solenoid test
 - 13 - Learn bill parameters
 - 14¹ - Exit shutter test
 - 15¹ - Presenter bill motor test
 - 16¹ - Presenter clamp test
 - 17 - Sensor test
 - 1B - Disable Hamming codes.
 - 1C¹ - Pre-present
 - 22² - Divert Gate test

M_DATA - Dispenser

The M_DATA returned under the dispenser format are associated with firmware level commands as follows:

- Stacking dispensers - commands 01, 04, and 13
- Spray dispensers - commands 01, 02, 05, and 17

For SELF_TEST and DIAG_SELF-TEST, the status bytes refer to pick position in the order, top, 2nd, 3rd, bottom instead of virtual cassette types 1, 2, 3, and 4, otherwise the status bytes are identical.

The following format of M_DATA is for pick by virtual type response (the normal case). If the response is for a pick by position command (such as SELF_TEST) then the data has a slightly different format. In this case M_DATA bytes 2, 3, 4, and 5 are auxiliary status codes for the top, second, third, and bottom positions, and M_DATA bytes 6, 7, 8, and 9 are divert reasons for the top, second, third, and bottom positions.

- Byte 1 - Main Dispenser Status:
 - Bit 7 - 1 Virtual cassette type 3 is low on bills
 - Bit 6 - 1 Virtual cassette type 4 is low on bills
 - Bit 5 - 0 Always
 - Bit 4 - 1 Pick fail occurred
 - Bit 3 - 1 Fatal malfunction INOP flag set
 - Bit 2 - 1 Self-Test command
 - Bit 1 - 1 Virtual cassette type 1 is low on bills
 - Bit 0 - 1 Virtual cassette type 2 is low on bills
- Byte 2 - Auxiliary Status Virtual Cassette Type 1:
 - 02 - Purge bin not installed
 - 03 - Pick failure (status 1 - 10H). Picking is tried three times after the initial attempt, each attempt being four or five pick cycles. If a bill has not been picked, the motor is shut off. After a time delay, the motor is turned on and picking is tried again. The cycle, motor off, delay, motor on, pick, is repeated up to three times. If all attempts fail, this status is generated
 - 04 - Invalid pick interrupt. A bill enters the transport from the wrong cassette or one of the pick sensors is faulty
 - 05² - Divert gate not in dispense position during a dispense operation when an attempt is made to dispense a bill into the note tray
 - 06² - Divert gate not in the reject position on receipt of a dispense command or when an attempt is made to reject a bill
 - 09 - Pick sensor failed clear during operation. A bill is seen at the PRE_LVDT/NTS sensor when it is not expected. This may be the result of a faulty pick sensor or a drag-out separating in the transport
 - 0C - Purge bin overfill or SOH active initially or at end of clear
 - 10 - Exit shutter not closed or closed sensor failed indicating not closed, or the I2C comms has failed (only applicable if the dispenser uses an I2C shutter) or open sensor failed indicating open
 - 11 - Pick sensor blocked initially or SOH active initially or at end of clear - occurs on a non-zero dispense.

OR

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CURRENCY DISPENSER

- Bill did not clear pick sensor - occurs when a picked bill jams under a pick sensor or the pick sensor is malfunctioning
- 14 - PRE_LVDT/NTS sensor blocked or SOH active initially or at end of clear
 - 21 - PRE_LVDT/NTS sensor failed clear during operation. A bill seen by the pick sensor has not reached the PRE_LVDT/NTS sensor in time
 - 22 - Pre_LVDT/NTS sensor failed blocked /bill jam during operation
 - 27¹ - Presenter clamp not in home position initially or clamp home sensor failed blocked
 - 40 - Virtual cassette type not installed. A command has been received to pick bills from cassette which is not installed
 - 41¹ - TSEN 1 sensor blocked or SOH active initially on stack or at end of clear (purge path)
 - 41² - TSEN 1 (FEED) sensor blocked or SOH active initially upon initiating a dispense command or on completion of a clear
 - 42¹ - TSEN 2 sensor blocked or SOH active initially upon initiating a dispense or on completion of a clear (clamp position)
 - 42² - TSEN 2 (EXIT) sensor blocked or SOH active initially on stack or at end of clear
 - 43¹ - TSEN 3 sensor blocked or SOH active initially on stack or at end of clear
 - 44¹ - TSEN 4 sensor blocked or SOH active initially on stack or at end of clear
 - 45¹ - TSEN 5 sensor blocked or SOH active initially on stack or at end of clear (exit sensor for 5674/5 rear access and all 5670 dispensers and Personas 86)
 - 46¹ - TSEN 6 sensor blocked or SOH active initially on stack or at end of clear
 - 47¹ - TSEN 7 sensor blocked or SOH active initially on stack or at end of clear (exit sensor for 5674/5 front access)
 - 48¹ - TSEN 8 sensor blocked or SOH active initially on stack or at end of clear (exit sensor for 568X through the wall)
 - 49¹ - I2C fault initially or a Pick Interface fault was detected before the operation started

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- 4A¹ - I2C failure during operation
- 4B - Main transport timing disk too slow
- 4C - Reserved
- 4D - Main transport timing disk failed
- 4E - Pick sensor failed clear initially
- 4F - Pick sensor failed blocked/bill jam during operation
- 50 - Pre LVDT/NTS sensor failed clear initially
- 51¹ - TSEN 1 failed clear initially on stack or at end of clear
- 51² - TSEN 1 (FEED) sensor failed clear initially on dispense or at end of clear
- 52¹ - TSEN 2 failed clear initially on stack or at end of clear
- 52² - TSEN 2 (EXIT) failed clear initially on dispense or at end of clear
- 53¹ - TSEN 3 failed clear initially on stack or at end of clear
- 54¹ - TSEN 4 failed clear initially on stack or at end of clear
- 55¹ - TSEN 5 failed clear initially on stack or at end of clear
- 56¹ - TSEN 6 failed clear initially on stack or at end of clear
- 57¹ - TSEN 7 failed clear initially on stack or at end of clear
- 58¹ - TSEN 8 failed clear initially on stack or at end of clear
- 59¹ - Purge bin overflow failed clear initially
- 5A - Gulp feed detected
- 5B¹ - Interlock open initially
- 5C¹ - Interlock open during operation
- 60¹ - Reference null reading too low LVDT #1 (right hand)
- 60² - Reference null reading too low on NTS
- 61¹ - Reference null reading too low LVDT #2 (left hand)
- 62¹ - Reference null reading too high LVDT #1 (right hand)
- 62² - Reference null reading too high on NTS
- 63¹ - Reference null reading too high LVDT #2 (left hand)

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CURRENCY DISPENSER

- 64 - LVDT/NTS fault - too few timing wheel interrupts received on execution processor within the time for one bill
- 65 - LVDT/NTS fault - execution processor did not receive a response to an LVDT request within the required time
- 70 - Cassette already FATAL because of: too many pick fails, too many rejects, cassette empty, or could not learn from the cassette
- 71 - Too many interrupts at pick sensor or pre-LVDT/NTS sensor
- 72 - Bill parameters were not configured. The singularity at this position was 0FFH. Either bill parameters were not configured after initial power up, or NVRAM was corrupted and reinitialized. Bill widths, singularities and stack order should be checked
- 73¹ - Total thickness of bills picked on a bill learn has reached the limit that can safely be purged
- 75² - TSEN 1 (FEED) sensor failed clear during operation or bill jam before sensor. May be due to a bill jam between the NTS sensor and the FEED sensor
- 76² - TSEN 1 (FEED) sensor failed blocked during operation or bill jam at sensor. May be due to a bill jam at the FEED sensor
- 77² - TSEN 2 (EXIT) sensor failed clear during operation or bill jam before sensor. May be due to a bill jam between the FEED sensor and the EXIT sensor
- 78² - TSEN 2 (EXIT) sensor failed blocked during operation or bill jam at sensor. May be due to a bill jam at the EXIT sensor
- Byte 3 - Auxiliary Status Virtual Cassette Type 2: (Byte 3 has the same description as Byte 2)
- Byte 4 - Auxiliary Status Virtual Cassette Type 3: (Byte 4 has the same description as Byte 2)
- Byte 5 - Auxiliary Status Virtual Cassette Type 4: (Byte 5 has the same description as Byte 2)
- Byte 6 - Divert Reason, Virtual Cassette Type 1:
 - Bit 7 - 1 LVDT transducer could not make an accurate assessment of the number of bills which

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have passed under it, or unable to learn bill parameters during learn bill parameters command

- Bit 6² - 0 always, also for Personas P86 style and USB S1
 - Bit 6¹ - 1 Thermistor indicates cold (below 10 deg. C)
 - Bit 5 - 1 A cassette with the correct ID was not found installed in the dispenser unit. This may be caused by the cassette installed bit being a logic 1 or the correct ID (three bits) was not found.
- Note:** If there is a fault reported against the equivalent auxiliary M_DATA, an error has occurred from a module that does not have a cassette present (or invalid cassette ID).
- Bit 4 - 1 At least one extra bill picked
 - Bit 3 - 1 At least one long bill detected
 - Bit 2 - 1 At least one short bill detected
 - Bit 1 - 1 At least one multiple (double) bill detected
 - Bit 0 - 1 At least one overlength bill detected
 - Byte 7 - Divert Reason, Virtual Cassette Type 2: (Byte 7 has the same description as Byte 6)
 - Byte 8 - Divert Reason, Virtual Cassette Type 3: (Byte 8 has the same description as Byte 6)
 - Byte 9 - Divert Reason, Virtual Cassette Type 4: (Byte 9 has the same description as Byte 6)

M_DATA - Divert Gate

If a divert gate error occurs during a SELF_TEST operation, the following M_DATA information will be returned:

- Byte 0 - 22 divert gate test command
- Byte 1 - Reject position sensor state when divert gate solenoid energised (dispense position)
 - 00 - Not at reject position
 - 01 - At reject position
- Byte 2 - Reject position sensor state when divert gate solenoid de-energised (reject position)
 - 00 - At reject position
 - 01 - Not at reject position
- Byte 3 - Time taken for gate to move from reject position to present position in ms
- Byte 4 - Time taken for gate to move from present position to reject position in ms

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Divert gate M_DATA bytes 3 or 4 may return a value of FFH. This indicates that the divert gate did not reach the desired position before the firmware timed out waiting on the event to occur.

M_DATA - Presenter

The M_DATA returned under presenter format are associated with firmware level commands 02, 03, 06, 0F and 1C.

- Byte 0 - Firmware Level Commands
- Byte 1 - Presenter Status:
 - 00 - Operation successful
 - 01 - Shutter jammed closed during operation or open sensor failed indicating not open
 - 02 - Shutter jammed open during operation or closed sensor failed indicating not closed
 - 04 - Clamp not in position initially
 - 06 - Clamp jammed moving home or home sensor failed showing not home
 - 07 - Clamp jammed moving to present or present sensor failed showing not present
 - 0C - Overfill sensor blocked initially
 - 11 - Overfill sensor failed clear
 - 13 - Overfill sensor blocked at end of purge operation
 - 14¹ - TSEN 1 sensor blocked or SOH line active initially on present or at end of purge
 - 15 - TSEN 2 sensor blocked or SOH line active initially on present or at end of purge
 - 16 - TSEN 3 sensor blocked or SOH line active initially on present or at end of purge
 - 17 - TSEN 4 sensor blocked or SOH line active initially on present or at end of purge
 - 18 - TSEN 5 sensor blocked or SOH line active initially on present or at end of purge
 - 19 - TSEN 6 sensor blocked or SOH line active initially on present or at end of purge
 - 1A - TSEN 7 sensor blocked or SOH line active initially on present or at end of purge
 - 1B¹ - TSEN 8 sensor blocked or SOH line active initially on present or at end of purge
 - 1C - TSEN 1 sensor failed clear initially on present or at end of purge

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- 1D - TSEN 2 sensor failed clear initially on present or at end of purge
- 1E - TSEN 3 sensor failed clear initially on present or at end of purge
- 1F - TSEN 4 sensor failed clear initially on present or at end of purge
- 20 - TSEN 5 sensor failed clear initially on present or at end of purge
- 21 - TSEN 6 sensor failed clear initially on present or at end of purge
- 22 - TSEN 7 sensor failed clear initially on present or at end of purge
- 23¹ - TSEN 8 sensor failed clear initially on present or at end of purge
- 24¹ - TSEN 1 sensor failed clear during operation
- 25 - TSEN 2 sensor failed clear during operation
- 26 - TSEN 3 sensor failed clear during operation
- 27 - TSEN 4 sensor failed clear during operation
- 28 - TSEN 5 sensor failed clear during operation
- 29¹ - TSEN 6 sensor failed clear during operation
- 2A¹ - TSEN 7 sensor failed clear during operation
- 2B¹ - TSEN 8 sensor failed clear during operation
- 2C¹ - TSEN 1 sensor or SOH input failed blocked or bill jam during operation
- 2D - TSEN 2 sensor or SOH input failed blocked or bill jam during operation or clamp present sensor failure
- 2E - TSEN 3 sensor or SOH input failed blocked or bill jam during operation
- 2F - TSEN 4 sensor or SOH input failed blocked or bill jam during operation
- 30 - TSEN 5 sensor or SOH input failed blocked or bill jam during operation
- 31¹ - TSEN 6 sensor or SOH input failed blocked or bill jam during operation
- 32¹ - TSEN 7 sensor or SOH input failed blocked or bill jam during operation
- 33¹ - TSEN 8 sensor or SOH input failed blocked or bill jam during operation
- 34 - Presenter transport timing wheel failed to operate correctly
- 35 - Reserved

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- 36 - Interlock open initially
- 37 - Interlock open during operation
- 38 - Clamp jammed in home position (attempting to move to present)
- 39 - Clamp jammed in present position (attempting to move to home)
- 3A - Clamp home sensor failed indicating home
- 3B - Clamp present sensor failed indicating present
- 3C - Shutter jammed closed or closed during operation
- 3D - Shutter jammed open or opened during operation
- 3E - Shutter open sensor failed indicating openopen or the I2C comms has failed (only applicable if the dispenser uses an I2C shutter)
- 3F - Shutter closed sensor failed indicating closed
- 40 - Purge bin not installed initially on present or at end of purge
- 41 - Bill jam has caused presenter motor to fail
- Byte 2:
 - 00 - Bills seen at purge/overflow sensor during the operation (purge only)
 - 01 - Bills not seen at purge/overflow sensor during the initial purge operation
 - 02 - Bills are still in the presenter transport (purge only)
 - 03 - An unknown present occurred (present only)

M_DATA - Main Motor

- Byte 0 - 11 - Main transport test
- Byte 1:
 - 00 - Good operation
 - 01 - Timing wheel slow
 - 02 - Timing wheel fast
 - 03 - Timing wheel failed
 - 04¹ - Timing wheel too slow
 - 05¹ - Timing wheel too fast
- Byte 2 - Number of timing wheel ticks seen during the test

M_DATA - Exit Shutter

- Byte 0 - 14 - Exit shutter
- Byte 1:
 - 00 - Operation OK
 - 01 - Shutter closed sensor indicated closed when it should have been not closed
- Byte 2:
 - 00 - Operation OK
 - 01 - Shutter open sensor indicated open when it should have been not open
- Byte 3:
 - 00 - Operation OK
 - 01 - Shutter closed sensor indicated not closed when it should have been closed
- Byte 4:
 - 00 - Operation OK
 - 01 - Shutter open sensor indicated not open when it should have been open

M_DATA - Presenter Bill Motor

- Byte 0 - 15 - Presenter bill motor test
- Byte 1 - Forwards Operation Status Slow Speed:
 - 00 - Good operation
 - 01 - Timing wheel slow
 - 02 - Timing wheel fast
 - 03 - Timing wheel failed
 - 04 - Timing wheel too slow
 - 05 - Timing wheel too fast
- Byte 2 - Forwards Operation Status High Speed: (status is the same as Byte 1)
- Byte 3 - Reverse Operation Status Slow Speed: (status is the same as Byte 1)
- Byte 4 - Reverse Operation Status High Speed: (status is the same as Byte 1)
- Byte 5 - No. of timing wheel ticks slow forwards
- Byte 6 - No. of timing wheel ticks fast forwards
- Byte 7 - No. of timing wheel ticks slow reverse
- Byte 8 - No. of timing wheel ticks fast reverse

M_DATA - Presenter Clamp

- Byte 0 - 16 - Presenter clamp test

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CURRENCY DISPENSER

- Byte 1:
 - 0 - Operation successful
 - 1 - Clamp home indicated home when it should have been not home
 - 2 - Bill aligner home indicated home when it should have been not home
- Byte 2:
 - 0 - Operation successful
 - 1 - Clamp present indicated present when it should have been not present
 - 2 - Bill aligner forward indicated forward when it should have been not forward

- Byte 3:
 - 0 - Operation successful
 - 1 - Clamp home indicated not home when it should have been home
 - 2 - Bill aligner home indicated not home when it should have been home
- Byte 4:
 - 0 - Operation successful
 - 1 - Clamp present indicated not present when it should have been present
 - 2 - Bill aligner forward indicated not forward when it should have indicated forward

M_DATA - Sensor/Switch

The M_DATA returned for the sensor/switch test describe the state of the each sensor, where:

- 00 - Sensor clear and healthy
- 01 - Sensor blocked and healthy
- 02 - Sensor clear and unhealthy
- 03 - Sensor blocked and unhealthy

1. All transport sensors give a low reading when blocked and, unless otherwise stated, all other sensors give a high reading when blocked.

2. The state of health for the pick sensors is on the I2C bus/pick interface (see bytes 29 to 32 inclusive).

The clamp sensors have the following meanings:

Clamp Position	Home Sensor	Present Sensor
Home	High	High
Between	Low	High
Present	Low	Low

The following M_DATA are returned for the sensor/switch test:

- Byte 0 - 17H - Device firmware command code
- Byte 1 - A/D reading of zero reference (always 0 on spray type)
- Byte 2 - Pre-LVDT/NTS sensor (reading from execution

CURRENCY DISPENSER

processor) or for P86-Style and USB S1 Dispensers:

Note alignment mechanism sensor (high - home).

- Byte 3 - A/D reading of LVDT No. 1 (stacking) or NTS (spray)
- Byte 4 - A/D reading of LVDT No. 2 (always 0 on spray type)
- Byte 5 - Presenter clamp home position/divert gate reject position sensor
- Byte 6 - Presenter clamp present position sensor (always 0 on spray type)
- Byte 7 - Exit shutter open sensor (high - open) (always 0 on spray type)
- Byte 8 - Exit shutter closed sensor (high - closed) (always 0 on spray type)
- Byte 9 - A/D reading of 80H reference
- Byte 10 - Main transport timing disk sensor
- Byte 11 - Presenter transport timing disk sensor (always 0 on spray type)
- Byte 12 - Purge bin in/reject bin present sensor
- Byte 13 - Purge bin overfill sensor (always 0 on spray type, Personas P86 style and USB S1)
- Byte 14 - TSEN1 (purge path/FEED) sensor
- Byte 15 - TSEN2 (clamp position/EXIT) sensor
- Byte 16 - TSEN3 (always 0 on spray type)
- Byte 17 - TSEN4 (always 0 on spray type)
- Byte 18 - TSEN5 (exit sensor - interior R/A, always 0 on spray type)
- Byte 19 - TSEN6 (always 0 on spray type)
- Byte 20 - TSEN7 (exit sensor for interior 5674/5 F/A) (always 0 on spray type)
- Byte 21 - TSEN8 (exit sensor for TTW) (always 0 on spray type)
- Bytes 22 and 23 have different interpretation according to dispenser type
- Byte 22 - 56XX style dispensers Config1:
 - 0 - F/A
 - 1 - R/A (not applicable if Config2 - 0)
- Byte 23 - - 56XX style dispensers Config2:
 - 0 - TTW
 - 1 - Interior

- Byte 22 and 23 for P86 style and USB S1 dispensers

Byte 22	Byte 23	P86 style dispenser	USB S1 style dispenser
0	0	Long Nose Through the Wall (P86)	Rear access long nose
0	1	Front access Interior and UF-NID	Front access
1	0	Short Nose Through the Wall (P87)	Rear access mid nose
1	1	UF-P86	Rear access short nose

- Byte 24 - Interlock switch:
 - 0 - Open, 24V and ac disconnected (always 0 on spray type)
- Byte 25 - Top pick sensor
- Byte 26 - 2nd pick sensor
- Byte 27 - 3rd pick sensor
- Byte 28 - Bottom pick sensor
- Byte 29 - Pick Interface - Top pick module:
 - Bit 7 - For 'Integrated electronics' and 'Pre-integrated electronics' bunch dispensers Pick sensor LED (1 - on, 0 - off). Always 0 for spray dispenser and all other dispenser types.
 - Bit 6 - Media low sensor (0 - low notes)
 - Bit 5 - Cassette reed switch ID4 (0 - magnet present)
 - Bit 4 - Cassette reed switch ID3
 - Bit 3 - Cassette reed switch ID2
 - Bit 2 - Cassette reed switch ID1 (cassette present)
 - Bit 1 - Thermistor (0 - low temperature) (always 1 on spray type)
 - Bit 0 - Pick sensor (1 - unhealthy) (always 0 on spray type)
- Byte 30 - Pick Interface - 2nd pick module: (Bit information is the same as Byte 29)
- Byte 31 - Pick Interface - 3rd pick module: (Bit information is the same as Byte 29)
- Byte 32 - Pick Interface - bottom pick module: (Bit information is the same as Byte 29)
- Byte 33 - Gulp feed sensor (always 0 on spray type)

CURRENCY DISPENSER

- Byte 34 - SDC command processor switch pack setting (Bit 0 is b0)
- Byte 35 - Execution processor switch pack setting (Bit 0 is e0)
- Byte 36 - Dispense enable switch status (value 00 or 01)
- Byte 37 - Internal LVDT compensation factor (always 0 on spray type)

1. LVDT No. 1 is the right-hand LVDT when facing the dispense control board. LVDT No. 2 is the left-hand LVDT.

2. Transport sensors 6, 7 and 8 will always report 0 for interior 5674/5 R/A, 5670, P86 style dispensers and USB S1 dispensers.

3. Transport sensor 8 will always report 0 for interior 5674/5 F/A.

4. The Pick Interface status will be 0 if the pick module is not configured and FFH if a Pick Interface error is detected.

T_DATA - CURRENCY DISPENSER SELF-TEST

- Byte 0 - Purge bin present/not present:
 - 00 - Not present
 - 01 - Standard bin present
 - 02 - Intelligent bin present

- Byte 1 - Physical currency cassette type in top pick module:
 - 00 - No cassette installed
 - 01 - Cassette type #1 installed
 - 02 - Cassette type #2 installed
 - 03 - Cassette type #3 installed
 - 04 - Cassette type #4 installed
 - 5–255 - Intelligent cassette type installed
- Byte 2 - State of currency cassette in top pick module:
 - 00 - A cassette is not installed
 - 01 - Cassette state is not low
 - 02 - Cassette low
 - 03 - Cassette is empty
 - 04 - Cassette FATAL
 - 05 - Cassette position disabled for replenishment
- Byte 3 - Physical currency cassette type in second pick module (same as Byte 1)
- Byte 4 - State of currency cassette in second pick module (same as Byte 2)
- Byte 5 - Physical currency cassette type in third pick module (same as Byte 1)
- Byte 6 - State of currency cassette in third pick module (same as Byte 2)
- Byte 7 - Physical currency cassette type in bottom pick module (same as Byte 1)
- Byte 8 - State of currency cassette in bottom pick module (same as Byte 2)

Learn Bill Parameters

T_DATA returned for the learn bill parameters test are:

- Byte 0 - Bill width in millimetres (0 if operation not successful)
- Byte 1 - Singularity (0 if operation not successful)
- Byte 2 - The width currently configured
- Byte 3 - The singularity currently configured

CURRENCY DISPENSER TI

The test offered on the Currency Dispenser TI Diagnostic menu is the Tamper Indicator test.

TAMPER INDICATION

The Tamper Indication test displays the status of the dispenser tamper indicator as M_DATA as follows:

- Byte 0 - Purge bin
 - 00 - In
 - 01 - Out
 - 02 - Cannot determine status
- Byte 1 - Top cassette (status as for byte 0)
- Byte 2 - 2nd cassette (status as for byte 0)
- Byte 3 - 3rd cassette (status as for byte 0)
- Byte 4 - Bottom cassette (status as for byte 0)

USB BIOMETRIC FINGER PRINT READER

M_STATUS

M_STATUS	Meaning
0	Device is fully operational.
1	Device is faulty.
2	Device is busy.
3	A communications error has occurred between PC and device.
4	There have been too many bad reads.
5	A finger is present.
6	Finger was removed.
7	The driver was unable to obtain a valid image from the device.
8	Device is not authorised
9	Finger was not removed after imaging process finished
10	Device initialisation failed during start-up or device recovery

M_DATA

- Byte 0 - Command Code:
 - 00H - Reset
 - 00H - Report Tally Mnemonics
 - 94H - Get Configuration
 - 97H - Set Configuration
 - 01H - Collect Biometric
 - 02H - Cancel Collect Biometric
 - 03H - Identify
 - 04H - Disable
 - 05H - Determine State
- Byte 1 - Device Command Code
- Byte 2 - Device Command Parameter
- Byte 3 - 6 Device Error Code
 - See the table below
- Byte 7 - 10 NCR Communications Library Status & Errors

□ See the table below

Device Error Codes

Error Code	Description
00H 00H 00H 00H	Success
00H 00H 00H 01H	ErrAlreadyOpen
00H 00H 00H 02H	ErrInvalidHandle
00H 00H 00H 03H	ErrIndexCorrupt
00H 00H 00H 05H	ErrDeviceBusy
00H 00H 00H 06H	ErrBadParameter
00H 00H 00H 07H	ErrSerialNotFound
00H 00H 00H 0AH	ErrCommandUnimplemented
00H 00H 00H 0BH	ErrInvalidCommand
00H 00H 00H 0DH	ErrFingerNotPresent
00H 00H 00H 11H	ErrInvalidBufferLength
00H 00H 00H 12H	ErrFrameNotAvailable
00H 00H 00H 1AH	ErrBadRotation
00H 00H 00H 1EH	ErrBufferIsEmpty
00H 00H 03H E8H	ErrDataNotAvailable
00H 00H 13H 88H	errNullSerialOrName
00H 00H 13H 89H	errNotA2600Device
00H 00H 13H 8AH	errGetInfoFailed
00H 00H 13H 8BH	errUnknownInterface
00H 00H 13H 8CH	errInvalidPID
00H 00H 13H 8DH	errInvalidVID
00H 00H 13H 8EH	errWritingCommand
00H 00H 13H 8FH	errReadingData
00H 00H 13H 90H	errUnknownChip
00H 00H 13H 91H	errInvalidChipRevision
00H 00H 13H 92H	errTransmissionLost
00H 00H 13H 93H	errCalibrationFailed
00H 00H 13H 94H	errCantReadCalibration
00H 00H 13H 95H	errResCalibFailed
00H 00H 13H 96H	errCapCalibFailed
00H 00H 13H 97H	errFingerPresent
00H 00H 15H 7DH	INVALID_DEVICE_HANDLE
00H 00H 15H 7EH	DEVICE_NOT_FOUND

Error Code	Description
00H 00H 15H 7FH	DEVICE_NOT_OPEN
00H 00H 15H 80H	IO_ERROR
00H 00H 15H 81H	INSUFFICIENT_RESOURCES
00H 00H 15H 82H	INVALID_PARAMETER
00H 00H 15H 83H	INVALID_BAUD_RATE
00H 00H 15H 84H	DEVICE_NOT_OPENED_FOR_ERASE
00H 00H 15H 85H	DEVICE_NOT_OPENED_FOR_WRITE
00H 00H 15H 86H	FAILED_TO_WRITE_DEVICE
00H 00H 15H 87H	EEPROM_READ_FAILED
00H 00H 15H 88H	EEPROM_WRITE_FAILED
00H 00H 15H 89H	EEPROM_ERASE_FAILED
00H 00H 15H 8AH	EEPROM_NOT_PRESENT
00H 00H 15H 8BH	EEPROM_NOT_PROGRAMMED
00H 00H 15H 8CH	INVALID_ARGS
00H 00H 15H 8DH	NOT_SUPPORTED
00H 00H 15H 8EH	OTHER_ERROR

NCR Communications Library Status & Errors

Description	Error Code
Port Communication Errors	
CE_PE_BASE	0x0000FF00
ALREADY_INIT	CE_PE_BASE + 1
ALREADY_OPEN	CE_PE_BASE + 2
ALREADY_CLOSED	CE_PE_BASE + 3
PORTCLOSE	CE_PE_BASE + 4
DEVICE_DISCONNECTED	CE_PE_BASE + 5

Description	Error Code
Port Write Errors	
CE_WE_BASE	0x00FF0000
LARGEPACKET	CE_WE_BASE + 1
OPERATIONPENDING	CE_WE_BASE + 2
Port Read Errors	
CE_RE_BASE	0x00FF0064
TRANSMISSIONERROR	CE_RE_BASE + 1
Miscellaneous	
Communication Errors	
CE_ME_BASE	0xFF000000
READTIMEOUT	CE_ME_BASE + 1
WRITETIMEOUT	CE_ME_BASE + 2
DATA_INCOMPLETE	CE_ME_BASE + 3
BUFFER_TOO_SMALL	CE_ME_BASE + 4
UNKNOWN_READERROR	CE_ME_BASE + 5
UNKNOWN_WRITEERROR	CE_ME_BASE + 6
SAME_FIRMWARE	CE_ME_BASE + 7
FIRMWARE_UPDATE_REQUIRED	CE_ME_BASE + 8
MISSING_FIRMWARE_FILE	CE_ME_BASE + 9
FIRMWARE_FILE_ERROR	CE_ME_BASE + 10
FIRMWARE_WRITE_ERROR	CE_ME_BASE + 11
COMMAND_SENT	CE_ME_BASE + 12
COMMAND_CANCELED	CE_ME_BASE + 13
DEVLIB_DOESNOT_SUPPORT	CE_ME_BASE + 14

NLX PC CORE - MISC I/F BOARD

M_STATUS

M_STATUS	Meaning
0	Good
3	Error in test

M_DATA

- Byte (0-7) - Test state of I/O line (0-7):
 - 30H - Good
 - 31H - Stuck high
 - 32H - Stuck low

NLX PC CORE - NIGHTSAFE DEPOSITORY

M_STATUS

M_STATUS	Meaning
0	Bag drop switch is open
1	Bag drop switch is closed
2	Deposit not done and bag drop switch is open

M_DATA

M_DATA is only returned for the enhanced version of the nightsafe depository as follows:

- Bit 0:
 - 0 - Door closed
 - 1 - Door open
- Bit 1:
 - 0 - Bag drop switch open
 - 1 - Bag drop switch closed

NLX PC CORE - ALARMS

M_STATUS

M_STATUS	Meaning
00	No error
03	Turnaround test
146	SDC link failed

M_DATA

For M_STATUS 00, the M_DATA is as follows:

- Byte 0 - Sensor number 0 (Composite):
 - 30 - Inactive
 - 31 - Active
- Byte 1 - Sensor number 1 (Safe Door):
 - 30 - Inactive
 - 31 - Active
- Byte 2 - Sensor number 2 (Silent Alarm):
 - 30 - Inactive
 - 31 - Active
- Byte 3 - Sensor number 3 (Tamper):
 - 30 - Inactive
 - 31 - Active
- Byte 4 - Sensor number 4 (Not Allocated):
 - 30 - Inactive
 - 31 - Active
- Byte 5 - Sensor number 5 (Not Allocated):
 - 30 - Inactive
 - 31 - Active

For M_STATUS 03, the turnaround tests for the miscellaneous interface devices are run from the INDICATORS menu.

NLX PC CORE - MEDIA ENTRY INDICATORS

M_STATUS

M_STATUS	Meaning
0	GOOD
3	Turnaround test failed

UPS

Unless stated otherwise, the following applies to both the generic control device and the HID power device.

M_STATUS

M_STATUS	Meaning
00	Operations successful
01	Command not Supported
02	Battery low
03	Device failure
04	Battery failure
05	Command failure
06	Communications error (RS232 comms failure in data mode)
20	Device inoperative
95	RS232 Turnaround failure

M_DATA

- Byte 0 - Command code
- Byte 1 - Mode identifier:
 - 30 - Control mode
 - 31 - Data mode
- Byte 2 - Hardware Identifier:
 - 00 - Generic Device
 - 01 - HID device
- Byte 3 - Main supply out status indicator:
 - 30 - UPS does not indicate mains supply is out
 - 31 - UPS indicates mains supply is out
- Byte 4 - Low status indicator:
 - 30 - Battery not indicating low
 - 31 - Battery indicating low

UPS

- Bye 5 - Detailed status indicator (HID device only)
 - 30 - Good
 - 31 - Overload
 - 32 - Over temperature
 - 33 Internal failure
 - 34 - Communications lost

USB MISC I/F BOARD

M_STATUS

M_STATUS	Meaning
00	Good
01	Communications Failure
03	Not Authorised - Device hardware has changed

USB NIGHTSAFE DEPOSITORY

M_STATUS

M_STATUS	Meaning
00	Good
00	Bag drop switch is open
01	Bag drop switch is closed
02	Deposit not done error and bag drop switch is open
03	Not Authorised - Device hardware has changed

M_DATA

M_DATA is only returned for the enhanced version of the night safe as follows:

- Bit 0:
 - 0 - Door closed
 - 1 - Door open
- Bit 1:
 - 0 - Bag drop switch open
 - 1 - Bag drop switch closed

USB ALARMS

M_STATUS

M_STATUS	Meaning
00	Good
03	Not Authorised - Device hardware has changed

M_DATA

For M_STATUS 00, the M_DATA is as follows:

- Byte 0 - Sensor number 0:
 - 30H - Inactive
 - 31H - Active
- Byte 1 - Sensor number 1:
 - 30H - Inactive
 - 31H - Active
- Byte 2 - Sensor number 2:
 - 30H - Inactive
 - 31H - Active
- Byte 3 - Sensor number 3:
 - 30H - Inactive
 - 31H - Active
- Byte 4 - Sensor number 4:
 - 30H - Inactive
 - 31H - Active
- Byte 5 - Sensor number 5:
 - 30H - Inactive
 - 31H - Active
- Byte 6 - Sensor number 6:
 - 30H - Inactive
 - 31H - Active
- Byte 7 - Sensor number 7:
 - 30H - Inactive
 - 31H - Active

USB INDICATORS

M_STATUS

M_STATUS	Meaning
00	Good
03	Not Authorised - Device hardware has changed

USB MEDIA ENTRY INDICATORS

M_STATUS

M_STATUS	Meaning
00	Good
03	Not Authorised - Device hardware has changed

USB AUDIO

M_STATUS

M_STATUS	Meaning
00	Good
03	Not Authorised - Device hardware has changed

M_DATA

- Byte 0:
 - 0 - IO_COMMAND
 - 1 - DIAG_COMMAND
- Byte 1:
 - Command Code
- Byte 2:
 - 0 - No Jack Plug in Jack Socket
 - 1 - Jack Plug in Jack Socket
- Byte 3:
 - 0 - PUBLIC output channel active
 - 1 - PRIVATE output channel active
- Byte 4:
 - 0 - Jack Socket event monitoring disabled
 - 1 - Jack Socket event monitoring enabled
- Byte 5:
 - 0 - Semi-automatic Channel Switching Mode disabled
 - 1 - Semi-automatic Channel Switching Mode enabled
 - 2 - Automatic Channel Switching Mode
- Byte 6:
 - 0 - No switch pack
- Byte 7:
 - 0 - No switch pack

USB MINI MISCELLANEOUS INTERFACE

M_STATUS

M_STATUS	Meaning
00	No error
01	Unable to communicate with MiniMisc device hardware

M_DATA

None

USB BARCODE READER

M_STATUS

M_STATUS	Meaning
00	Operation Successful
01	Barcode Data Error or Timeout
02	Device Inoperative
03	USB Communication Error
04	Barcode Read Error or Timeout (≥ 30 consecutive errors)
05	Device Scanning
07	No Data

M_DATA

- Byte 0: Command Class
- Byte 1: Command Code
- Byte 2:
 - 0 - Not Connected
 - 1 - Connected

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